

Bonds and Bond Markets

What you own, where the price comes from, why it moves, and what it does to a portfolio

What this is. A first-principles tutorial on bonds: the instrument, the markets it trades in, the arithmetic that prices it, the forces that move that price, and the decisions that face someone putting money into it. Bonds are usually taught either as a spreadsheet exercise — here is the present-value formula, now compute duration — or as a vague reassurance that they are the safe part of a portfolio. Both are bad preparation. The formula tells you nothing about why a yield is 4% rather than 7%, and the reassurance was falsified in 2022, when a portfolio of the safest bonds in the world lost more than it ever had in a calendar year.

The claim of this document is that one decomposition organises the whole subject. A bond's yield is a sum of components, each of which is payment for carrying a specific risk. Which components are present tells you what market you are in; how large they are tells you what you are being paid; how they move tells you what your return will be. Government bonds, corporate bonds, emerging-market debt, mortgage securities and inflation-linked bonds are not five subjects. They are one sum with different terms switched on.

Who it is for. Someone numerate and technically strong who is not a fixed-income specialist and who wants to understand bonds well enough to make decisions — to judge whether a yield is attractive, to know what a fund actually holds, to predict how a position behaves when the central bank moves, and to recognise the specific ways bond investors lose money. No prior fixed-income vocabulary is assumed. Intuition comes first everywhere: every formula is preceded by what it means and followed by a number.

How to read it. Seven parts, and they are genuinely sequential — Part III uses Part I's vocabulary, and Part V is Part III applied.

- Part I (§1–§2) — the object. What a bond is, the mental models that mislead, the three identities that carry the rest of the document, and the life of a bond from issuance to repayment or default.
- Part II (§3–§5) — the map. Government bond markets and the risk-free curve; sovereign borrowing outside the United States; corporate credit and where a bond sits in a company's capital structure.
- Part III (§6–§9) — valuation and risk. Discounting and yield; duration and convexity; the family of spread measures and which of them mean the same thing; credit risk, default probability and recovery.
- Part IV (§10–§12) — where yields come from. The macroeconomics that set the level of rates, the shape of the curve and what it does and does not forecast, and the supply-and-demand facts that academic models leave out.
- Part V (§13–§15) — behaviour. Why bonds sell off and rally, with the major episodes as worked examples; the bond–equity relationship and why it flipped sign; and the relationship to currencies.
- Part VI (§16) — default. What actually happens when a borrower cannot pay, for companies and for countries.
- Part VII (§17–§19) — investing. Constructing a bond portfolio, a consolidated list of the ways people lose money in this asset class, and a synthesis.

§20 is the reference list, grouped by kind. Appendix A collects the concepts the main text leans on without fully developing, ordered as a build-up.

If you read four things, read §1.4 (the three identities the document is built on), §6.4 (why yield to maturity is not your expected return), §7.5 (the cushion: how much of a sell-off a bond's yield can absorb), and §14 (what bonds do and do not do for an equity portfolio).

Relationship to the other notes. Simple and Log Returns covers the return conventions used throughout. Stochastic Processes develops the Brownian machinery behind the term-structure models mentioned in §11.4. Portfolio Construction and the Covariance Matrix is the right frame for §17's allocation questions, and Market Regimes and Machine Learning for §14's correlation-regime discussion. Dealer Hedging and Gamma Exposure explains the option machinery that §7.6 borrows for callable bonds and mortgages. Each note stands alone.

Epistemic tags. Claims are flagged where the status changes what you should do:

- [Fact] — replicated across independent datasets or implementations, with broad agreement among people who have looked.
- [Contested] — documented, but with live disagreement about magnitude, robustness, or cause.
- [Hypothesis] — a proposed mechanism, not decisively tested.
- [Practice] — practitioner convention. May well be right; the evidence is private or absent.

Untagged sentences are definitions, arithmetic, or institutional description. Numbers computed from the models in this document are labelled [Computed]; the generating code is committed alongside as `figures/bd_*.py`, and each script prints the numbers quoted in the text, so you can change the assumptions and rerun.

Notation. A bond promises a schedule of payments. C_t is the cash flow at time t , c the annual coupon rate as a fraction of face value, and F the face (par, principal, redemption) amount — 100 by convention, so prices are quoted per 100 of face. τ is time remaining to maturity in years, T the maturity date.

P is the price. Unless a passage says otherwise it means the dirty price, the full amount that changes hands; P^{clean} is the quoted price, which excludes accrued interest (§6.6).

Symbol	Name	What it is
y	yield to maturity	The single rate that makes discounted cash flows equal the price (§6.3)
$z(t)$	spot or zero rate	The rate for a single payment at t , with no intervening coupons (§6.2)
$f(t_1, t_2)$	forward rate	The rate for borrowing between two future dates, implied by today's curve
$Z(t)$	discount factor	Today's value of 1 unit paid at t ; $Z(t) = (1 + z(t))^{-t}$
D_{mac}	Macaulay duration	Cash-flow-weighted average time to payment, in years (§7.1)
D	modified duration	Percentage price change per unit change in yield (§7.2)
DV01	dollar value of a basis point	Price change per 0.01% yield move, in currency (§7.2)
$\mathcal{C}$	convexity	Curvature of price in yield; the second derivative, scaled (§7.4)
s	spread	Yield in excess of a reference rate (§8)
λ	hazard rate	Instantaneous probability of default per unit time (§9.2)
R	recovery rate	Fraction of face value creditors receive in default (§9.3)
r	real interest rate	The rate net of inflation (§10.1); r^* is its long-run level
π^e	expected inflation	Inflation the market expects over a stated horizon
TP	term premium	Extra yield for holding a long bond rather than rolling short ones (§10.4)
EL	expected loss	Default probability times loss given default, per unit of exposure (§9.1)
CRP	credit risk premium	Extra spread demanded for bearing default risk beyond its expected cost (§9.2)
ℓ	liquidity premium	Extra yield for a bond that is harder to sell (§3.3)
g	growth rate	Real growth rate of the economy, compared with r in debt dynamics (§10.6)
o	option cost	Yield given up for an option the borrower holds against you, such as a call or prepayment right (§7.6)

A basis point (bp) is one hundredth of a percentage point: 100bp = 1%. Yields and spreads are quoted in basis points almost universally, and this document follows that. Δy is a change in yield, always in the same units as y itself (so a 100bp move is $\Delta y = 0.01$).

The running example is a 10-year bond with a 4% annual coupon paid semi-annually, issued at par, so its price is 100 and its yield is 4.00%. It has a modified duration of 8.18 years and a convexity of 78.9. When a passage says “the bond” without qualification, this is the one.

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Part I — The object

1. What a bond is

1.1 The promise

A bond is a loan that has been cut into tradeable pieces.

That sentence contains everything structural. Someone needs money — a government, a company, a city, a pool of mortgages. Rather than borrow it from one bank under one contract, they write a standardised promise: I will pay the holder of this piece of paper 2 currency units every six months for ten years, and 100 units at the end. Then they sell tens of thousands of identical copies of that promise to whoever will buy them. Each copy is a bond. The buyer is a lender; the seller is a borrower.

Two consequences follow immediately, and they are the two facts that make bonds behave the way they do.

The payments are fixed in advance. Not guaranteed — the borrower may fail — but specified. The bond says 2 units every six months, and if the borrower's profits triple, the bond still pays 2. If the borrower's profits collapse, the bond still demands 2, and the borrower must find it or default. A shareholder's claim floats with the fortunes of the business; a bondholder's claim does not. This is the fundamental asymmetry of debt: fixed upside, and downside only when the promise breaks.

The promise is transferable. Because the pieces are standardised and identical, you can sell yours to someone else without renegotiating anything. The original borrower does not need to know or approve. That is the difference between a bond and a bank loan, and it is what creates a market — a continuously updating price for the same promise as the world's opinion about it changes.

Now hold those two facts together and you can already deduce the central puzzle of the asset class. The payments are fixed. The price is not. So the entire variation in a bond's price is variation in what the market will pay for a fixed schedule of payments. Nothing about the bond itself changes when its price falls 20%; the coupon is the same, the maturity is the same, the borrower is the same. What changed is the price of money, or the market's belief that the promise will be kept.

That is worth stating as the first principle, because almost every mistake in fixed income is a failure to keep it in view:

A bond is a fixed schedule of promised payments. Its price is what that schedule is currently worth. Only two things can move the price: the rate at which future money is discounted, or the market's belief about whether the payments will arrive. Everything in this document is one of those two channels.

1.2 The intuitions to discard

Four mental models about bonds are common, intuitive, and wrong in ways that cost money. It is worth killing them explicitly before building anything.

"Bonds are safe." Safe against what? A US Treasury bond carries essentially no risk that you will fail to be repaid — the government prints the currency it owes. But in 2022 the Bloomberg US Aggregate index, dominated by Treasuries and high-quality corporates, returned about -13%, its worst calendar year in the half-century the index has existed. Nothing defaulted. Long Treasuries did far worse: a 30-year bond bought at 2% and sold at 4% loses roughly a third of its value, which is equity-sized. [Fact] The safety of a government bond is safety of repayment, and that is a different thing from safety of price. Confusing the two is the single most expensive error available in this asset class, and it is enshrined in the phrase "risk-free asset", which means free of default risk and nothing else.

"Bonds pay a fixed rate, so I know my return." You know your return only if you hold to maturity, reinvest every coupon at exactly the original yield, and the borrower pays in full. Fail any of those and your realised return differs, sometimes dramatically. §6.4 makes this precise: yield to maturity is an internal rate of return, and an IRR is a quoted number, not a forecast.

"A bond fund is a bond." It is not, and the difference matters. An individual bond has a maturity date at which, absent default, you get 100 back. That date anchors your outcome: hold it and price movements in between are unrealised. A bond fund has no maturity date. It holds a rolling population of bonds, sells them before they mature to keep its duration constant, and therefore never "pulls to par". If rates rise and stay risen, the individual bondholder eventually gets their money back; the fund holder's loss is permanent in price terms and is only recovered through the higher yields the fund now earns. Both end up in a similar place over a horizon equal to the duration (§17.3), but the paths and the psychology are completely different.

"Higher yield means a better investment." Yield is compensation for risk. A bond yielding 9% when Treasuries yield 4% is not offering you 5% of free money; it is offering you 5% in exchange for the possibility that you will not be paid, plus the possibility that you cannot sell it when you want to. Over a full cycle, high-yield bonds have historically delivered a net premium after default losses — but a considerably smaller one than the headline spread suggests, because a meaningful fraction of the spread is expected loss rather than reward (§9.5). The instinct to reach for yield is the most reliably punished instinct in fixed income, and it is measurable: corporate bond funds systematically tilt toward bonds yielding more than their benchmark, most of all when rates are low, because it attracts investor flows — and the tilt raises raw returns but delivers negative risk-adjusted ones (Choi & Kronlund, 2018).

1.3 What a bond is not

Defining the object against its neighbours sharpens it. Each row below is a claim on someone's money; the differences are where bond behaviour comes from.

	Payments	If the borrower fails	Maturity	Upside
Bond	Fixed and specified	Legal claim, senior to equity	Fixed date	Capped at the promised payments
Bank loan	Fixed or floating	Same claim, but privately negotiated	Fixed, often amortising	Capped; usually not traded
Bank deposit	Interest at the bank's discretion	Deposit insurance up to a limit	None	Capped

	Payments	If the borrower fails	Maturity	Upside
Preferred stock	Fixed dividend, but skippable	Junior to all debt	Often none	Capped
Common equity	Whatever is left over	Last in line, usually zero	None	Unlimited
Convertible bond	Fixed, plus a share conversion right	Bond claim	Fixed date	Participates in equity upside

Two rows deserve a second look.

Preferred stock is the instructive comparison. It looks like a bond — a fixed periodic payment, no maturity, no share in growth. But the issuer can skip the payment without triggering default. That single difference moves it from the debt side of the balance sheet to the equity side, and its price behaves accordingly: in a crisis, preferreds trade like equity, not like bonds. The definitional core of a bond is not “fixed payments” but “fixed payments you can sue over.”

A convertible bond is genuinely two instruments. It is a bond plus a call option on the issuer’s equity, and it can usefully be priced as exactly that. This is the first appearance of a theme that recurs throughout: many fixed-income instruments that look like distinct products are a plain bond plus or minus an option, and finding the decomposition is how you understand them (§7.6, §8.4).

1.4 Three identities

Here is the spine. Three equations carry this entire document, and they are three views of the same object: what the bond is worth, what its yield is made of, and what your return will be.

Identity 1 — the price. A bond is worth the present value of what it pays, weighted by the chance of being paid:

$$P = \sum_t \underbrace{C_t}_{\text{promised cash flow}} \times \underbrace{Z(t)}_{\text{discount factor}} \times \underbrace{Q(t)}_{\text{probability-weighted chance of payment}}$$

C_t is the promise and comes from the bond’s legal documents. $Z(t)$ is the price today of one unit of currency delivered at time t , and it comes from the market for government debt. $Q(t)$ captures the probability-weighted chance the payment arrives, and comes from the market’s view of the borrower. For a US Treasury, $Q(t) = 1$ and the whole thing reduces to discounting. For a distressed corporate bond, $Q(t)$ is doing most of the work.

Everything in Part III is the arithmetic of this identity. Everything in Parts IV and V is about what makes $Z(t)$ and $Q(t)$ move.

Identity 2 — the yield decomposition. Rather than working with Z and Q directly, the market quotes a single number, the yield, and decomposes it into additive pieces:

$$y = \underbrace{r + \pi^e}_{\text{expected short rate path}} + \underbrace{\text{TP}}_{\text{term premium}} + \underbrace{\text{EL} + \text{CRP}}_{\text{credit spread}} + \underbrace{\ell}_{\text{liquidity premium}} + \underbrace{o}_{\text{option cost}}$$

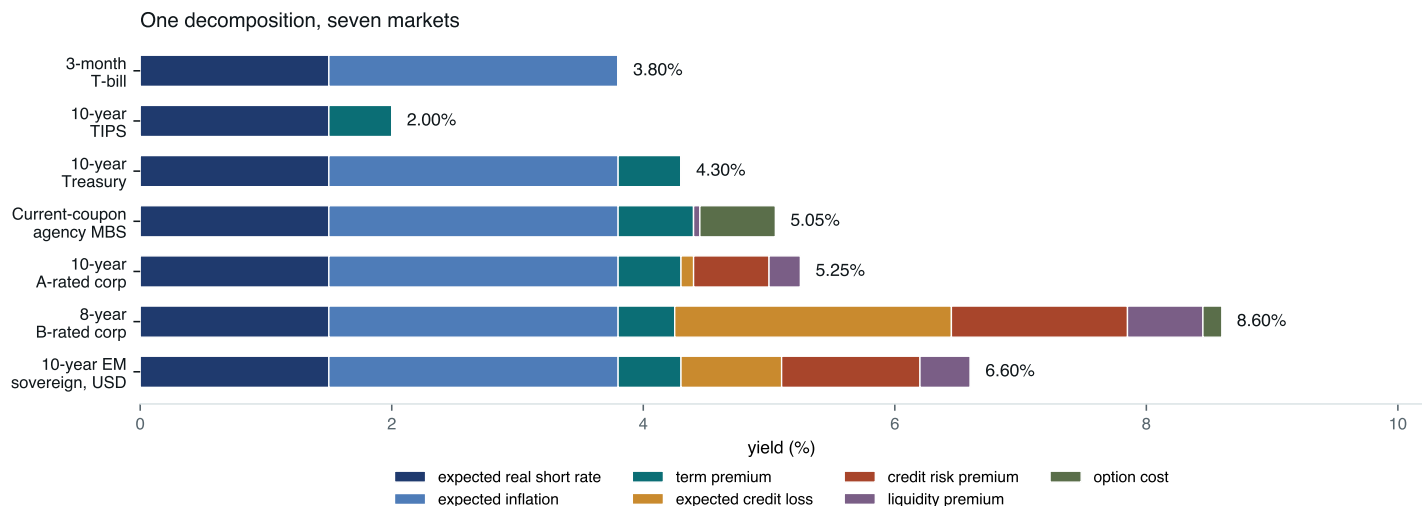
Reading left to right: r is the expected average real short-term interest rate over the bond’s life; π^e is expected inflation over the same period; together they are what you would earn rolling short-term government paper. TP is the term premium, the extra yield for committing for a long time instead. These first three terms are the government curve. Then EL is expected credit loss — default probability times loss given default — and CRP is the credit risk premium, the extra return demanded for bearing default risk beyond its expected cost. ℓ compensates for not being able to sell easily. o is the value of any options the borrower holds against you, such as the right to repay early.

This is the map of the entire asset class, and the most useful half-page in this document. Every bond market in the world is this sum with a different subset of terms switched on:

Instrument	r	π^e	TP	EL+CRP	ℓ	o
3-month Treasury bill	✓	✓	—	—	—	—
10-year Treasury note	✓	✓	✓	—	—	—
10-year inflation-linked (TIPS)	✓	—	✓†	—	small	—
Investment-grade corporate	✓	✓	✓	small	small	—
High-yield corporate	✓	✓	✓	large	✓	✓
Emerging-market sovereign, hard currency	✓	✓	✓	✓	✓	—
Agency mortgage pass-through	✓	✓	✓	—	small	large
Callable corporate	✓	✓	✓	✓	✓	✓
Corporate floating-rate note	✓	✓	≈0‡	✓	small	—
Leveraged loan	✓	✓	≈0‡	large	✓	✓§

† A real term premium, for committing to a long real rate. ‡ A floater's coupon resets to the prevailing short rate, so it earns essentially no term premium and carries almost no duration (§3.2). § A loan's option is the borrower's right to repay at par, exercised when its credit improves rather than when rates fall (§5.4).

The figure below draws the same table with representative magnitudes.



Three things to take from the picture. The TIPS bar is short not because inflation-linked bonds are poor value but because expected inflation has been removed from the number — a real yield and a nominal yield are quoted in different units, and comparing them directly is a category error. About half of the B-rated corporate bar's spread is expected loss rather than premium — compensation for defaults you should expect to suffer — which is the difference between a spread and a return. And the mortgage bar's excess over Treasuries is almost entirely option cost — the homeowner's right to refinance — so an investor who buys agency mortgages for "spread" is being paid to sell options, not to take credit risk.

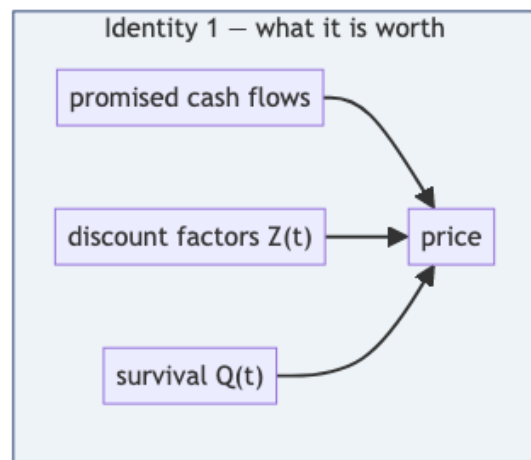
Identity 3 — the return. Identity 2 says what the yield is made of. This one says what happens to your money. Over a holding period, to a good approximation:

$$r_{\text{holding}} \approx \underbrace{y \Delta t}_{\text{carry}} + \underbrace{\text{roll-down}}_{\text{aging down the curve}} - \underbrace{D \Delta y}_{\text{duration effect}} + \underbrace{\frac{1}{2} \mathcal{C} (\Delta y)^2}_{\text{convexity}} - \underbrace{\bar{L}}_{\text{realised credit losses}}$$

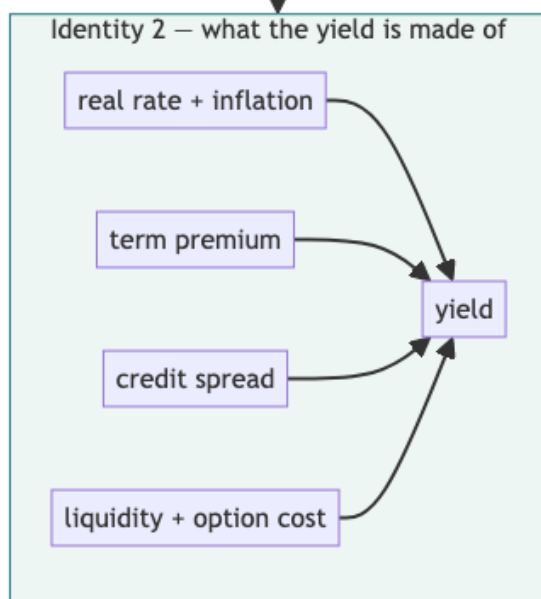
The first two terms are the return you get for doing nothing, and you know them the day you buy. The third is the bet: duration D converts a change in yield into a change in price, with a minus sign, because yields up means prices down. The fourth is a small correction that is almost always in your favour for a normal bond, and against you for a mortgage or a callable bond. The last is what the borrower failed to pay.

Every question of the form "why did my bonds lose money?" is answered by naming a term in this identity. Every bond strategy is a bet on one of them. The whole of Part V is this identity applied to history.

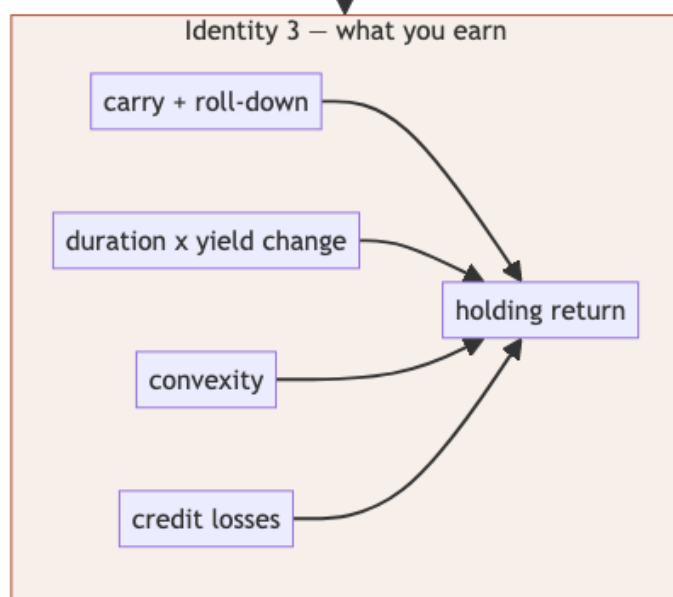
The three identities connect like this:



invert to a single rate



differentiate in time and yield



Identity 1 is the machinery. Identity 2 is Identity 1 inverted into a single quoted rate and then split into the risks that rate is paying for. Identity 3 is Identity 2 differentiated — what happens to your money when the components move. Know the pieces of a yield and you know both the risks you are running and the return you can expect. That is the whole thesis.

1.5 Why anyone issues bonds

Half of the market's behaviour comes from the borrower's side, and it is usually skipped. Issuers are not passive suppliers of investment product; they are optimisers, and what they optimise shapes what is available to buy.

Governments issue because they spend more than they tax and must fund the difference. They are also, uniquely, the manufacturer of the currency they borrow in, which is why a government borrowing in its own currency is in a different risk category from every other borrower (§4.2). Beyond funding, governments issue because the market wants the product: safe, liquid collateral is a raw material for the financial system, and there is persistent evidence that investors pay a premium for Treasury securities over and above their cash flows — a “convenience yield” worth perhaps 70 basis points on average (Krishnamurthy & Vissing-Jorgensen, 2012). [Fact]

Companies issue because debt is cheaper than equity, for two reasons. Interest is tax-deductible in most jurisdictions and dividends are not, which is a direct subsidy. And debt is a less risky claim, so investors require less return for it. The counterweight is that debt must be serviced regardless of circumstances, so more of it raises the probability of financial distress. The classical treatment of this trade-off runs from Modigliani & Miller (1958) through the agency-cost literature (Jensen & Meckling, 1976; Myers, 1977), and is summarised in §5.1.

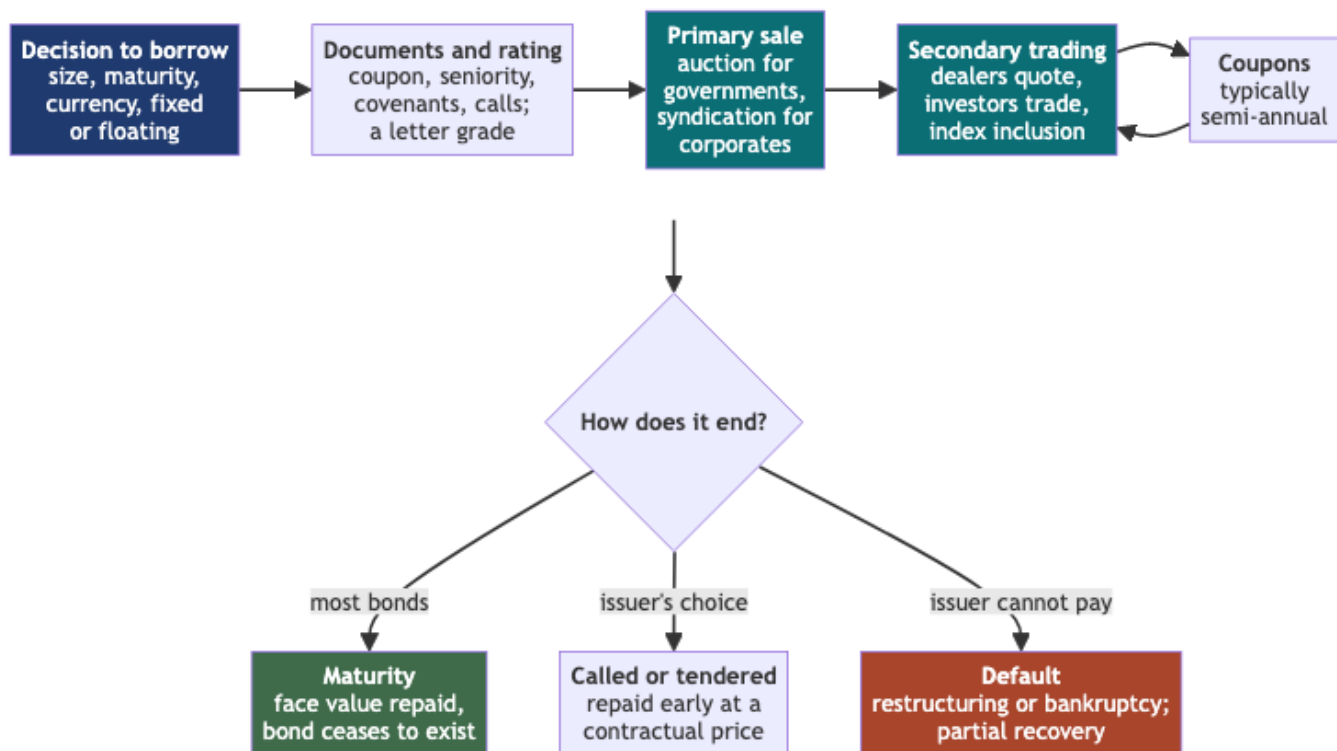
The timing matters to you as an investor. Companies issue when it is cheap to issue, which means supply is heaviest exactly when spreads are tight and credit standards are loose. Greenwood & Hanson (2013) show that when the quality of issuers coming to market deteriorates — measured by the share of issuance from low-rated firms — subsequent excess returns on corporate bonds are low. [Fact] The composition of new supply is a usable signal precisely because issuers are better informed about the cycle than buyers are. This is a general and slightly uncomfortable principle: in a market where one side chooses when to transact, that side has an edge.

§1 Key takeaways

1. A bond is a fixed schedule of promised payments, made tradeable. All price variation is variation in what the market pays for that fixed schedule.
2. Only two channels move a bond's price: the discount rate applied to future money, and the belief that the money will arrive. Every mechanism in this document routes through one of them.
3. “Risk-free” means free of default risk and nothing else. A long Treasury can and does lose a third of its value.
4. The definitional core of a bond is not fixed payments but enforceable fixed payments — that is what separates it from preferred stock.
5. Identity 2, the yield decomposition, is the map of the asset class: every bond market is the same sum with a different subset of terms switched on.
6. Identity 3, the return decomposition, answers every “why did my bonds move” question; carry and roll are known in advance, duration is the bet.
7. Issuers choose when to sell. Supply is heaviest when it is cheapest to issue, which is when future returns are worst.

2. The life of a bond

Following one bond from birth to death is the fastest way to meet the institutions. The vocabulary introduced here — indenture, auction, on-the-run, TRACE, make-whole — recurs throughout, and each term is easier to learn attached to a stage than as a glossary entry.



2.1 Birth: the terms of the promise

Before a bond exists, someone decides its shape. Four decisions determine almost everything about how it will behave.

Maturity. How long until the principal is repaid. This is the single biggest determinant of price volatility: a 30-year bond moves roughly nine times as much per unit of yield change as a 2-year bond (§7.2). Issuers pick maturity by trading off the cost of long-term funding against the risk of having to refinance at a bad moment — and the choice is consequential for the market as a whole, since the government's maturity choice changes the supply of duration that investors must hold (Greenwood, Hanson & Stein, 2015).

Coupon. The periodic payment, usually set so the bond prices at or near 100 on issue day. A bond issued when 10-year yields are 4% will carry roughly a 4% coupon. This is why a bond's coupon is a fossil record of the rate environment at its birth, and why bond portfolios accumulated over decades contain wildly heterogeneous coupons. Some bonds pay no coupon at all and are sold at a discount to face — zero coupon bonds, of which Treasury STRIPS are the standard example.

Currency. Whoever bears the exchange-rate risk is determined here, and for sovereign borrowers it is nearly the whole story (§4.3).

Seniority, security and covenants. Where the lender stands if things go wrong. A senior secured bond has a claim on specific assets and is paid before others; a subordinated bond is paid after senior creditors are made whole. Covenants are contractual promises by the borrower — limits on additional debt, on asset sales, on dividends — that give lenders rights if the borrower's condition deteriorates. They are worth real money: covenant violations transfer control rights to creditors and demonstrably change corporate behaviour, with investment falling sharply after a breach (Chava & Roberts, 2008). [Fact]

All of this is written into an indenture (for a corporate bond under US law) or a prospectus, a document that runs to hundreds of pages and that almost no investor reads in full. Practically, investors rely on summaries and on covenant-scoring services. [Practice] This is a genuine information asymmetry, and it is largest exactly where it matters most — in high-yield issuance, where documentation quality varies enormously and has deteriorated over the past two decades.

2.2 The primary market: how bonds are first sold

Governments and companies sell their bonds in structurally different ways, and the difference tells you something about both markets.

Governments auction. The US Treasury announces in advance that it will sell, say, \$42 billion of 10-year notes on a particular date. Bidders — mostly primary dealers, a designated group of banks obliged to participate, plus direct institutional and

retail bidders — submit either competitive bids specifying a yield and size, or non-competitive bids accepting whatever yield clears. The Treasury fills from the lowest yield upward until the issue is sold, and everyone pays the same clearing yield. This “uniform price” or Dutch format replaced the older pay-your-bid format after Treasury experiments in the 1990s found it reduced bidder caution and improved revenue (Malvey & Archibald, 1998). [Contested] — the theoretical comparison between formats is ambiguous, and the empirical gains measured were small.

The auction calendar is public, regular, and enormous, and it has a measurable price footprint: yields tend to rise into an auction and fall after it, consistent with dealers demanding compensation for temporarily absorbing supply (Lou, Yan & Zhang, 2013). [Fact] This is a pure inventory effect in the most liquid market on earth, which should calibrate your expectations about how much price pressure supply can create in less liquid ones.

Companies syndicate. A company hires banks, who sound out investors, announce “initial price thoughts” — an indicative spread — take orders, and then tighten the spread if the book is oversubscribed. Allocation is discretionary: the syndicate decides who gets what. The new bond typically prices at a small concession to the issuer’s existing bonds, the new issue concession, which is the sweetener that gets the deal done and which widens sharply when markets are stressed. In a bad week, the corporate primary market simply closes — no deals price at all — which is one of the cleaner real-time indicators of credit stress. [Practice]

2.3 The secondary market: dealers, not exchanges

Here is the institutional fact that surprises people coming from equities: bonds overwhelmingly do not trade on exchanges. There is no central limit order book for the 10-year Treasury note the way there is for Apple stock. Bonds trade over-the-counter, bilaterally, through dealers who quote a price at which they will buy (bid) and a price at which they will sell (ask) and who hold inventory in between.

Two reasons for this, one structural and one a consequence.

The structural reason is fragmentation. Apple has one common share. A large company may have forty outstanding bonds with different maturities, coupons and covenants, most of which do not trade on a given day. There are tens of thousands of distinct corporate bond issues in the US market against a few thousand listed stocks. You cannot maintain a continuous two-sided market in an instrument that trades twice a month, so a search-and-bargaining structure emerges instead — which is exactly the setting analysed by Duffie, Gârleanu & Pedersen (2005), whose central result is that investors with worse outside options get worse prices. [Fact] Retail investors in corporate bonds pay dramatically more than institutions for the same bond on the same day, and, unusually, they pay more per bond on small trades — the opposite of the equity market’s pattern (Edwards, Harris & Piwowar, 2007).

The consequence is opacity, which US regulators addressed in 2002 by requiring corporate bond trades to be reported to TRACE within minutes. The natural experiment this created is one of the cleanest results in market microstructure: transaction costs fell substantially for bonds brought into the system, with no measurable damage to liquidity (Bessembinder, Maxwell & Venkataraman, 2006; Goldstein, Hotchkiss & Sirri, 2007). [Fact] Transparency helped investors and cost dealers, which is why dealers opposed it.

Since roughly 2015 the corporate bond market has been electronifying — moving from telephone negotiation to request-for-quote platforms where an investor pings several dealers simultaneously, and increasingly to all-to-all venues where investors trade with each other. The economics are the ones you would expect from an auction replacing a search: more competition, tighter prices, and the largest benefit in the liquid part of the market (Hendershott & Madhavan, 2015). [Fact] Portfolio trading — executing a basket of hundreds of bonds in one negotiated transaction — has grown rapidly and is tied closely to the ETF market’s ability to price baskets (§17.4). [Practice]

The on-the-run distinction. In government markets, the most recently issued bond of a given maturity is on-the-run; everything older is off-the-run. The on-the-run bond is far more liquid, concentrates the trading and hedging activity, and therefore trades at a lower yield — you give up a few basis points for the ability to transact in size. The gap is a clean measure of liquidity value and widens sharply in crises (Amihud & Mendelson, 1991; Longstaff, 2004). [Fact] It is also the source of a famous trade: buy the cheap off-the-run bond, sell the expensive on-the-run, wait for convergence. The trade is correct on average and was a major component of what destroyed Long-Term Capital Management in 1998, because convergence trades funded with borrowed money can require more capital precisely when capital is scarce.

2.4 Living: coupons, ratings and index membership

Between issue and maturity, a bond mostly just pays coupons. Three things can happen to it that matter.

Coupon payments arrive, usually semi-annually in the US and UK, annually in much of Europe. Between payment dates the buyer owes the seller the interest accrued so far, which is why quoted prices and settled prices differ (§6.6).

Ratings change. Agencies — Moody's, S&P, Fitch — assign letter grades that map to a broad ordering of credit quality. Ratings migrate: an issuer downgraded from BBB- to BB+ crosses the investment-grade boundary and becomes a “fallen angel”, which matters far more than one notch should, because a large population of investors is contractually forbidden from holding sub-investment-grade paper. The resulting forced selling is a genuine, documented price effect: insurers subject to ratings-based capital rules sell downgraded bonds, and those sales push prices below fundamental value temporarily (Ellul, Jotikasthira & Lundblad, 2011). [Fact] It is also an opportunity — fallen-angel bonds have historically been bought cheaply by investors without the constraint.

Ratings themselves deserve scepticism. They are explicitly through-the-cycle ordinal rankings, not probability estimates, they lag market prices, and the issuer-pays business model creates a conflict that shows up in the data: ratings are more favourable when competition among agencies is more intense (Becker & Milbourn, 2011). [Fact] Use them as a coarse sorting device and a description of who is allowed to own the bond, not as a risk measure.

Index membership changes. When a bond is added to or dropped from a major index — for falling below investment grade, for having less than a year to maturity, for shrinking below a minimum size — every index-tracking fund must trade it. Index rules are therefore a significant, mechanical source of demand, and index construction is one of the underappreciated facts of the asset class (§12.5).

2.5 Early endings: calls, tenders and exchanges

Many bonds do not run to maturity, because the issuer has reserved the right to end them early or negotiates to do so.

A call provision gives the issuer the right to repay at a stated price on stated dates. The issuer will exercise it when refinancing is cheaper — that is, when rates have fallen or their credit has improved. This is straightforwardly bad for the holder: you get your money back exactly when reinvesting it is least attractive. The compensation is a higher yield, which is the α term of Identity 2, and the consequences for price behaviour are severe enough to warrant their own treatment (§7.6).

A make-whole call is a gentler variant common in investment-grade issuance: the issuer may call, but must pay the present value of remaining cash flows discounted at a Treasury yield plus a small spread. Because the make-whole price rises as rates fall, it removes most of the harm — a make-whole bond behaves almost like a non-callable one. [Practice] The distinction between a hard call schedule and a make-whole provision is one of the highest-value five minutes of document reading available.

Tender offers and exchanges. An issuer may simply offer to buy its bonds back, at a premium if it wants to retire expensive debt, or at a deep discount if it is distressed. A distressed exchange — swapping old bonds for new ones worth less — is default by another name, and rating agencies classify it as such. It has become the dominant form of corporate default resolution: most high-yield defaults now happen through negotiated liability management rather than bankruptcy filing (§16.2). [Practice]

2.6 Death: maturity or default

Most bonds simply mature. The issuer pays the face value, the bond ceases to exist, and the holder's return over the whole life was determined by the purchase yield, the reinvestment of coupons, and nothing else.

The rest default. §16 covers what happens then in detail; the summary is that default is not a binary wipeout but the start of a process that ends with creditors owning something — cash, new bonds, or equity in the reorganised business — worth on average perhaps 40% of face for a senior unsecured corporate bond, with enormous variation around that average and a systematic pattern of lower recoveries in exactly the years when defaults are most numerous (Altman & Kishore, 1996; Acharya, Bharath & Srinivasan, 2007). [Fact] That co-movement — many defaults and low recoveries arriving together — is what makes credit risk a systemic exposure rather than a diversifiable one, and it is the reason a credit portfolio's loss distribution has such a long tail.

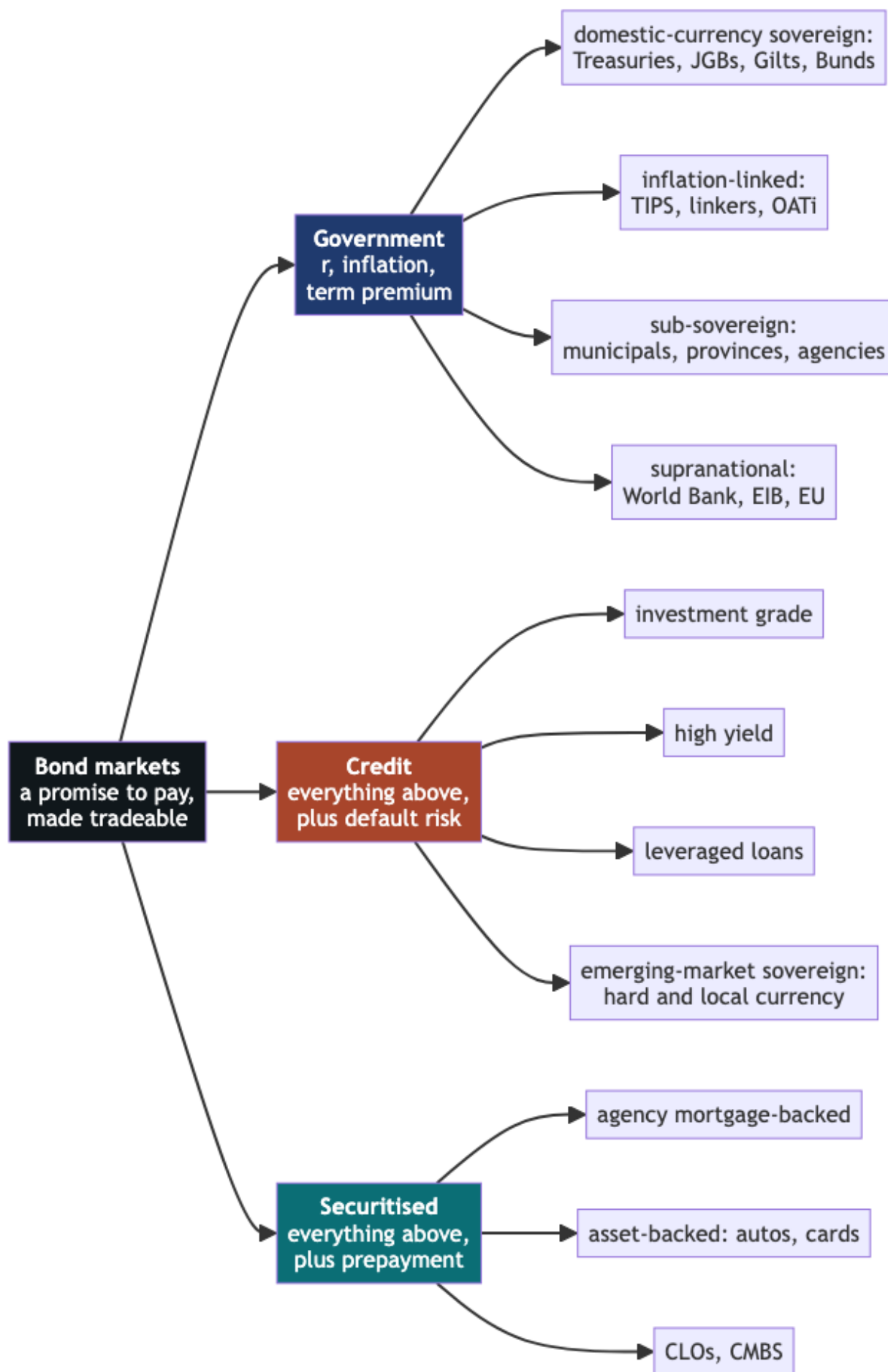
§2 Key takeaways

1. Four birth decisions — maturity, coupon, currency, seniority — determine most of how a bond will behave for its whole life.
2. Governments auction on a public, regular calendar; companies syndicate opportunistically. The difference shows up as predictable supply pressure in one market and closed windows in the other.
3. Bonds trade over-the-counter through dealers, not on exchanges, because there are an order of magnitude more bond issues than stocks and most do not trade on a given day.

4. In a search market the worse-informed and smaller counterparty gets a worse price. Retail investors pay dramatically more for corporate bonds than institutions, and pay more per bond on small trades.
 5. Post-trade transparency (TRACE) lowered transaction costs materially without damaging liquidity — one of the cleanest natural experiments in microstructure.
 6. The on-the-run/off-the-run yield gap is the price of liquidity, and it widens in crises exactly when a convergence trade needs it not to.
 7. Ratings are through-the-cycle ordinal rankings with a known issuer-pays conflict. Their main investment significance is determining who is permitted to hold the bond, and the forced selling that follows a downgrade is a real, measurable price effect.
 8. Check whether a call is a hard schedule or make-whole. The first materially changes the bond's risk; the second nearly does not.
-

Part II — The map of the market

The global bond market is on the order of \$140 trillion, larger than the capitalisation of global equity markets, and it is not one market but a nested set of them. Understanding the structure means understanding which of Identity 2's terms is doing the work in each. This part walks the three main territories — government, international sovereign, and corporate — in that order, because each adds a component to the yield that the previous one lacked.



3. Government bonds and the risk-free curve

3.1 Why this market is the reference

Every other bond in the world is priced relative to a government curve, and it is worth being precise about why, because the reason is not that governments are morally trustworthy.

A government that borrows in a currency it issues cannot be forced into nominal default. It can always create the units it owes. This does not make the debt riskless in economic terms — the creditor may be repaid in devalued money, which is a real loss — but it removes the credit term from Identity 2 and leaves only the rate terms. What remains is the purest available observation of the price of time and the price of inflation risk. That is what makes it a reference.

Three further properties make the US Treasury market the global reference in particular:

- Size and homogeneity. Around \$29 trillion of marketable debt outstanding in the mid-2020s, in a small number of standardised, fungible instruments issued on a predictable calendar.
- Liquidity. The on-the-run 10-year note trades with bid-ask spreads measured in fractions of a basis point, and at volumes no other fixed-income instrument approaches.
- Collateral status. Treasuries are the dominant collateral in the repo market (§3.4), which means they are not merely an investment but a money-like asset used to fund positions across the entire financial system.

That last property has a price. Because Treasuries do a job beyond paying their cash flows, investors accept a lower yield than the cash flows alone justify. Krishnamurthy & Vissing-Jorgensen (2012) estimate this convenience yield at roughly 70 basis points on average, and show it varies inversely with the supply of Treasury debt — when the government issues more, the premium shrinks. [Fact] A closely related literature finds that the “true” risk-free rate implied by option markets sits above the Treasury yield, which is the same observation from another angle (van Binsbergen, Diamond & Grotteria, 2022). [Fact] The practical consequence: the Treasury curve is the market’s reference rate, but it is not a clean measurement of the risk-free rate. It is the risk-free rate minus a time-varying convenience premium. Practitioners who need a genuine discount curve increasingly use the overnight indexed swap (OIS) curve instead (§8.2).

3.2 The instrument set

The US Treasury issues a deliberately small menu. Other developed sovereigns issue close analogues, and the vocabulary transfers.

Instrument	Maturity at issue	Coupon	What it is for
Bills	4 to 52 weeks	None; sold at a discount	Cash management; the front of the curve
Notes	2, 3, 5, 7, 10 years	Semi-annual fixed	The core of the market
Bonds	20 and 30 years	Semi-annual fixed	Long duration for liability matchers
TIPS	5, 10, 30 years	Semi-annual on an inflation-adjusted principal	Real, not nominal, returns
FRNs	2 years	Quarterly, resets to the 13-week bill rate	Near-zero duration
STRIPS	Any	None	Separated single cash flows; pure zero-coupon exposure

Two of these are worth dwelling on.

STRIPS are what you get when a dealer takes a coupon bond and sells each of its payments separately. A 30-year bond becomes sixty small zero-coupon claims plus one large principal claim. Since a coupon bond is definitionally a portfolio of zero-coupon bonds, stripping and reconstitution are near-costless and the prices must line up. This matters conceptually: there is nothing primitive about a coupon bond. The zero-coupon curve is the primitive object, and coupon bonds are portfolios of it. §6.2 builds everything on that footing. Practically, a 30-year STRIP has a duration of 30 years and is the most rate-sensitive liquid instrument available, which makes it the natural tool for pension funds matching very long liabilities.

FRNs are the opposite. Their coupon resets to the prevailing short rate every quarter, so their price barely moves when rates move — they have duration measured in weeks. A floating-rate note is economically cash plus a credit spread, and recognising this dissolves a lot of confusion about bank loans and other floating instruments (§5.4).

3.3 Inflation-linked bonds, and the breakeven trap

Inflation-linked bonds are the single most useful instrument for building correct intuition about what a yield means, and the most commonly misread.

The mechanism. A TIPS has a principal amount that is adjusted upward with the consumer price index. The coupon rate is fixed, but it is applied to the adjusted principal, so both coupon payments and the final redemption grow with inflation. If you buy a 10-year TIPS at a 2% real yield and inflation runs at 3%, you receive roughly 5% in nominal terms. If inflation runs at 8%, you receive roughly 10%. The real return is locked; the nominal return floats. A conventional bond is the exact mirror: the nominal return is locked and the real return floats.

This is the cleanest possible illustration of Identity 2. A nominal Treasury yield contains $r + \pi^e + \text{TP}$; a TIPS yield contains $r + \text{TP}^{\text{real}}$ and no π^e at all. Subtract them:

$$\underbrace{y^{\text{nominal}} - y^{\text{real}}}_{\text{breakeven inflation}} = \pi^e + \underbrace{\text{IRP}}_{\text{inflation risk premium}} - \underbrace{(\ell^{\text{TIPS}} - \ell^{\text{nominal}})}_{\text{relative liquidity}}$$

The difference is called breakeven inflation, and it is very widely quoted as “what the market expects inflation to be.” It is not. It is expected inflation plus the premium investors demand for bearing inflation uncertainty minus the extra liquidity discount on TIPS, which are a far smaller and less liquid market than nominal Treasuries. [Fact] Both correction terms are material and both move around. In the autumn of 2008, TIPS breakevens briefly implied deflation of several percent per year for a decade — an implausible forecast, and in fact mostly a collapse in TIPS liquidity as leveraged holders were forced to sell (Gürkaynak, Sack & Wright, 2010; Fleckenstein, Longstaff & Lustig, 2014).

The Fleckenstein–Longstaff–Lustig result deserves emphasis because it is remarkable: a TIPS combined with an inflation swap replicates a nominal Treasury’s cash flows almost exactly, and yet the two packages persistently differ in price, sometimes by more than \$20 per \$100 of face. [Fact] That is a large, documented violation of the law of one price in the world’s most liquid market, and it survives because arbitraging it requires balance sheet, which is scarce exactly when the gap is widest. The general lesson recurs throughout Part V: relative-value gaps in fixed income are usually funding constraints wearing a disguise.

The practical guidance:

- Use breakevens as a market-implied inflation compensation, not a forecast. When the two need to be distinguished, survey measures of expectations and model-based decompositions do the job better. [Practice]
- Treat breakeven moves during liquidity events as noise about inflation and signal about funding stress.
- For a real-money investor — someone whose liabilities are real, like a pension or a retiree — the inflation-linked bond is the risk-free asset and the nominal bond is the risky one. The whole convention of calling nominal Treasuries “risk-free” is an artefact of measuring returns in nominal units.

3.4 The plumbing: repo, and why a government bond is a funding instrument

You cannot understand government bond markets without understanding repo, because most of the positions in them are financed rather than paid for.

A repurchase agreement is a sale with a promise to buy back. I sell you a Treasury bond today for \$99 and agree to repurchase it tomorrow for \$99.01. Economically this is a one-day secured loan: I have borrowed \$99, you hold the bond as collateral, and the price difference is the interest. The rate implied — the repo rate — is one of the most important prices in finance, because it is the cost of funding a bond position.

Three consequences follow, and each explains a chunk of market behaviour.

Leverage is cheap and therefore ubiquitous. If I can finance a Treasury position at close to the risk-free rate, the capital required to hold it is only the haircut — often 1–2% for Treasuries. A hedge fund can therefore run a \$50 billion Treasury position on \$1 billion of capital. This is not pathological; it is how the market makes the tiny spreads on very safe instruments worth capturing. But it means that positioning in this market is far larger than the capital behind it, and that forced deleveraging can move prices violently (§13.4).

Specific bonds can go “special”. If many people want to short a particular bond — typically the on-the-run issue, because it is the hedging instrument of choice — then the bond becomes scarce as collateral, and lenders of it can demand a lower repo rate. A bond “on special” effectively earns its holder extra income, and that income is capitalised into a higher price and lower yield (Duffie, 1996). [Fact] Part of the on-the-run premium is precisely this.

The cash–futures basis trade. Treasury futures usually trade slightly rich to the underlying bonds, because many investors prefer the capital efficiency of futures. A levered fund can sell the future, buy the bond, finance the bond in repo, and collect the small difference. The gap is a few basis points, so the trade is run at very high leverage, and it has grown to a very large scale (Barth & Kahn, 2025). It is generally stabilising — it links the futures and cash markets — but in March 2020 the unwind of these positions was a significant contributor to Treasury market dysfunction, and it remains a standing concern of financial stability authorities. [Contested] — the size of the trade is well documented; its precise contribution to March 2020 is debated.

3.5 When the safest market breaks

It is tempting to treat the Treasury market as a frictionless benchmark. Twice in recent memory it has not been.

October 2014 “flash rally”. The 10-year yield moved roughly 37 basis points intraday, most of it in a few minutes, with no news to explain it. The official post-mortem found no single cause, and pointed to changes in market structure — principal trading firms, self-trading, and the withdrawal of dealer liquidity — as contributing conditions.

March 2020. In the most consequential episode, the demand for cash became so intense that Treasuries — normally the asset everyone flees to — were sold heavily and their yields rose during the worst of the equity collapse. The mechanism was a combination of foreign official selling, mutual fund redemptions, and the forced unwind of levered relative-value positions, all hitting dealer balance sheets that were constrained by post-crisis leverage rules. He, Nagel & Song (2022) document the resulting “inconvenience yield” — Treasuries trading cheap to their own derivatives, the convenience premium inverting. [Fact] The Federal Reserve resolved it by buying roughly \$1 trillion of Treasuries in a matter of weeks (Vissing-Jorgensen, 2021).

The lesson is not that Treasuries are dangerous. It is that liquidity is a property of the market’s intermediation capacity, not of the instrument. The bonds were exactly as safe on 18 March 2020 as on 18 February. What changed was the balance sheet available to stand between buyers and sellers. Every model that treats liquidity as an attribute of a security rather than a state of the system will be wrong at precisely the moment it matters. Whether the current structure remains adequate as debt outstanding grows is an active policy debate (Duffie, 2020).

3.6 Government-adjacent markets

Two large markets sit beside the sovereign curve and are worth knowing exist.

Agency and municipal debt. In the US, government-sponsored enterprises (Fannie Mae, Freddie Mac, the Federal Home Loan Banks) issue debt with an implicit — and since 2008, effectively explicit — federal backstop, trading at a small spread to Treasuries. Municipal bonds, issued by states and cities, are the more interesting case: their interest is generally exempt from federal income tax, so their yields are lower than taxable equivalents, and comparing them to Treasuries requires converting to a tax-equivalent basis. The naive conversion $y^{\text{taxable-equivalent}} = y^{\text{muni}} / (1 - \text{tax rate})$ is a decent first pass, but the market does not price municipals as if all investors faced the same tax rate, and the apparent cheapness of long municipals partly reflects the tax treatment of capital gains on bonds bought at a discount (Ang, Bhansali & Xing, 2010). [Fact] Municipals also carry genuine credit risk — Detroit and Puerto Rico both defaulted in the 2010s — though default rates for general-obligation debt of US states and large cities have historically been far below corporate rates at equivalent ratings.

Supranationals. The World Bank, the European Investment Bank, and since 2020 the European Union itself issue highly rated bonds backed by member-state commitments. They are a small, high-quality corner of the market whose main significance is as a benchmark for how a multi-sovereign credit prices.

§3 Key takeaways

1. A government borrowing in its own currency cannot be forced into nominal default. That removes the credit term from Identity 2 and is what makes its curve the reference.
2. Treasuries carry a convenience yield of roughly 70bp on average because they function as collateral, not just as an investment. The Treasury curve is therefore the risk-free rate minus a time-varying premium, not the risk-free rate itself.
3. Zero-coupon bonds are the primitive; coupon bonds are portfolios of them. STRIPS make this literal and tradeable.
4. A TIPS locks the real return and floats the nominal; a conventional bond does the reverse. For an investor with real liabilities, the inflation-linked bond is the safe asset.

5. Breakeven inflation is expected inflation plus an inflation risk premium minus a relative liquidity discount. Reading it as a forecast produced absurd conclusions in 2008 and will again.
6. Most positions in this market are financed through repo, so leverage is large relative to capital and the market's liquidity depends on dealer balance sheet.
7. In March 2020 Treasuries sold off during an equity crash and traded cheap to their own derivatives. Liquidity is a state of the intermediation system, not a property of the bond.

4. The rest of the world: sovereign and international markets

The United States is roughly two-fifths of the global bond market. The other three-fifths are instructive precisely because they violate the assumptions that make Treasuries simple. Three deviations organise everything here, and each corresponds to switching a term of Identity 2 back on:

1. The issuer does not control the currency. Then credit risk returns, even for a sovereign. This is the euro area.
2. The issuer borrows in someone else's currency. Then the debt burden moves with the exchange rate, which is the emerging-market hard-currency problem.
3. The investor's currency differs from the bond's. Then currency risk swamps interest-rate risk, and the hedging decision matters more than the bond decision. This is §15.

4.1 The large developed markets

Market	Benchmark name	Distinguishing feature
Japan	JGBs	Enormous stock, extraordinary central bank ownership, decades at the zero bound
Euro area	Bunds (Germany) as benchmark; OATs, BTPs, Bonos	Many issuers, one currency, no single risk-free curve
United Kingdom	Gilts	Unusually long maturities, driven by pension demand
Canada, Australia	GoCs, ACGBs	Commodity-linked economies; smaller, well-run markets
China	CGBs	Large, increasingly index-included, partially open to foreigners

Japan is the natural experiment in what happens when rates stay at zero for a very long time. The Bank of Japan's asset purchases, and from 2016 its explicit yield curve control — pinning the 10-year yield in a band — resulted in the central bank owning more than half of all outstanding JGBs. [Fact] Two lessons generalise. First, a determined central bank can hold a nominal yield essentially wherever it wants for a long time, because it can print the money to buy the bonds; the cost shows up elsewhere, in the currency and in the functioning of the market rather than in the yield. Second, when the pin is eventually removed, the adjustment is abrupt, since the accumulated pressure has nowhere else to go.

The euro area is the important structural case, treated next.

The UK gilt market is shaped by an unusual demand structure: defined-benefit pension funds with very long liabilities are large, and they want very long assets. The result is a gilt curve that is often inverted at the long end — 50-year yields below 30-year yields — because of demand concentration rather than any expectation about rates. It is the cleanest live demonstration of the preferred-habitat mechanism of §12.1, and it set up the 2022 crisis described in §13.4.

4.2 The euro area: sovereign credit without a printing press

When Italy issues a bond denominated in euros, Italy does not control the euro. The European Central Bank does, and it is prohibited from monetary financing of member states. So Italian government debt has something that US, Japanese and UK government debt in their own currencies does not: a genuine possibility of default, because Italy can run out of euros in a way the US cannot run out of dollars.

This is why euro-area government bonds are quoted as a spread to Bunds, and why that spread behaved during 2010–2012 like a credit spread rather than a rate differential — widening violently, correlating with equity market stress, and responding

to fiscal news. The spread contains two things that are hard to separate: the risk that Italy does not pay in euros, and the risk that Italy stops using euros and pays in something else (redenomination risk).

The episode also produced the single clearest demonstration in modern markets that a sovereign bond price can be a self-fulfilling equilibrium. If investors believe Italy will default, they demand a higher yield; the higher yield raises Italy's interest burden; the burden makes default more likely. Two equilibria exist for the same fundamentals. The ECB's announcement in July 2012 that it would do "whatever it takes," followed by the never-used Outright Monetary Transactions programme, collapsed spreads without a single bond being purchased — which is the behaviour you expect if the market had been sitting in the bad equilibrium of a multiple-equilibrium model. [Hypothesis] — the multiple-equilibrium reading is the standard one and fits the facts, but it is not decisively separable from a story in which the ECB simply revealed private information about its reaction function.

The general principle worth extracting: whether a government controls the currency it borrows in is a more important fact about its debt than its debt-to-GDP ratio. Japan, with debt above 200% of GDP in its own currency, has never faced a solvency crisis. Greece, at a lower ratio in a currency it did not control, did.

4.3 Original sin and the two emerging markets

Emerging-market sovereign debt splits into two markets that share a name and almost nothing else.

Hard-currency debt is issued in dollars or euros. It carries genuine default risk, because the issuer cannot print the currency of repayment, and its yield decomposes as the US Treasury curve plus a credit spread. An investor holding it takes credit risk but no direct currency risk.

Local-currency debt is issued in the country's own currency. Default risk is lower for the same reason it is lower for the US — the government can print — but the investor bears the exchange rate, and the yield contains the country's own real rate, its own (usually higher and more volatile) inflation expectation, and a term premium reflecting that volatility.

The historical inability of most developing countries to borrow abroad in their own currency was named "original sin" by Eichengreen and Hausmann, and its consequences are set out in Eichengreen, Hausmann & Panizza (2003). The mechanism is vicious: because the debt is in dollars and the revenue is in local currency, a currency depreciation increases the real debt burden exactly when the economy is weakening. A shock that would be absorbed by a floating exchange rate instead becomes a solvency problem. This is the mechanism behind most emerging-market crises of the 1980s and 1990s.

The encouraging development is that it has substantially receded. Many major emerging economies now issue most of their debt domestically in local currency, with developed inflation-targeting frameworks and deeper domestic investor bases. [Fact] The corollary for investors is that the local-currency index is now the larger opportunity set, and that its risk is predominantly currency risk, not credit risk — a distinction that governs how it should be sized in a portfolio (§15.4).

A further inconvenient fact: sovereign credit spreads across countries are driven far more by global factors — US risk appetite, global volatility — than by country-specific fundamentals (Longstaff, Pan, Pedersen & Singleton, 2011). [Fact] A portfolio of twenty emerging-market sovereigns is much less diversified than it looks, because they are mostly one trade: global risk appetite.

4.4 Why sovereigns repay at all

This is a genuinely deep question with a satisfying literature, and it changes how you read a sovereign bond price.

When a company defaults, creditors can seize assets through bankruptcy courts. When a country defaults, they mostly cannot: a creditor cannot repossess Argentina. So why does any country ever repay?

The first answer was reputation (Eaton & Gersovitz, 1981): default means losing access to credit markets, and a country that expects to want to borrow again will pay to preserve that access. Bulow & Rogoff (1989) then demonstrated that reputation alone is insufficient: a country that defaults and loses access to future borrowing can instead save what it would have repaid — somewhere no creditor can reach — and draw on those savings exactly as it would have drawn on new loans, so losing its reputation costs it nothing once self-insurance is available. Repayment must therefore be sustained by direct costs: trade sanctions, seizure of assets abroad, disruption of trade credit and the domestic banking system. [Fact] The modern quantitative models build on this, generating default as an optimal decision that becomes attractive when output is low (Arellano, 2008).

Three things follow that matter practically:

- Sovereign default is a choice, not an event. It happens when the cost of paying exceeds the cost of not paying, which is a political calculation, not an accounting one. This is why sovereign credit analysis is more political-economy than spreadsheet, and why debt ratios predict default far less well than people expect.

- Default is usually partial and negotiated. The outcome is a restructuring with a haircut, and the size of that haircut is the key variable. Cruces & Trebesch (2013) show that larger haircuts are followed by significantly higher borrowing spreads and longer exclusion from markets — so countries face a real trade-off, and the market does punish aggressive restructurings. [Fact]
- The legal architecture matters. Argentina's 2001 default was followed by fifteen years of litigation in which holdout creditors used a *pari passu* clause to block payments to the restructured majority. The response was the widespread adoption of collective action clauses, which let a supermajority of bondholders bind the minority. When buying a sovereign bond, the governing law and the CAC threshold are part of what you are buying. [Practice]

§4 Key takeaways

1. Three deviations from the Treasury case generate everything in international markets: the issuer does not control the currency, the issuer borrows in a foreign currency, or the investor's currency differs from the bond's.
2. Whether a government controls the currency it borrows in matters more than its debt-to-GDP ratio. Japan at 200%+ has never had a solvency crisis; Greece at less did.
3. Euro-area sovereign spreads are credit spreads, and they exhibit multiple-equilibrium behaviour: the 2012 OMT announcement collapsed them without a single purchase.
4. Emerging-market debt is two unrelated asset classes. Hard-currency debt is a credit exposure; local-currency debt is predominantly a currency exposure.
5. Sovereign credit spreads co-move strongly with global risk appetite, so a diversified basket of sovereigns is far less diversified than it appears.
6. Sovereigns repay because default carries direct economic costs, not because of reputation alone. Default is therefore a political choice, and debt ratios forecast it poorly.
7. Larger restructuring haircuts are followed by higher spreads and longer market exclusion — the market does price the borrower's revealed willingness to impose losses.

5. Corporate credit and the capital structure

Corporate bonds add the credit term to Identity 2, and with it a whole second dimension of analysis: not just when you get paid but whether, and if not, how much. This section builds the structure — who gets paid in what order — and then gives the single most useful conceptual tool in credit, which is that a corporate bond is a government bond minus an insurance policy you have written.

5.1 How much a company borrows, and why

A company choosing its capital structure faces a trade-off that is easy to state and hard to optimise.

For debt: interest is tax-deductible in most jurisdictions, so each dollar of interest is subsidised by the corporate tax rate. Debt is also a less risky claim than equity, so it costs less. And debt imposes discipline — a manager who must service debt has less freedom to spend cash unwisely, which is the “free cash flow” argument of Jensen (1986).

Against debt: it must be serviced regardless of circumstances. More of it raises the probability of financial distress, which is expensive even short of bankruptcy — customers leave, suppliers tighten terms, talent departs. And a heavily indebted firm suffers debt overhang (Myers, 1977): profitable investments may go unmade, because the gains accrue to creditors while shareholders fund the outlay.

For a bond investor, the useful implication is that a company's leverage is a choice that tells you about management's intentions, and that the direction of travel matters more than the level. A firm deliberately leveraging up to buy back stock is transferring value from you to shareholders; the bond market prices this as “event risk,” and it is why covenants restricting dividends and buybacks are worth paying for.

5.2 The capital stack

If the company fails, claims are paid in a strict order. Everything above is paid in full before anything below receives anything — in principle.

Rank	Claim	Typical recovery	Notes
1	Secured bank debt / first-lien loans	60–80%	Claim on specific collateral
2	Second-lien loans	30–50%	Same collateral, behind the first lien
3	Senior unsecured bonds	35–45%	The typical “corporate bond”
4	Senior subordinated	20–30%	Contractually behind senior
5	Junior subordinated / hybrids	5–20%	Often with coupon deferral rights
6	Preferred stock	near 0	Dividends skippable without default
7	Common equity	near 0	Residual claim

Recovery figures are long-run averages from rating-agency studies and vary enormously by industry, cycle and jurisdiction; treat them as orders of magnitude, not estimates (§9.3). [Fact]

Three refinements matter more than the table suggests.

Structural subordination. A bond issued by a holding company ranks behind debt issued by the operating subsidiaries that actually own the assets, even if both are called “senior unsecured,” because the subsidiary’s creditors are paid from the subsidiary’s assets before anything flows up to the parent. Two bonds from the same group with identical labels can have very different claims. [Practice] This is one of the highest-return items in a bond prospectus to check.

The stack is a contract, not a law of nature. In March 2023, Credit Suisse’s Additional Tier 1 (AT1) bonds — a form of bank capital designed to absorb losses — were written down to zero while shareholders received shares in UBS. Holders reasonably expected to rank above equity; the contractual terms and the Swiss resolution framework said otherwise. [Fact] The episode is the definitive reminder that seniority is whatever the documents and the applicable resolution law say it is, and that in bank capital especially, the documents are unusual.

Covenants determine whether the stack holds. A borrower with weak covenants can move assets outside the reach of existing creditors, issue new debt that ranks ahead, or transfer collateral to a new entity — practices grouped under the informal heading of “liability management,” which have become common in the leveraged finance market. [Practice] The value of a senior claim depends on whether the borrower can create a more senior one.

5.3 The investment-grade boundary

Ratings map to a ladder, and one rung on that ladder matters far more than the others.

	Moody’s	S&P / Fitch	Character
Investment grade	Aaa to Baa3	AAA to BBB–	Institutional core holdings
High yield	Ba1 to C	BB+ to C	“Junk”; a different investor base
Default	D	D	In or near default

The BBB–/BB+ boundary is an economic discontinuity rather than a small change in credit quality, because a large population of investors — insurers under capital rules, many pension mandates, index funds tracking IG benchmarks — is contractually restricted to investment grade. A downgrade across the line therefore triggers mechanical selling into a market with a different and smaller buyer base. This produces measurable price pressure beyond fundamentals (Ellul, Jotikasthira & Lundblad, 2011). [Fact]

Two practical consequences. The growth of the BBB segment — the lowest IG rung — is a standing systemic concern, since a recession that downgrades a large share of it forces a lot of paper into a market too small to absorb it. And fallen angels have historically been a good place to buy, precisely because the selling is forced rather than informed. [Contested] — the effect is documented, but it is not free money: the same bonds are also genuinely deteriorating, and the excess return net of subsequent defaults is smaller than the price dislocation suggests.

5.4 The instrument zoo

Instrument	Rate	Seniority	The defining feature
Senior unsecured bond	Fixed	Senior	The reference corporate instrument
Leveraged loan	Floating	Senior secured	Near-zero duration; private documentation; prepayable at par
Convertible bond	Fixed, low	Senior	Holder may convert to equity
Hybrid / junior subordinated	Fixed, then resets	Deeply junior	Coupon deferrable; treated partly as equity by agencies
Bank AT1 / CoCo	Fixed, resets	Junior to all debt	Converts or writes down on a capital trigger
Private credit	Floating	Senior secured	Not traded; marked by the lender

The two extremes teach the most.

Leveraged loans are floating-rate and therefore carry almost no interest-rate duration — an investor in loans is taking credit risk nearly pure. They are also callable at par at any time, which caps their upside: when a borrower's credit improves, they refinance, so the loan cannot trade far above 100. A loan portfolio is thus structurally short optionality, with limited price upside and full downside. [Practice]

Convertible bonds are the clearest instance of the “bond plus option” decomposition. A convert is a straight bond plus a call on the issuer's equity; the issuer accepts a lower coupon in exchange for granting it. Their pricing, and the convertible arbitrage strategy built around buying them and shorting the equity, are an options problem more than a credit problem.

5.5 The key equivalence: credit is a short put on the firm

Here is the idea that makes credit intelligible, due to Merton (1974).

Consider a firm whose assets are worth V and which owes a single payment of F at time T . At maturity, one of two things happens. If $V_T \geq F$, the debt is paid in full and shareholders keep $V_T - F$. If $V_T < F$, the firm defaults, creditors take the assets, and shareholders get nothing.

Two definitions make the next step readable. A call option with strike K is the right, but not the obligation, to buy an asset for K at expiry; it pays $\max(S_T - K, 0)$ — the amount by which the asset's final value S_T exceeds the strike, or nothing. A put option is the right to sell for K and pays $\max(K - S_T, 0)$. Selling either one earns a premium today in exchange for that payoff tomorrow.

Now write down the payoffs:

$$\text{equity}_T = \max(V_T - F, 0), \quad \text{debt}_T = \min(V_T, F) = F - \max(F - V_T, 0)$$

The first is exactly a call option on the firm's assets with strike F . The second is the face value of the debt minus a put option on the firm's assets with the same strike. So:

$$\underbrace{\text{risky corporate bond}}_{\text{what you own}} = \underbrace{\text{risk-free bond}}_{\text{the promise}} - \underbrace{\text{put on firm assets}}_{\text{the insurance you wrote}}$$

You own a Treasury and you have sold the shareholders a put. The credit spread is the premium on that put, expressed as an annual yield. That single sentence explains most of credit's behaviour, and each consequence below is a direct reading of it:

- Credit spreads should widen with asset volatility. A put is worth more when the underlying is more volatile. This is why equity volatility predicts credit spreads, and why credit and equity-volatility markets move together.
- Credit spreads should widen with leverage. More debt means a higher strike relative to the asset value, so the put is further in the money.
- Credit is short optionality, hence negatively skewed. Selling a put gives you a small premium most of the time and a large loss occasionally. A credit portfolio's return distribution looks like a short-volatility strategy's, because in a precise sense it is one.

- A “distance to default” measure follows immediately: how many standard deviations of asset value separate the firm from the default boundary. This is the basis of the commercial Moody’s/KMV framework and of most quantitative default prediction.
- Equity and credit should be hedgeable against each other, which is the foundation of capital structure arbitrage.

The model’s empirical failures are as informative as its successes, and they define the credit spread puzzle (§9.5): Merton-type models calibrated to match observed default rates and recovery rates predict spreads far below what investment-grade bonds actually yield, especially at short maturities (Huang & Huang, 2012). [Fact] Resolving that gap is the central empirical question in credit, and the answer turns out to matter for whether you should own credit at all.

5.6 Securitised credit, in one page

A large market sits beside corporate bonds and works differently: instead of lending to a company, you buy a claim on a pool of loans held in a legal entity created for the purpose.

Agency mortgage-backed securities are the largest and most important. A pool of US residential mortgages, guaranteed against default by Fannie Mae, Freddie Mac or Ginnie Mae, is sold as a pass-through security. Credit risk is essentially absent; the entire risk is that homeowners prepay. They refinance when rates fall and stay put when rates rise, so the investor receives their money back exactly when reinvestment is worst and is stuck exactly when it is best. That is a short option position, it is the *o* term of Identity 2, and it makes agency MBS the canonical negatively-convex instrument (§7.6).

The consequences reach beyond the asset class, because MBS investors hedge their changing duration by trading Treasuries. When rates fall, mortgages shorten, and a holder targeting a fixed duration must buy duration to replace what it lost; when rates rise, mortgages extend, and it must sell. Buying into rallies and selling into sell-offs amplifies moves in both directions. This convexity hedging is a documented driver of Treasury yield volatility (Hanson, 2014; Malkhozov, Mueller, Vedolin & Venter, 2016). [Fact] It is one of the clearest cases in markets of one market’s hedging demand being another market’s price dynamics.

Non-agency securitisations — CLOs holding leveraged loans, ABS holding auto loans or credit card receivables, CMBS holding commercial mortgages — take real credit risk and slice it into tranches of differing seniority. The senior tranche absorbs losses last and is rated AAA; the equity tranche absorbs them first. The critical insight is that tranching does not remove risk, it concentrates correlation risk in the senior tranche: a AAA tranche is safe unless losses are widely correlated, in which case it is not safe at all. That is precisely what happened to subprime securitisations in 2007–08, and the rating agencies’ correlation assumptions were the mechanism of failure, not their default assumptions.

§5 Key takeaways

1. A company’s leverage is a choice about the tax shield, distress costs and discipline. For a bondholder, the direction of travel matters more than the level, and it is why covenants have value.
2. Seniority is contractual, not natural. Structural subordination, liability management and resolution law can all invert the apparent ordering — Credit Suisse’s AT1 wipeout in 2023 is the reference case.
3. The BBB–/BB+ boundary is an economic discontinuity because a large investor population is contractually banned from crossing it. The forced selling is real and measurable.
4. Leveraged loans are credit risk with almost no duration, capped at par by the borrower’s right to prepay — a structurally short-optionality position.
5. The key equivalence: owning a corporate bond is owning a Treasury and having sold a put on the firm’s assets. Everything else about credit behaviour follows from that sentence.
6. Credit is therefore a short-volatility exposure with negative skew, and its spread should and does widen with asset volatility and leverage.
7. Agency MBS carries no credit risk and enormous prepayment risk; its holders’ hedging amplifies Treasury moves in both directions.
8. Tranching does not remove risk; it concentrates correlation risk in the senior tranche. That is what failed in 2007–08.

Part III — Valuation and risk

6. What a bond is worth

6.1 The one idea

A bond is worth the present value of what it pays. That is the whole of valuation; everything else is detail about how to compute the present value and how to quote the answer.

The reason present value is the right concept is worth stating rather than assuming. Money available today can be lent out and will grow; money promised for later cannot. So a payment of 100 in five years is worth less than 100 now, and the amount less is determined by what you could earn in the meantime. If safe five-year money earns 4% a year, then $100/(1.04)^5 = 82.19$ invested today becomes 100 in five years. Anyone offering that future 100 for more than 82.19 is offering you a worse deal than the alternative, and you should decline.

Doing this for each of the bond's payments and adding up gives Identity 1. With no credit risk:

$$P = \sum_t C_t Z(t), \quad Z(t) = \frac{1}{(1 + z(t))^t}$$

$Z(t)$ is the discount factor: today's price of one currency unit delivered at time t . $z(t)$ is the corresponding spot rate or zero rate. Note the subscript: there is a different discount rate for each maturity, because money for two years and money for ten years genuinely have different prices.

6.2 The curve, and three ways to say the same thing

The set of discount factors $\{Z(t)\}$ is the fundamental object. It can be quoted three ways, all containing identical information, and confusion between them is a standing source of error.

Spot (zero) rates $z(t)$ — the annualised rate for a single payment at t . Forward rates $f(t-1, t)$ — the rate for borrowing between two future dates, implied by today's prices. Par yields — the coupon rate that would make a bond of that maturity price at exactly 100.

They convert into each other exactly:

$$(1 + z(t))^t = (1 + z(t-1))^{t-1} (1 + f(t-1, t)), \quad y^{\text{par}}(T) = \frac{1 - Z(T)}{\sum_{t \leq T} Z(t)}$$

The first says that lending for t years must equal lending for $t-1$ years and then rolling into the forward — otherwise there is an arbitrage. The second says a par bond's coupon is set so that the present value of the coupons plus the discounted principal comes to 100.

A worked example makes the relationship concrete. Take spot rates of 3.00%, 3.40% and 3.70% for one, two and three years:

	Year 1	Year 2	Year 3
Spot rate $z(t)$	3.000%	3.400%	3.700%
Discount factor $Z(t)$	0.97087	0.93532	0.89673
Forward rate $f(t-1, t)$	3.000%	3.802%	4.303%
Par yield to t	3.000%	3.393%	3.684%

Three observations, all of which generalise. [Computed]

Forwards exceed spots when the curve rises. The 3-year spot rate of 3.70% is an average of the three one-year forwards (3.000%, 3.802%, 4.303%); for the average to be rising, the later terms must exceed it. This is the arithmetic behind a fact people find surprising: an upward-sloping curve does not say rates are expected to rise by the amount of the slope — it says the forwards are above today's spot, which is a much weaker statement (§11.2).

Par yields sit below spots on a rising curve. The 3-year par yield is 3.684% against a 3-year spot of 3.700%, because a coupon bond's early payments are discounted at the lower short rates.

A coupon bond's yield depends on its coupon. Pricing a 3-year 5% bond off this same curve gives 103.688 and a yield to maturity of 3.679% — different from both the 3-year spot rate and the par yield, purely because its cash flows are weighted more toward the early, cheaper years. Two bonds maturing on the same day, with the same issuer and identical credit, can and do have different yields to maturity. That is not a mispricing; it is an artefact of the yield measure.

6.3 Yield to maturity: what it actually is

Quoting a whole curve is cumbersome, so the market compresses a bond's valuation into a single number. Yield to maturity is the one discount rate that, applied to every cash flow, reproduces the observed price:

$$P = \sum_t \frac{C_t}{(1 + \frac{y}{m})^{mt}}$$

where m is the coupon frequency (2 for a semi-annual bond). There is no closed form for y ; it is found numerically, and it always exists and is unique for a bond with positive cash flows.

The crucial recognition: yield to maturity is an internal rate of return. It is the constant rate that equates a stream of payments to a price. That gives it two properties, one convenient and one dangerous.

Convenient: it is a single, comparable, monotone summary. Price up means yield down, always. Every bond in the world can be quoted on it.

Dangerous: an IRR is a description of a price, not a prediction of a return, and it carries a silent assumption.

6.4 Why yield to maturity is not your expected return

Solve the equation above for what you actually end up with, and the assumption becomes visible: the yield to maturity is the return you earn only if every coupon is reinvested at the yield to maturity itself. The formula discounts each cash flow at y , which is the same as assuming each cash flow compounds forward at y .

That assumption is almost never true, and the error is not small. Take the running example — a 10-year 4% bond bought at par — and vary only the rate at which coupons are reinvested: [Computed]

Reinvestment rate	Terminal wealth per 100	Realised annual return
0%	140.00	3.42%
2%	144.04	3.72%
4% (= the yield)	148.59	4.04%
6%	153.74	4.39%
8%	159.56	4.78%

A bond advertised as “yielding 4%” delivers anywhere from 3.42% to 4.78% depending on something entirely outside the bond — a range of 136 basis points. And the effect grows dramatically with maturity, because there are more coupons and they compound for longer. For a 30-year 4% bond, the same exercise spans 2.66% to 6.01% — a range of 335 basis points, wider than most people's entire expected equity risk premium.

The reason is visible in the composition of the terminal wealth. For the 30-year bond held with coupons reinvested at 4%, you end with 328.10 per 100 invested. Of the 228.10 of income, 120 is coupons and 108.10 — 47% — is interest earned on the coupons. [Computed] Nearly half the “fixed income” of a long bond is not fixed at all: it depends on rates you will encounter over the next thirty years.

Three implications:

- A quoted yield is a price, not a forecast. Use it to compare bonds, not to project wealth. If you need an expected return, model the reinvestment path explicitly.
- Zero-coupon bonds are the exception. With no coupons there is nothing to reinvest, so a zero's yield is its locked-in return if held to maturity. This is why liability-matching investors prefer them, and it is the honest version of “I have locked in 4%.”
- Reinvestment risk and price risk offset. If rates rise, your price falls but your reinvestment improves. At one particular horizon they cancel exactly, and that horizon is the duration (§7.3). This is not a coincidence; it is the deepest fact about duration and the reason the concept exists.

6.5 Pull to par

A bond's price converges to its face value as maturity approaches, regardless of what happens in between, because at maturity the bond is simply a claim on 100 tomorrow. This is pull to par, and it has a consequence people often find counter-intuitive.

A bond trading at 108 is not “expensive”. It is a bond with an above-market coupon, and you will lose 8 points of price over its remaining life, offset by the higher coupons you collect along the way. A bond trading at 92 is not “cheap”; you will gain 8 points, offsetting its below-market coupon. Price levels relative to 100 tell you about the coupon relative to current yields, and nothing about value.

Where it does matter is tax, and the details depend on the jurisdiction. Where the pull-to-par gain is taxed as a capital gain, more lightly than coupon income, a low-coupon bond bought below par is worth more after tax than an otherwise-identical high-coupon bond; UK gilts, whose capital gains are exempt for individuals, are the clean case. US rules are less generous: a discount on a bond bought in the secondary market is generally taxed as ordinary income when realised, unless it is very small. Either way the effect is real and shows up in relative pricing (§17.6).

6.6 The conventions, briefly

These are boring, they are where implementation errors live, and each is a one-line idea.

Clean and dirty price. A bond accrues interest continuously but pays it twice a year. A buyer between payment dates owes the seller the interest accrued so far. Markets quote the clean price, which excludes it, so that the quote does not sawtooth upward between coupons and drop on payment date. The amount actually paid is the dirty price:

$$P^{\text{dirty}} = P^{\text{clean}} + \text{accrued interest}$$

All the analytics in this document — yield, duration, convexity — are computed from the dirty price, because that is the actual investment.

Day counts. How “the fraction of a period elapsed” is measured. US Treasuries use actual/actual; US corporates and municipals use 30/360, which pretends every month has 30 days; money markets often use actual/360, which quietly makes a quoted rate about 1.4% higher than it appears. The conventions differ by market for historical reasons and nothing else, but using the wrong one produces small, persistent, hard-to-find errors.

Settlement. Treasuries settle T+1, corporates T+2. Accrued interest is computed to the settlement date, not the trade date.

Compounding frequency. A 4% semi-annual yield is not 4% a year; it is $(1.02)^2 - 1 = 4.04\%$ effective. Comparing an annual-pay European bond to a semi-annual-pay US bond without converting introduces a 4-basis-point error at these levels, and more at higher yields.

6.7 What actually matters

Readers consistently misallocate effort here, so the ordering is worth stating explicitly. In descending order of how much it affects your result:

1. Which cash flows, and whether you get them. Seniority, covenants, call schedules, and default risk dominate everything below.
2. The discount curve you use. Whether you discount off Treasuries, OIS, or the issuer's own curve changes the answer far more than any convention.
3. The reinvestment assumption, if you are projecting a return rather than quoting a price.
4. Compounding and day-count conventions. Small but systematic.
5. The root-finding algorithm for the yield. Irrelevant; any method converges.

The fifth item is where textbooks spend their time and where practitioners spend none.

§6 Key takeaways

1. A bond is worth the present value of its payments. The set of discount factors is the primitive object; spot rates, forward rates and par yields are three lossless encodings of it.
2. An upward-sloping curve makes forward rates exceed spot rates as a matter of arithmetic. It does not, by itself, say the market expects rates to rise.

3. Two bonds with the same issuer and maturity can have different yields to maturity if their coupons differ. That is an artefact of the yield measure, not a mispricing.
 4. Yield to maturity is an internal rate of return: it describes the price and silently assumes every coupon is reinvested at that same rate.
 5. That assumption is material. A 10-year 4% bond realises 3.42%–4.78% depending on reinvestment; a 30-year realises 2.66%–6.01%. For the 30-year, 47% of total income is interest on interest.
 6. Only a zero-coupon bond held to maturity truly locks in its quoted yield.
 7. Price relative to 100 says something about the coupon, not about value — except after tax, where the split between income and capital gain is real.
 8. Effort belongs on which cash flows you will receive and which curve you discount with, not on conventions and not at all on the root-finder.
-

7. Interest-rate risk: duration and convexity

Duration is the most useful number in fixed income and the most frequently misunderstood, because it has three different definitions that turn out to be the same number, and people learn one of them without being told about the others. This section builds all three, shows why they coincide, and then turns to the cases where the whole framework quietly breaks.

7.1 Duration as a balance point

Start with the simplest reading. A bond pays a sequence of amounts at a sequence of dates. Macaulay duration is the average of those dates, weighted by the present value of what arrives on each, with every cash flow discounted at the bond's own yield y (which by definition reproduces the price):

$$D_{\text{mac}} = \frac{\sum_t t \cdot C_t (1 + y/m)^{-mt}}{\sum_t C_t (1 + y/m)^{-mt}} = \sum_t t \cdot w_t, \quad w_t = \frac{C_t (1 + y/m)^{-mt}}{P}$$

(Discounting each cash flow off the zero curve instead, with $Z(t)$, gives a very slightly different number, the Fisher–Weil duration. On a flat curve they coincide.)

The weights w_t sum to one, so this is a genuine weighted average, measured in years. Picture the bond's cash flows as weights placed along a timeline: duration is where you would put the fulcrum to balance it.

For the running example — a 10-year 4% semi-annual bond at par — the Macaulay duration is 8.34 years. [Computed] It is less than 10 because the coupons arrive earlier than the principal and pull the balance point left. Three immediate corollaries:

- A zero-coupon bond's duration equals its maturity, exactly. There is only one cash flow, so the average date is that date. This is why a 30-year STRIP is the longest-duration instrument available.
- A higher coupon means shorter duration. More weight arrives early.
- A higher yield means shorter duration. Discounting at a higher rate shrinks the distant cash flows more, moving the balance point left. A 30-year bond is a much longer-duration instrument at 2% yields than at 8%.

7.2 Duration as price sensitivity

The balance-point picture is elegant, but the question people actually ask is more direct: if yields move 1%, how much does the price move? Answering it means taking the derivative of price with respect to yield. Differentiating $P = \sum_t C_t (1 + y/m)^{-mt}$ term by term gives

$$\frac{dP}{dy} = -\frac{1}{1 + y/m} \sum_t t C_t \left(1 + \frac{y}{m}\right)^{-mt},$$

and the sum on the right is $P \times D_{\text{mac}}$ — exactly the numerator of the weighted average from §7.1. Divide both sides by P :

$$\frac{1}{P} \frac{dP}{dy} = -\frac{D_{\text{mac}}}{1 + y/m} \equiv -D$$

The quantity $D = D_{\text{mac}}/(1 + y/m)$ is modified duration. For our bond, $D = 8.34/1.02 = 8.18$, so a 100 basis point rise in yield costs roughly 8.18% of value.

The practical units:

$$\frac{\Delta P}{P} \approx -D \Delta y, \quad \text{DV01} = D \times P \times 0.0001$$

DV01 — the dollar value of one basis point — is what trading desks actually use, because it is additive across positions in currency terms and does not require everything to be expressed as a percentage. A \$10 million position in our bond has a DV01 of about \$8,180: each basis point of yield move is worth roughly eight thousand dollars. Risk limits, hedge ratios and portfolio construction are all done in DV01.

The maturity scaling is the number worth memorising: [Computed]

Bond (4% coupon, at par)	Macaulay	Modified	DV01 per \$10m	Convexity
2-year	1.94	1.90	\$1,904	4.6
10-year	8.34	8.18	\$8,176	78.9
30-year	17.73	17.38	\$17,380	420.8

A 30-year bond carries roughly nine times the interest-rate risk of a 2-year, per dollar invested. That ratio, not the difference in yield, is the dominant fact when choosing where on the curve to sit.

7.3 Duration as the horizon where risk cancels

This third definition is the least known and the most illuminating, and it explains why the same number serves two apparently unrelated purposes.

§6.4 noted that rising yields hurt your price and help your reinvestment. Those two effects run in opposite directions, so there must be a holding period at which they exactly offset — a horizon at which your terminal wealth is insensitive to what happens to rates. That horizon is the Macaulay duration. The result is due to Redington (1952), who called the resulting strategy immunisation.

The demonstration is worth seeing numerically. Take the 10-year bond, shock yields immediately to some level, reinvest all coupons at that level, and record terminal wealth per 100 invested at various horizons: [Computed]

Horizon	@2%	@3%	@4%	@5%	@6%	Range
5.00 years	130.40	126.02	121.90	118.03	114.40	16.00
7.00 years	135.69	133.75	131.95	130.28	128.76	6.93
8.34 years	139.36	139.19	139.14	139.19	139.36	0.23
9.00 years	141.20	141.96	142.82	143.81	144.92	3.72
10.00 years	144.04	146.25	148.59	151.09	153.74	9.70

At the five-year horizon, a 400 basis point range of outcomes produces a 16-point spread in terminal wealth and you are hurt by rising rates. At the ten-year horizon you get a 9.7-point spread and are helped by rising rates. At 8.34 years — the duration — the spread collapses to 0.23 points, essentially nothing.

Notice also that 8.34 years is a minimum: terminal wealth is 139.14 if nothing happens and slightly higher for any move in either direction. That small upward curvature is convexity, and it means an immunised portfolio is not merely hedged but marginally long volatility.

So the three definitions are one number:

The equivalence. Macaulay duration is (i) the weighted-average time to receive the bond's cash flows, (ii) up to the factor $1/(1 + y/m)$, the percentage price sensitivity to yield, and (iii) the holding period over which price risk and reinvestment risk exactly cancel. These are the same quantity, not three quantities that happen to be close.

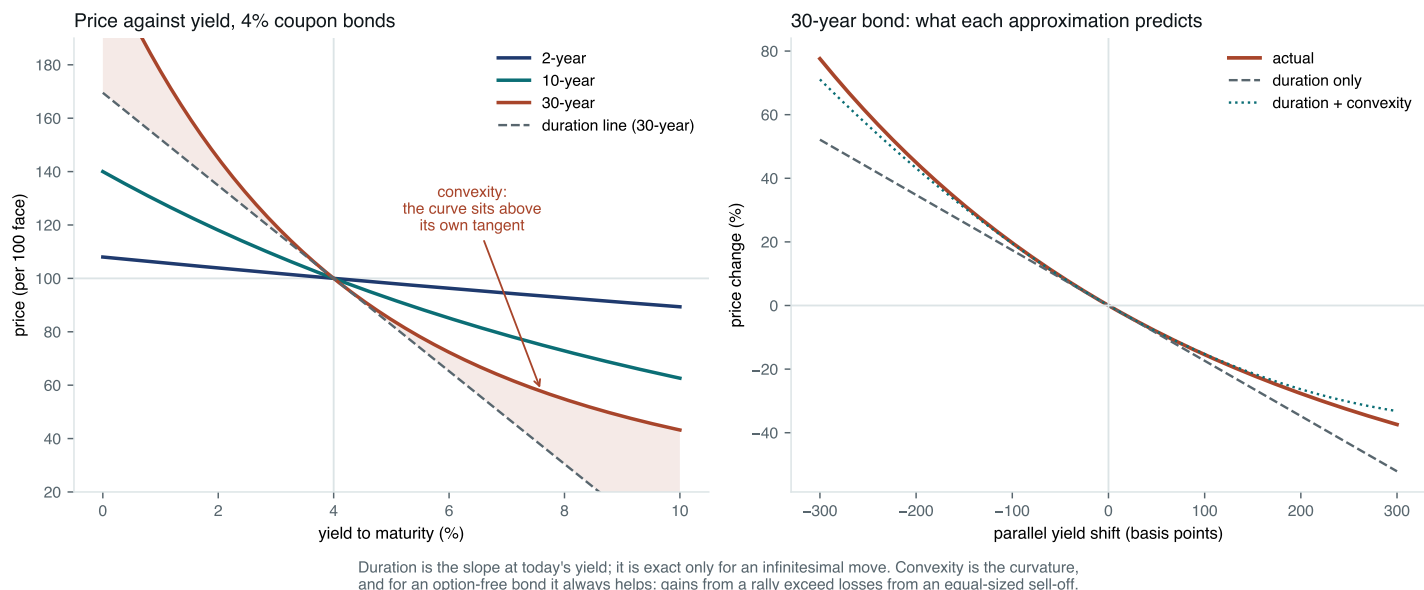
The third reading is what makes duration the organising concept of institutional bond investing. A pension fund with liabilities of duration 15 does not want a particular yield; it wants assets of duration 15, because that is the portfolio whose value moves with its obligations. Everything under the heading of liability-driven investing is this observation applied with leverage (§12.3, §13.4).

7.4 Convexity

Duration is a first derivative, so it is a straight-line approximation to a curved relationship — the first term of a Taylor expansion (A.3). The next term, the curvature, is convexity:

$$\mathcal{C} = \frac{1}{P} \frac{d^2 P}{dy^2}, \quad \frac{\Delta P}{P} \approx -D \Delta y + \frac{1}{2} \mathcal{C} (\Delta y)^2$$

Because the second term carries $(\Delta y)^2$, it is positive whichever way yields move. For an ordinary bond, convexity is always in the holder's favour: you gain more in a rally than you lose in an equal-sized sell-off.



The magnitudes: [Computed]

Move	30-year actual	Duration only	Error	Duration + convexity	Error
-100bp	+19.69%	+17.38%	-2.31	+19.48%	-0.21
+100bp	-15.45%	-17.38%	+1.93	-15.28%	+0.18
-300bp	+77.59%	+52.14%	-25.45	+71.08%	-6.51
+300bp	-37.42%	-52.14%	+14.72	-33.20%	+4.21

Three readings.

Convexity is worth real money in large moves. The 30-year bond gains 19.69% on a 100bp rally and loses 15.45% on a 100bp sell-off — an asymmetry of 4.24 percentage points in your favour, for free. At 300bp the asymmetry is enormous: +77.6% against -37.4%.

Duration alone is dangerously wrong for long bonds in big moves. It predicts a 52% loss on a 300bp sell-off; the truth is 37%. A risk system using duration only overstates the downside of long bonds and — much worse — understates their upside, which mis-sizes every trade.

Convexity is a convenience, not an edge. It is priced. A bond with more convexity yields slightly less, all else equal, which is exactly why long-dated and low-coupon bonds look expensive on simple yield comparisons. You buy convexity with yield.

The rule of thumb that follows: convexity matters little for short bonds and small moves, and a great deal for long bonds and large moves. For the 2-year bond, duration alone is accurate to two basis points on a 100bp move. For the 30-year at 300bp, it is off by fifteen percentage points.

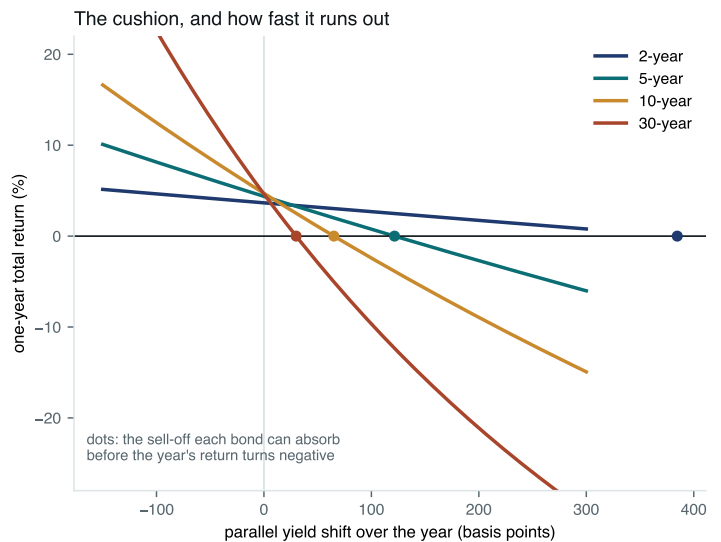
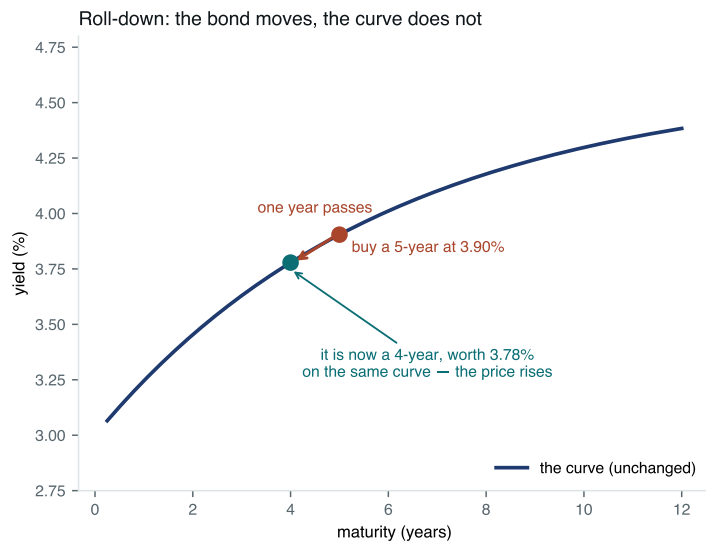
7.5 The cushion: how much of a sell-off a yield can absorb

Combine duration with Identity 3 and you get the most practically useful calculation in bond investing — one that reframes “is this yield attractive?” into a question with an answer.

Over one year you collect the yield (carry) and you roll down the curve. Against that, a yield rise costs you roughly $D \times \Delta y$. Set the total to zero:

$$\Delta y^{\text{breakeven}} \approx \frac{\text{carry} + \text{roll}}{D}$$

This is how much yields can rise before your year's return turns negative. It is the bond's cushion, and it collapses with maturity in a way that the yield curve's modest upward slope entirely disguises.



Carry and roll-down are the return you get for doing nothing, and they are known the day you buy. Duration converts a yield move into a price move. A long bond has more yield and far less protection: its cushion is measured in weeks of patience.

On a plain upward-sloping curve — 3.45% at two years rising to 4.59% at thirty: [Computed]

Bond	Yield	Roll-down	Carry + roll	Breakeven rise	Return if +100bp	Return if -100bp
2-year	3.45%	+0.20%	3.66%	385bp	+2.69%	+4.64%
5-year	3.90%	+0.46%	4.37%	122bp	+0.76%	+8.14%
10-year	4.30%	+0.41%	4.70%	65bp	-2.42%	+12.47%
30-year	4.59%	+0.03%	4.62%	30bp	-9.66%	+22.56%

Read the two ends against each other. The 30-year offers 114 basis points more yield than the 2-year — a 33% higher income. For that, its cushion falls from 385 basis points to 30. You are paid a third more to take thirteen times less protection.

That is the trade-off at the heart of every maturity decision, and stating it this way makes the answer situational rather than doctrinal. If you have a strong view that yields will fall, the long bond is where the leverage is: +22.6% against +4.6% on a 100bp rally. If you have no view, the short bond is being paid to wait, and can be wrong by nearly 4% before it costs anything.

Notice also what the roll-down column shows: it is largest in the middle of the curve, not at the long end. The 5-year picks up 46 basis points of price from rolling down a steep part of the curve; the 30-year picks up 3, because the curve is flat out there. The intuition is that roll-down is the slope of the curve times duration, and the slope dies out at the long end while duration keeps growing. This is the quantitative reason practitioners talk about the “belly” of the curve as the carry-efficient place to sit. [Practice]

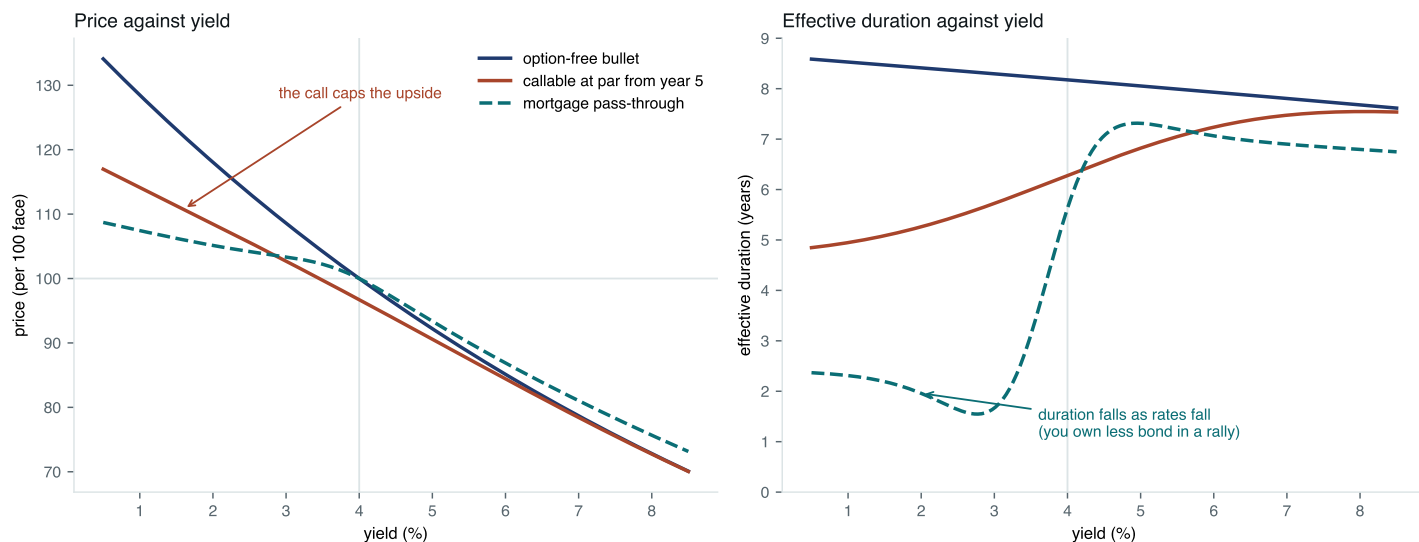
7.6 When convexity turns against you

Everything above assumed the bond’s cash flows are fixed. For a large part of the bond market they are not, because the borrower holds an option to change them — and borrowers exercise options in their own favour, which is by construction against yours.

Two cases, one mechanism:

- A callable bond. The issuer may repay early. They will, when rates have fallen and they can refinance cheaply.
- A mortgage pass-through. Homeowners may prepay. They do, when rates have fallen and they can refinance cheaply.

In both cases the bond’s upside is capped: as rates fall, the probability of early repayment rises, and the bond stops appreciating. §5.5’s decomposition applies again, with a different option — you own a bond and you have sold an option on interest rates.



Positive convexity means duration lengthens into a rally and shortens into a sell-off — the holder is automatically on the right side. Negative convexity reverses both. Stylised instruments; the shapes are the point, not the levels.

The asymmetry reverses completely: [Computed]

Instrument	-100bp	+100bp	Asymmetry
Option-free bullet	+8.58%	-7.79%	+0.79pp (helps you)
Callable from year 5	+6.18%	-6.35%	-0.16pp
Mortgage pass-through	+3.37%	-6.63%	-3.27pp (hurts you)

The pass-through loses roughly twice as much in a sell-off as it gains in an identical rally. That is the signature of a short option position, and it is the economic content of “negative convexity.”

The right-hand panel of the figure shows the consequence that trips people up most. The pass-through’s effective duration is about 5.6 years at a 4% yield, 1.7 years at 3%, and 7.3 years at 5%. [Computed] Your position size changes without you trading. A rally shortens you exactly when being long would have paid; a sell-off lengthens you exactly when being long hurts. A risk report showing “duration 5.6” is describing a number that will be 7.3 after a bad week — which is why practitioners quote effective duration, computed by actually repricing the instrument under shifted curves rather than by differentiating a formula that assumes fixed cash flows.

Two consequences worth carrying:

- Never use analytic duration for an instrument with embedded options. The formula in §7.2 assumes C_t does not depend on y . For a callable bond or an MBS it does, and the formula is not approximately wrong but structurally wrong.
- The spread on these instruments is option premium, not risk premium. An agency MBS yielding 75bp over Treasuries is not paying you for credit risk — there is none — it is paying you for having written an option to millions of homeowners. Whether that is a good trade depends on whether the premium exceeds the option’s fair value, which is what option-adjusted spread is designed to answer (§8.4).

7.7 What duration does not capture

Duration answers one question: what if the whole curve moves in parallel? Curves do not move in parallel. Three extensions cover most of the gap.

Key-rate durations. Instead of one number, compute sensitivity to a shift at each of several points on the curve — 2, 5, 10, 30 years. A portfolio can have zero total duration and large exposure to the curve steepening, which key-rate durations reveal and aggregate duration hides. Empirically, curve movements decompose almost entirely into three factors — level, slope and curvature, accounting for the great majority of variance (Litterman & Scheinkman, 1991) [Fact] — so three numbers usually suffice.

Spread duration. For a credit bond, sensitivity to its spread moving is distinct from sensitivity to the underlying government curve moving, even though both are computed the same way, because the two move for different reasons and often in

opposite directions. A 10-year corporate bond might have 7 years of rate duration and 7 years of spread duration, and lose on one while gaining on the other.

The refinement that matters here is duration times spread (DTS): the empirical regularity is that spreads move proportionally rather than in parallel — a bond at 500bp moves about five times as much in basis points as a bond at 100bp — so the right risk measure for credit is spread duration multiplied by spread level, not spread duration alone (Ben Dor, Dynkin, Hyman, Houweling, van Leeuwen & Penninga, 2007). [Fact] This is one of the most useful practical results in credit portfolio management, and it means that risk budgeting on spread duration alone systematically understates the risk of low-quality holdings.

Empirical duration. For high-yield bonds especially, the measured sensitivity to Treasury yields is much lower than analytic duration implies, because credit spreads tend to tighten when government yields rise — both being driven by improving growth. A high-yield bond with 4 years of analytic duration may behave as though it had one or two. [Contested] — the effect is well documented but unstable, and it reverses in inflation-driven sell-offs, which is exactly when you need it not to. 2022 was such a period: rates rose and spreads widened together, and high-yield investors got the full duration they thought they had hedged away.

§7 Key takeaways

1. Duration has three definitions — weighted-average time to cash flow, percentage price sensitivity, and the horizon where price and reinvestment risk cancel — and they are the same number.
2. The third definition is why institutions match duration to liabilities. A 10-year bond's terminal wealth at its 8.34-year duration horizon varies by 0.23 points across a 400bp range of yields, against 16 points at a 5-year horizon.
3. A 30-year bond carries about nine times the rate risk of a 2-year per dollar. That ratio dominates the yield difference in any maturity decision.
4. Convexity always helps an option-free bond: the 30-year gains 19.7% on a 100bp rally and loses 15.5% on a 100bp sell-off. It is priced, so you buy it with yield.
5. The cushion. Breakeven yield rise $\approx$ (carry + roll) / duration. On a normal curve that is 385bp for a 2-year and 30bp for a 30-year: a third more yield for a thirteenth of the protection.
6. Roll-down is largest in the belly of the curve, not at the long end, because slope dies out while duration keeps growing.
7. Callable bonds and mortgages are short options: their upside is capped, their convexity is negative, and their duration shortens in a rally. A pass-through can lose twice what it gains on a symmetric move.
8. Never use analytic duration on an instrument with embedded options — the formula assumes cash flows independent of yield, and there they are not.
9. For credit, use duration times spread, not spread duration alone: spreads move proportionally, so a 500bp bond carries roughly five times the risk of a 100bp bond at equal duration.

8. Spreads: one idea, eight names

8.1 The idea, and why there are so many versions

A spread is a bond's yield in excess of some reference. That is the whole idea. It exists because the level of interest rates is common to every bond and tells you nothing about the issuer — strip it out and what remains is the market's price for this particular borrower's risk.

The proliferation of spread measures comes from three legitimate questions the simple version dodges. Excess over what — a government bond, a swap, a whole curve? Measured how — one rate applied to everything, or a curve-consistent calculation? And with embedded options handled how?

Getting this wrong is the commonest unforced error in credit analysis, because different desks quote different measures, all of them called “the spread”, and the differences are large enough to reverse a relative-value conclusion.

8.2 The family

Measure	Reference	Definition	What it is good for
Nominal / G-spread	One government yield	Bond YTM minus the yield of a matched government bond	Quick quoting; the newspaper number
I-spread	One swap rate	Bond YTM minus the matched-maturity swap rate	Comparing to a bank's funding cost
Z-spread	The whole curve	The constant addition to every zero rate that reproduces the price	The correct curve-consistent measure
ASW	Floating index	The floating spread you receive after swapping the bond's fixed coupons	What a leveraged buyer actually earns
Discount margin	Floating index	The Z-spread of a floating-rate note	Loans and FRNs
OAS	The whole curve, with volatility	Z-spread minus the value of embedded options	Anything callable or prepayable
CDS spread	—	The annual premium to insure the issuer against default	Pure credit, no funding or curve
Spread to worst	One government yield	Spread to the least favourable call date	Callable high yield, as a convention

A note on the swap curve, since the I-spread and the asset swap spread both use it. With LIBOR retired, the swap curve in the major currencies is now referenced to an overnight rate — SOFR in dollars, €STR in euros, SONIA in sterling — and is therefore an overnight indexed swap (OIS) curve: the fixed rate that exchanges for the compounded overnight rate over the swap's life (Schrimpf & Sushko, 2019). Because an overnight loan carries almost no credit risk and the swap exchanges no principal, the OIS curve is the closest thing available to a clean risk-free discount curve, free of the convenience premium embedded in Treasury yields (§3.1).

Two of these deserve the reasoning spelled out.

Z-spread exists because a nominal spread compares a bond's single yield to a government bond's single yield, and single yields depend on coupon (§6.2). Two bonds with identical credit risk and maturity but different coupons will show different nominal spreads. Z-spread fixes this by discounting each cash flow at the matching zero rate plus a constant s , and solving for s :

$$P = \sum_t \frac{C_t}{(1 + (z(t) + s)/m)^{mt}}$$

How much does it matter? On a steep curve rising from 2% to 6.5%, take two bonds that genuinely carry an identical Z-spread of 100 basis points: [Computed]

Bond	Price	Z-spread	Nominal spread	Gap
30-year, 8% coupon	115.41	100.0bp	89.2bp	−10.8bp
30-year, 1% coupon	24.34	100.0bp	127.4bp	+27.4bp
10-year, 8% coupon	115.46	100.0bp	91.2bp	−8.8bp
10-year, 1% coupon	61.90	100.0bp	114.4bp	+14.4bp

The two 30-year bonds are equally risky by construction, and the nominal spread says one is 38 basis points cheaper than the other. That is the entire spread differential between adjacent rating categories, manufactured out of nothing but the curve's slope and the coupon. The gap shrinks as the curve flattens and vanishes when it is flat — which is why the sloppy measure survives: it is usually close enough, and then suddenly is not.

Asset swap spread answers the leveraged investor's question. Buy the bond, enter a swap paying its fixed coupons and receiving a floating rate plus a spread, and you have converted a fixed-rate credit bond into a floating-rate credit exposure. The spread you receive is the asset swap spread, and it is what a bank funding at floating rates actually earns for taking the credit risk. The subtlety is that the standard “par asset swap” structures 100 notional of swap against a bond that may be priced at 115 or 85, which introduces a mismatch — so ASW and Z-spread agree closely for bonds near par and diverge as the price moves away from it. [Practice]

8.3 The equivalences

The table below is the one to keep. It says which of these measures are the same thing, which are approximately the same and under what conditions, and which merely share a name.

Relationship	Status	Condition / size of the gap
Z-spread = nominal spread	Exact	Only if the zero curve is flat
Z-spread $\approx$ nominal spread	Approximate	Within a few bp for near-par bullets on a normal curve; tens of bp for off-par bonds on a steep curve
OAS = Z-spread	Exact	Only for an option-free bond
OAS = Z-spread — option cost	Exact, by definition	The option cost is model-dependent, so OAS inherits that dependence
Discount margin = Z-spread	Exact in concept	The same calculation applied to a floater; quoted margins also depend on the convention for projecting future coupons
ASW $\approx$ Z-spread	Approximate	Equal at par; diverges roughly in proportion to (price — 100)
CDS spread $\approx$ bond spread over risk-free	Approximate	The difference is the basis (§8.5); driven by funding, deliverability and bond price
Spread $\approx \lambda(1 - R)$	Approximate	The credit triangle; exact in continuous time with a flat hazard rate (§9.2)
Spread duration = rate duration	Different	Same formula, different risk; they often move oppositely
Spread to worst = spread to maturity	Different	Equal only for a non-callable bond or one trading well below its call price
Index OAS = average of constituent OAS	Different	Index spreads are market-value weighted, so they are dominated by the largest issuers

The last three rows are where the real damage happens, because they share names with things they are not.

8.4 Option-adjusted spread

OAS is what is left of a bond's spread after paying for the options the borrower holds against you. Conceptually:

$$\text{OAS} = \text{Z-spread} - \underbrace{\text{option cost}}_{\text{annualised}}$$

Computing it requires a model of how interest rates evolve, because the value of the borrower's option depends on the volatility of rates, not just their level. The standard procedure simulates many rate paths, computes the cash flows along each (exercising calls or prepaying as a rule dictates), discounts at the path's rates plus a constant s , and solves for the s that averages to the market price.

Three things to know about the resulting number.

It is model-dependent, and the dependence is not small. OAS depends on the assumed volatility and, for mortgages, on the prepayment model. Two dealers can quote materially different OAS for the same bond because they assume different things. An OAS is a statement about a bond given a model, not a property of the bond. [Practice]

A negative OAS is meaningful, not an error. It says the bond is priced richer than the model thinks the options are worth — usually because investors value something the model omits, such as the instrument's liquidity or its role as collateral.

For mortgages, OAS is effectively a bet on the prepayment model. The largest source of return dispersion among MBS investors is not rate views but differences in prepayment forecasting. This is not a side issue: the whole asset class is a wager on household refinancing behaviour, and the spread you are quoted has already netted out someone's estimate of it.

8.5 CDS, and the basis

A credit default swap is insurance on a borrower. The buyer pays a fixed annual premium; if the borrower experiences a defined credit event, the seller compensates for the loss. It is the cleanest available measure of credit risk, because it contains no interest-rate exposure and requires no capital to fund a bond position.

The arbitrage relationship is straightforward in principle: owning a corporate bond and buying protection on the issuer should leave you with a risk-free position, so

$$\text{CDS spread} - \text{bond spread} = \underbrace{\text{the basis}}_{\text{should be zero}}$$

It is not zero. The CDS–bond basis is a persistent, time-varying quantity, and its drivers are a compact list of everything that breaks textbook arbitrage:

- Funding. The arbitrage requires financing the bond. If your funding cost exceeds the risk-free rate, you need the bond to yield more, pushing the basis negative.
- Balance sheet. The trade consumes capital. When capital is scarce — 2008, March 2020 — the basis goes deeply negative and stays there, which is the same phenomenon as the TIPS–Treasury gap of \$3.3.
- Cheapest-to-deliver. A CDS typically references a class of obligations, and the protection buyer can deliver the cheapest, which is a small option in their favour.
- Bond price relative to par. A bond trading at 60 has much less to lose in default than one at 100, so its spread and the CDS spread are not measuring the same loss.

The basis is therefore a useful barometer: large negative basis means capital is scarce, not that credit is cheap. [Practice] Reading it as a mispricing has destroyed more than one levered credit fund.

8.6 Which measure to use

Situation	Use
Quoting quickly, near-par bullet, normal curve	Nominal spread
Comparing bonds with different coupons or maturities	Z-spread
Anything callable, puttable or prepayable	OAS — and ask about the model
Floating-rate note or leveraged loan	Discount margin
A leveraged or bank buyer	ASW
Comparing to the derivative market, or isolating pure credit	CDS spread
Callable high yield trading near its call price	Spread to worst
Risk-budgeting a credit portfolio	Duration times spread (§7.7)

§8 Key takeaways

1. A spread strips out the common level of rates and leaves the market's price for this borrower. The many versions differ in the reference, the curve treatment, and the handling of options.
2. Z-spread is the curve-consistent measure. Nominal spread can manufacture a 38bp difference between two genuinely identical credits on a steep curve, purely from the coupon.
3. OAS equals Z-spread minus the annualised cost of the options the borrower holds against you. It equals the Z-spread exactly for an option-free bond, and never otherwise.
4. OAS is model-dependent. Quoting one without the volatility and prepayment assumptions behind it is quoting half a number.
5. CDS spread is the cleanest credit measure because it carries no rate exposure and needs no funding.
6. The CDS–bond basis measures the cost of balance sheet, not the mispricing of credit. A deeply negative basis is a signal about funding conditions.
7. “Spread to worst” and “spread to maturity” are different numbers, and index spreads are market-value weighted, so they describe the largest issuers rather than the typical one.

9. Credit risk: default, recovery, and what the spread pays for

9.1 The four numbers

Credit risk reduces to four quantities, and almost all of credit analysis is estimating them or arguing about them.

- Probability of default (PD) — the chance the borrower fails to pay over some horizon.
- Loss given default (LGD) — the fraction of your claim you do not recover. Equivalently $1 - R$, where R is the recovery rate.
- Exposure at default (EAD) — how much you are owed at the moment it happens. Trivial for a bullet bond; not for a revolving credit line, which borrowers draw down precisely as they deteriorate.
- Default correlation — the tendency of borrowers to fail together.

Expected loss is $PD \times LGD \times EAD$, and it is the easy part. The fourth number is what makes credit hard, because it determines the shape of the loss distribution rather than its mean, and because a portfolio's diversification is entirely a claim about it. A book of 200 loans with 2% expected loss each has a comfortable loss distribution if defaults are independent and a terrifying one if they are driven by a common factor — which they are.

9.2 The credit triangle

The single most useful relationship in credit connects a spread to a default rate, and it takes three lines to derive.

Model default as arriving at random with constant intensity λ — a hazard rate, so that the probability of defaulting in the next instant dt , given survival so far, is λdt . Chaining that survival requirement across every instant between now and t compounds to a survival probability of $e^{-\lambda t}$. Now hold a bond paying a spread s over the risk-free rate. Over an instant you earn $s dt$ extra. With probability λdt the borrower defaults and you lose $(1 - R)$ of your money. For the position to be fair:

$$\underbrace{s dt}_{\text{what you earn}} = \underbrace{\lambda dt (1 - R)}_{\text{what you expect to lose}} \implies \boxed{s = \lambda (1 - R)}$$

This is the credit triangle: spread, hazard rate and recovery are three corners of one relationship, and any two determine the third. Working it in the usual direction — from an observed spread to an implied default rate — gives: [Computed]

Spread	Assumed recovery	Implied hazard λ	Implied 5-year default probability
100bp	40%	1.67% / yr	8.0%
100bp	20%	1.25% / yr	6.1%
100bp	70%	3.33% / yr	15.4%
500bp	40%	8.33% / yr	34.1%
500bp	20%	6.25% / yr	26.8%

Three lessons, each of which matters in practice.

You cannot separate PD from recovery using a single spread. Only the product $\lambda(1 - R)$ is observed. A bond at 500bp is consistent with an 8.33%-a-year hazard and 40% recovery, or a 6.25%-a-year hazard and 20% recovery, and the market price cannot tell you which. Any statement of the form “the market is pricing a 34% chance of default” is really “the market is pricing a 34% chance of default conditional on my recovery assumption”, and the assumption is usually someone's convention rather than an estimate. [Practice]

Spreads translate into large-sounding default probabilities. 500 basis points — an ordinary high-yield spread — implies a one-in-three chance of default over five years. Whether that is the market's actual belief is the subject of §9.4, and the answer is no.

The relationship is why senior secured debt yields less. Higher recovery means a lower spread for the same default risk, mechanically.

9.3 Recovery is not a constant

The triangle treats R as a fixed number. It is not, and the way it varies is what makes credit a systematic risk rather than a diversifiable one.

Long-run averages from rating-agency studies give roughly: 60–80% for senior secured bank debt, 35–45% for senior unsecured bonds, 20–30% for subordinated. [Fact] But the dispersion around those averages is enormous, and it is not random. Recovery depends on:

- How much debt sits ahead of you, which is the dominant driver. A senior unsecured bond behind a large secured loan may recover almost nothing.
- Whether the assets have value outside the firm. An airline's aircraft have a liquid resale market; a software company's assets are mostly the people, who leave.
- The state of the industry. And this is the important one.

Acharya, Bharath & Srinivasan (2007) document that creditors recover significantly less when the defaulting firm's industry is distressed — because the natural buyers of its assets are themselves constrained. [Fact] Since industries tend to be distressed when many of their firms are defaulting, this produces a systematic negative correlation between default rates and recovery rates: the years with the most defaults are also the years with the worst recoveries. Altman, Brady, Resti & Sironi (2005) estimate this relationship directly. [Fact]

The consequence is that a credit portfolio's bad years are much worse than an independent-defaults model predicts, because two variables that a simple model treats as separate move together against you. Any credit risk model that assumes a fixed recovery rate is understating tail risk, and by a lot.

9.4 Risk-neutral and real-world default probabilities

This distinction is the most important conceptual point in the section, and skipping it leads directly to the wrong conclusion about whether credit is worth owning.

The probability implied by a spread is a risk-neutral probability: it is the default rate that would make the bond fairly priced if investors were indifferent to risk. They are not. Investors demand compensation for bearing default risk beyond its expected cost, and that compensation inflates the implied probability above the true one.

The gap is large. [Fact] Historical default rates for investment-grade issuers have averaged well below a quarter of a percent per year, while investment-grade spreads have averaged over 100 basis points — implying risk-neutral default rates several times the historical ones. The ratio is largest for the highest-quality credits: for Aaa and Aa names, implied default rates can exceed historical rates by an order of magnitude.

So the spread is doing two jobs at once, exactly as Identity 2 says:

$$s = \underbrace{EL}_{\text{what you actually expect to lose}} + \underbrace{CRP}_{\text{what you demand for bearing it}} + \underbrace{\ell}_{\text{illiquidity}} + \underbrace{\text{tax}}_{\text{in some jurisdictions}}$$

The tax term is the one piece not in Identity 2, because it belongs to the holder rather than the bond: US corporate coupons bear state income tax that Treasury coupons do not, so a taxable buyer needs extra yield to be indifferent, and when such buyers are marginal the requirement shows up in the price (§9.5).

Separating these is the central empirical problem in credit, and it has a name.

9.5 The credit spread puzzle

The puzzle. Structural models in the Merton tradition (§5.5), calibrated to match observed default rates, recovery rates and equity premia, predict credit spreads far below the observed ones — particularly for investment grade and particularly at short maturities. The canonical statement is Huang & Huang (2012), who find that credit risk accounts for only a modest fraction of the spread on investment-grade bonds, with the share rising as quality falls. [Fact]

A complementary decomposition by Elton, Gruber, Agrawal & Mann (2001) finds that expected default loss explains a surprisingly small part of the corporate-Treasury spread; a substantial chunk is US state taxes, which apply to corporate coupon income but not to Treasury income; and most of the rest behaves like a systematic risk premium, moving with the same factors that price equities. [Fact]

The candidate resolutions, each with a real case:

- Defaults cluster in bad states. Losing money on your bonds at the same moment your equities fall and your job is at risk is far worse than losing the same amount at random. Chen, Collin-Dufresne & Goldstein (2009) show that a model with the countercyclical risk aversion needed to explain the equity premium also generates realistic credit spreads — linking the two puzzles into one. This is currently the most persuasive strand. [Contested]

- Skewness and undiversifiability. A credit portfolio's return is negatively skewed by construction (§5.5), and the number of names needed to diversify it is far larger than for equities, so investors reasonably demand a premium (Amato & Remolona, 2003). [Contested]
- Illiquidity. Corporate bonds are genuinely hard to trade, and the illiquid component of the spread is measurable and large, especially in crises (Bao, Pan & Wang, 2011; Dick-Nielsen, Feldhütter & Lando, 2012). [Fact]
- There is no puzzle. A more recent line argues the puzzle is an artefact of calibration choices and largely disappears with better default-probability inputs. [Contested]

My read, and it is a read: the truth is a mixture, and the practically important part is not the decomposition but the residual. Over the long sweep of history, credit spreads have averaged roughly twice realised default losses (Giesecke, Longstaff, Schaefer & Strebulaev, 2011), which is to say credit has paid a genuine premium — but one considerably smaller than the headline spread, and earned in a shape that punishes leverage and forced selling. [Fact] Sizing a credit allocation off the quoted spread rather than off the spread net of expected loss is the single most common error in this asset class.

9.6 The default cycle

Defaults are not a constant background rate; they arrive in waves, and the waves are tightly linked to the credit cycle.

The mechanism is a feedback loop that runs the same way every time. Spreads are tight, so borrowing is cheap. Cheap borrowing attracts weaker borrowers, and lenders relax covenants to win business. Leverage builds. Then something changes — a recession, a rate shock, a commodity move — and the weakest of the recent cohort cannot refinance. Defaults rise, spreads widen, lending stops, and more borrowers fail because they cannot refinance rather than because their business failed.

That last step is the one people underweight. [Practice] Many corporate defaults are refinancing failures rather than business failures. A company with a viable business and a bond maturing in a closed market defaults; the same company with the same business and a five-year runway does not. This is why the maturity wall — the schedule of upcoming refinancings — is a genuinely useful forward-looking indicator, and why central bank actions that reopen credit markets reduce defaults so effectively.

Two usable regularities. Greenwood & Hanson (2013) show that the quality of issuers coming to market predicts subsequent credit excess returns: when junk issuance share is high, returns are subsequently low. [Fact] And spreads themselves are mean-reverting and mildly predictive of their own future returns, which is the credit analogue of the carry effect found across asset classes (Kojen, Moskowitz, Pedersen & Vrugt, 2018).

9.7 How practitioners actually estimate default risk

Approach	Core idea	Strength	Weakness
Ratings	Agency ordinal judgment	Comparable across issuers and time; drives mandates	Lags; through-the-cycle by design; issuer-pays conflict
Structural (Merton/KMV)	Default when asset value hits the debt boundary	Uses market data; updates continuously; economically interpretable	Needs unobservable asset value and volatility; underpredicts short-horizon spreads
Reduced-form	Default is a jump with an intensity fitted to prices	Fits market prices well; natural for derivatives pricing	Says nothing about why; no economic content
Accounting scores	Ratios combined into a discriminant score (Altman, 1968)	Simple, transparent, long track record	Backward-looking; poor for financials and asset-light firms
Market-implied	Read the CDS or bond spread directly	Fastest-updating; incorporates everything known	Contains a risk premium, so it is not a probability (§9.4)

The honest summary is that these are complements rather than competitors. Market-implied measures move first and contain a premium; structural models give an economic story; ratings determine who may hold the bond; accounting scores catch the deteriorations that markets have not yet noticed. A credit process using only one of them has a predictable blind spot. [Practice]

§9 Key takeaways

1. Credit risk is four numbers: default probability, loss given default, exposure, and correlation. The fourth is the hard one and determines the shape of the loss distribution.
 2. The credit triangle: $\text{spread} = \lambda(1 - R)$. Any two of spread, hazard rate and recovery determine the third.
 3. A spread cannot separate default probability from recovery — only their product is observable. Every “the market implies an X% default chance” claim hides a recovery assumption.
 4. Recovery rates fall in the years when default rates rise, because the buyers of distressed assets are themselves distressed. Fixed-recovery models understate tail risk substantially.
 5. Spread-implied default probabilities are risk-neutral and exceed historical default rates by a wide margin — by an order of magnitude for the highest-quality credits.
 6. The credit spread puzzle: structural models explain only part of investment-grade spreads. Taxes, illiquidity and a genuine risk premium for defaults clustering in bad states each account for some of the rest.
 7. Historically, credit spreads have averaged roughly twice realised default losses. Credit pays a real premium — about half the headline spread, in a shape hostile to leverage.
 8. Many defaults are refinancing failures rather than business failures, which is why the maturity wall is informative and why reopening credit markets prevents defaults.
-

Part IV — Where yields come from

Part III took yields as given and asked what follows. Part IV asks where they come from. The organising device remains Identity 2: this part walks its first three terms — real rate, expected inflation, term premium — and then turns to the supply and demand facts that determine them in practice rather than in theory.

10. The macro engine: real rates, inflation, and the central bank

10.1 The first split: real and nominal

A lender cares about purchasing power, not currency units. If you lend at 5% and prices rise 3%, you have gained 2% of real buying power. This gives the Fisher relation (Fisher, 1930):

$$\underbrace{y}_{\text{nominal}} \approx \underbrace{r}_{\text{real}} + \underbrace{\pi^e}_{\text{expected inflation}}$$

(Exactly, $(1 + y) = (1 + r)(1 + \pi^e)$; the additive version drops a cross-term that is negligible at ordinary rates and not at high ones.)

The split is the right first cut because the two halves are determined by entirely different forces. The real rate is set by the supply of savings and the demand for investment — the deep structure of the economy. Expected inflation is set by the central bank’s target and the market’s belief that it will be achieved. A bond yield’s movements can always be usefully sorted into “the real economy changed” and “the inflation outlook changed”, and the two have opposite implications for almost everything else you own (§14).

10.2 What sets the real rate

The real interest rate is the price that balances the desire to save against the desire to invest. If people want to save more than firms want to invest, the price of capital falls until they match.

Economists call the level that prevails when the economy is at full employment with stable inflation the natural rate or r^* . It is not observable — it must be inferred from the behaviour of output and inflation — and the standard estimation approach is due to Laubach & Williams (2003), extended internationally by Holston, Laubach & Williams (2017). Their central finding is the dominant macro-financial fact of the past forty years: r^* declined substantially and in nearly every developed economy simultaneously. [Fact] Estimates of the US r^* fell from around 3–4% in the 1980s to below 1% by the mid-2010s.

The candidate explanations — demographics raising desired saving, slower productivity growth reducing desired investment, rising inequality concentrating income among high savers, increased demand for safe assets — are surveyed by Rachel & Summers (2019), who make the additional point that without the offsetting rise in government debt and pension spending, the private sector’s natural rate would have fallen further still. [Contested] — the direction is agreed; the relative weights are not, and the measurement itself is imprecise enough that the confidence bands on r^* are embarrassingly wide.

For an investor, three things follow. The level of yields you grew up with is not a constant of nature. It is a slowly-moving equilibrium that can shift by percentage points over a decade. Estimates of r^* are too imprecise to trade on directly, but the direction of revisions matters, and in the early 2020s those revisions turned upward. And the whole framework is about the real rate, so inflation shocks do not move r^* — they move the other term.

10.3 What sets expected inflation

In a modern inflation-targeting regime, expected inflation is mostly the central bank's target plus the market's doubt about it. When a central bank is credible, long-horizon inflation expectations barely move in response to actual inflation — a property called anchoring, and it is the central bank's most valuable asset.

The 2021–23 inflation episode is the natural test. US inflation reached levels unseen since the early 1980s. Ten-year breakevens rose — but by far less than realised inflation, peaking around 3% against headline inflation above 8%. [Fact] That gap is anchoring working: the market believed the overshoot was temporary because it believed the Federal Reserve would act. Compare the late 1970s, when long-horizon expectations moved with realised inflation, requiring the Volcker disinflation to re-anchor them at enormous cost.

The practical reading: watch the level of long-horizon breakevens, not short ones. Short breakevens mechanically track energy prices and tell you nothing about regime. A move in five-year-forward five-year breakevens is a statement about the central bank's credibility, and it is one of the few genuinely high-information prices in macro.

10.4 The term premium

The first two terms explain the level of short rates. The third explains why long rates differ from the expected average of short rates.

The concept. You can earn the ten-year yield by buying a ten-year bond, or by rolling one-year bonds ten times. The two are not the same: the first locks in a known nominal return, the second exposes you to whatever short rates turn out to be. The term premium is the extra yield the long bond must offer to make investors indifferent:

$$y_{10} = \underbrace{\frac{1}{10} \mathbb{E} \left[\sum_{k=0}^9 y_1^{(t+k)} \right]}_{\text{expected average short rate}} + \text{TP}_{10}$$

Why it should be positive — and why it might not be. The standard argument: a long bond's price is volatile, so a risk-averse investor demands compensation. But that argument assumes the investor cares about short-horizon price volatility. A pension fund with thirty-year liabilities faces the opposite problem — for it, the risky asset is cash, because rolling short exposes it to a fall in rates that raises the value of its obligations. For such an investor the long bond is the hedge, and it might accept a negative term premium to hold it.

So the sign of the term premium depends on who is marginal and, more fundamentally, on whether bonds hedge or amplify an investor's other risks. That is the same question as the bond–equity correlation of §14, and it is not a coincidence. When the dominant macro shock is a demand shock — growth falls, inflation falls, bonds rally while equities fall — bonds are insurance and the term premium can be negative. When the dominant shock is a supply shock — inflation rises, growth falls, bonds and equities fall together — bonds are risk, and the term premium must be positive. The formal version of this argument is Campbell, Pflueger & Viceira (2020), and it is the most useful single idea in this part of the document.

Measurement, and its difficulties. The term premium is not observable: it is the residual after subtracting an expectation nobody can see. Every estimate is a model output. Two standard ones are the affine model of Kim & Wright (2005) — “affine” is unpacked in §11.4 — and the regression-based estimator of Adrian, Crump & Moench (2013), the latter maintained publicly by the New York Fed. They agree on the broad picture and disagree on levels by enough to matter.

The broad picture: the US ten-year term premium fell steadily from the 1980s and was estimated as negative for much of the period from 2016 to 2021. [Fact] A negative term premium means investors accepted less than the expected average short rate to own a ten-year bond — paying for duration rather than being paid for it. That is an extraordinary state of affairs, and the explanations for it are the subject of §12: central bank purchases, regulatory demand for safe assets, and a global savings glut all removed duration from the market. It reversed sharply in 2022–23.

Practical guidance. Treat term premium estimates as a coarse regime indicator, not a valuation signal. A negative term premium tells you the long end is being held up by something other than investors being paid to hold it, which is a fragile condition. But the estimates are too model-dependent and too heavily revised to trade directly. [Practice]

10.5 The central bank

Central banks set one price directly and influence the rest.

The policy rate is the overnight rate at which banks lend reserves to each other, and the central bank sets it by choosing the rate it pays on reserves. This anchors the very front of the curve exactly. Everything beyond overnight is the market's guess about the future path of that rate, plus the term premium.

The transmission runs through expectations. When a central bank raises rates by 25 basis points, the two-year yield may move 40 basis points or not at all, depending entirely on what the move tells the market about the path. The policy decision matters far less than the signal it sends about the trajectory — which is why central bank communication is treated as a policy instrument in its own right, and why the largest yield moves often happen on days with no policy change.

Quantitative easing is the purchase of long-dated bonds with newly created reserves. It works through two channels. The signalling channel: buying long bonds commits the bank to keeping short rates low, because raising them would inflict losses on its own portfolio. The portfolio-balance channel: removing duration from the market forces the remaining holders to hold less of it, and if investors have preferences over maturities rather than being indifferent arbitrageurs, that raises the price of what remains (Vayanos & Vila, 2021). The empirical literature broadly supports material effects (Gagnon, Raskin, Remache & Sack, 2011; Krishnamurthy & Vissing-Jorgensen, 2011; D'Amico & King, 2013), with estimates clustering around tens of basis points per few hundred billion of purchases. [Contested] — the sign and existence are well established; the magnitude varies by study and by market conditions, and the effects appear larger when markets are stressed and smaller when they are calm.

Quantitative tightening is the reverse, and the asymmetry is instructive: QE was announced with fanfare as a stimulus tool, QT is run quietly on autopilot, and the evidence suggests its effects are smaller and slower. [Contested] The plausible reason is that QE's largest channel operates through signalling and stress relief, neither of which has a mirror image in an orderly runoff.

10.6 Fiscal policy, and when it takes over

Government borrowing is the supply side of the government bond market, and for most of the past forty years its effect on yields was small enough to argue about. Two developments have made it live again.

The stock of debt has grown substantially across developed economies, and the composition of holders has shifted from price-insensitive official buyers toward price-sensitive private ones (§12.4).

The relationship between r and g has become the centre of the debate. Blanchard (2019) made the influential argument that when the interest rate on government debt is below the economy's growth rate, debt can be rolled indefinitely without ever being repaid, and the fiscal costs of debt are much lower than conventionally assumed. [Contested] — the arithmetic is correct and the policy conclusion is disputed, most sharply on the grounds that the condition $r < g$ is not guaranteed to persist and that its failure is precisely when you cannot afford it to.

The limiting case is fiscal dominance: the point at which the debt burden constrains the central bank, because raising rates far enough to control inflation would make the debt unsustainable. The classic analysis is Sargent & Wallace (1981), whose “unpleasant arithmetic” is that a government committed to deficits forces the central bank to eventually monetise them, so tight money today buys looser money tomorrow. The modern fiscal-theory literature (Cochrane, 2023) develops this into a full theory of the price level. [Contested] — a serious, internally consistent framework whose empirical discrimination against conventional monetary theory remains debated.

For an investor, the practical question is narrower and answerable: at what point does a deficit start to be priced as credit risk rather than as supply? The UK in September 2022 supplied the answer for an advanced economy (§13.4): when a fiscal announcement causes the currency to fall and long yields to rise and equities to fall simultaneously, the market has stopped treating the issuer as risk-free. The simultaneity is the signal — ordinary supply pressure raises yields while supporting the currency.

§10 Key takeaways

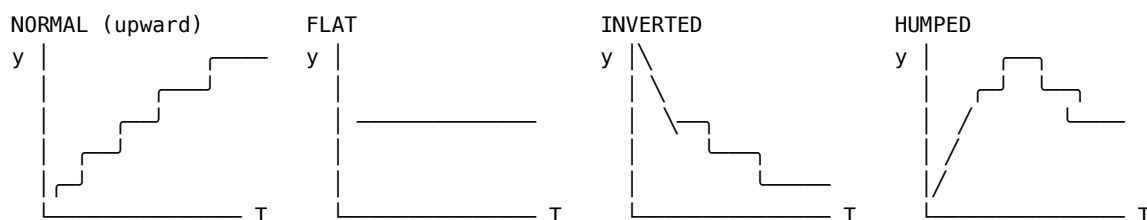
1. Split every yield move into “the real economy changed” and “the inflation outlook changed”. The two have opposite implications for the rest of a portfolio.
2. The natural real rate r^* fell by percentage points across the developed world over forty years. The level of yields is a slow-moving equilibrium, not a constant.
3. Anchored inflation expectations are why ten-year breakevens peaked near 3% while realised inflation exceeded 8%. Long-horizon breakevens measure central bank credibility; short ones measure energy prices.

4. The term premium is the extra yield for committing long rather than rolling short. Its sign depends on whether bonds hedge or amplify investors' other risks — the same question as the bond–equity correlation.
5. The US ten-year term premium was estimated as negative for much of 2016–2021: investors paid for duration rather than being paid for it. Estimates are model-dependent; treat them as a regime indicator, not a trade.
6. Policy rate decisions matter far less than what they signal about the path. The largest yield moves often occur with no policy change.
7. QE works through signalling and portfolio balance, with effects of tens of basis points and larger impact when markets are stressed. QT is not symmetric.
8. Fiscal risk becomes credit risk at an identifiable moment: when yields rise while the currency falls and equities fall, simultaneously.

11. The yield curve

11.1 The shapes and what they mean

Plot yield against maturity and you get the yield curve — the single most watched picture in macro finance. It takes four characteristic shapes.



Expansion, or a term premium for committing long. The usual state.	Transition, or a term premium offsetting expected rate cuts.	Policy is tight and expected to ease; a recession signal.	Near a turning point: hikes priced first, then cuts.
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The intuitive reading of an upward slope is “the market expects rates to rise”. That reading is mostly wrong, and understanding why is the substance of this section. The curve is normally upward-sloping — it has been for the great majority of observations — but rates have not risen most of the time. The slope is therefore mostly term premium, not expectation.

11.2 The expectations hypothesis and its failure

The hypothesis. In its pure form: long yields are the average of expected future short yields, so the term premium is zero and today's forward rates are unbiased forecasts of future spot rates. It is an appealing idea — it says the curve contains no compensation, only information.

It is false, and the way it fails is systematic. Two classic tests.

Fama & Bliss (1987) regressed a bond's excess return over the following year on the spread between its forward rate and the current spot rate. Under the expectations hypothesis, that spread should predict nothing — it is pure forecast. In fact it predicts strongly, with explanatory power rising with horizon. [Fact] When forward rates are high relative to spot, long bonds subsequently earn high excess returns.

Campbell & Shiller (1991) ran the complementary test: regress the change in long yields on the curve slope. The expectations hypothesis predicts a coefficient of +1 — a steep curve means long yields rise. The estimated coefficients are consistently negative. [Fact] When the curve is steep, long yields have historically tended to fall. This is the opposite of the textbook prediction, and it is one of the most robust anomalies in empirical finance.

Cochrane & Piazzesi (2005) tightened this considerably: a single linear combination of forward rates — a tent-shaped weighting across maturities — predicts one-year excess returns on bonds of every maturity, with R^2 of roughly a third. [Fact] One factor prices the whole curve's risk premium.

The critique, which matters. These are predictive regressions on overlapping annual returns from a short sample of genuinely persistent variables, which is the setting where standard inference is most misleading. Bauer & Hamilton (2018) show that once small-sample bias and the persistence of the regressors are handled properly, the evidence for several proposed

bond-return predictors weakens substantially, and some does not survive. [Contested] The core Fama–Bliss and Campbell–Shiller results are more robust than the later macro-based extensions, but the honest summary is that bond risk premia are time-varying and predictable, and we know this much less precisely than the published R^2 figures suggest.

What to take from it. The practically usable conclusion is modest and durable: a steep curve is, on average, compensation rather than forecast. A steep curve is therefore a reason to own duration, not to avoid it. The converse — that an inverted curve is a reason to avoid duration — is much weaker, because inversion carries a second meaning.

11.3 Inversion and recessions

An inverted curve — short yields above long — is the most reliable recession indicator in the standard macro toolkit, and also one of the most over-read.

The record. The spread between the ten-year and three-month Treasury yields has inverted before every US recession since the 1960s, with roughly one false positive (Estrella & Hardouvelis, 1991; Estrella & Mishkin, 1998). [Fact] It outperforms most alternatives, including surveys and equity-market signals.

The mechanism. Inversion is not a cause; it is the market saying that policy is tight now and will have to be eased. Short rates are set by the central bank, which has raised them to slow the economy. Long rates are an average of expected future short rates, so if the market expects those cuts, long rates sit below short. An inverted curve is a forecast of rate cuts, and rate cuts happen in recessions.

The four caveats, all of which have burned people:

- The lead time is long and variable. Historically anywhere from about six months to two years between inversion and recession. A signal that fires up to two years early is nearly useless for timing anything.
- Which spread you use changes the answer. The 10-year minus 3-month spread has the better record; the more widely quoted 10-year minus 2-year is noisier.
- The sample is small. Roughly eight or nine US recessions. Any claim of high reliability from nine observations should be discounted heavily. [Contested]
- The term premium contaminates the signal. If the curve is flat because the term premium has been compressed by central bank purchases rather than because the market expects cuts, the signal means something different. This was the substance of the live debate over the 2019 and 2022 inversions, and it remains unresolved. [Contested]

My read: inversion is real information about the stance of policy relative to the economy, and it is close to useless as a trading signal because of its timing variance. Treat it as one input to a regime assessment, not as a trigger.

11.4 Modelling the curve

Three levels of machinery, in increasing order of commitment.

Statistical decomposition. The classic result is that essentially all curve movement is three factors (Litterman & Scheinkman, 1991):

Factor	Shape	Interpretation	Share of variance
Level	All maturities move together	Inflation and r^*	Most of it
Slope	Short and long move oppositely	The policy cycle	Most of the rest
Curvature	The middle moves against the ends	Policy-path timing	Small

Together these explain the great majority of the variation in yield changes, with the level factor dominant. [Fact] This is why a portfolio's duration (its level exposure) is the first risk number anyone computes, and why key-rate durations beyond three points rarely add much.

Parametric fitting. To get a smooth curve from a scatter of bond prices, the standard is the Nelson & Siegel (1987) functional form and its Svensson (1994) extension — a small number of parameters producing level, slope and curvature components by construction. The Federal Reserve publishes daily fitted curves on this basis (Gürkaynak, Sack & Wright, 2007), and that dataset is the standard research input.

Arbitrage-free models. For pricing derivatives, curve fitting is not enough — the model must rule out arbitrage across maturities and over time. The lineage runs from the one-factor equilibrium models of Vasicek (1977) and Cox, Ingersoll & Ross (1985), through models calibrated to fit today's curve exactly (Ho & Lee, 1986; Hull & White, 1990), to the general framework

of Heath, Jarrow & Morton (1992), which models the entire forward curve's evolution. The affine class — where yields are linear in a set of state variables (Duffie & Kan, 1996) — is the workhorse, and Piazzesi (2010) is the standard survey.

The practical guidance for someone who is not building a derivatives book: you almost certainly need the first level and rarely the third. Level-slope-curvature plus a fitted curve answers most investment questions. Affine models are for pricing options on rates, and their track record at forecasting is unimpressive.

11.5 Curve trades

Because the curve moves in factors, positions can be constructed to isolate them.

Trade	Construction	The bet
Duration long/short	Own or short bonds outright	The level factor
Steepener	Long short-maturity, short long-maturity, duration-matched	The slope factor: curve steepens
Flattener	The reverse	Curve flattens
Butterfly	Long the wings, short the belly (or reverse)	The curvature factor
Carry/roll trade	Own the point with the best carry-plus-roll per unit of duration	Nothing happens

The one worth understanding as an investor rather than a trader is the last. §7.5 showed that a five-year bond on a normal curve earns 4.37% of carry and roll and can absorb 122 basis points of sell-off before losing money, while a thirty-year earns 4.62% and can absorb 30. Positioning in the belly is, in a precise sense, the highest-return-per-unit-of-risk place to sit if the curve does not move — and since the curve does not move most of the time, this is a defensible default. [Practice]

The two cautions. A duration-matched steepener is not risk-free: it has significant exposure to curvature, and duration-matching is only valid for small moves. And carry trades in bonds, like carry trades everywhere, have negatively skewed returns — they earn small amounts steadily and lose large amounts occasionally, which is the general property documented across asset classes by Kojien, Moskowitz, Pedersen & Vrugt (2018). [Fact]

§11 Key takeaways

1. The curve is usually upward-sloping, and rates usually do not rise. The slope is therefore mostly term premium, not expectation.
2. The expectations hypothesis fails systematically. When forwards are high relative to spot, long bonds subsequently earn high excess returns; when the curve is steep, long yields have historically fallen, not risen.
3. A single tent-shaped combination of forward rates predicts a third of the variation in one-year bond excess returns across all maturities.
4. That literature is weaker than its published statistics suggest once small-sample bias and regressor persistence are handled. Premia are predictable; the precision is overstated.
5. A steep curve is compensation, so it is a reason to own duration. An inverted curve is not the symmetric argument, because it also carries a forecast of cuts.
6. Inversion has preceded every US recession since the 1960s, with a lead time of six months to two years and a sample of eight or nine. Real information, useless timing.
7. Level, slope and curvature explain nearly all curve movement, with level dominant. Three numbers describe a portfolio's curve risk.
8. Affine term-structure models are for pricing rate options, not for forecasting yields. Most investment questions need only a fitted curve and three factors.

12. Supply, demand, and who actually owns the bonds

12.1 Why supply and demand should not matter, and does

In a frictionless model, the quantity of bonds outstanding does not affect their price. Bonds are claims on cash flows; if the government issues more ten-year notes, arbitrageurs short them against other maturities until yields are consistent again. Supply changes who holds what, not what things are worth.

That argument requires arbitrageurs with unlimited capital and no preferences over maturity. Neither holds. Real investors have strong maturity preferences arising from their liabilities and their regulators, and real arbitrageurs have finite balance sheets. Once you admit both, quantities matter.

This is the preferred habitat theory, proposed by Modigliani & Sutch (1966) and formalised by Vayanos & Vila (2021). Investors have preferred maturities; arbitrageurs connect the segments but are risk-averse and capital-constrained. The model's central prediction is that a change in the supply of bonds at one maturity moves yields there and, attenuated, elsewhere — exactly what QE was designed to exploit.

The empirical support is good. Greenwood & Vayanos (2014) show that when the government's debt is tilted toward long maturities, long bonds subsequently earn higher excess returns — the market must be paid more to absorb more duration. [Fact] Greenwood, Hanson & Stein (2015) develop the corresponding theory of optimal government debt maturity.

The general principle, which generalises well beyond bonds: when a large share of a market's holders are price-insensitive — buying for regulatory, mandate or policy reasons rather than because the price is attractive — prices can depart from fundamental value, and the departure persists until someone is paid enough to take the other side.

12.2 The holder base

Knowing who owns an asset tells you how it will behave under stress, because it tells you who is forced to sell and who is free to buy.

Holder	Why they hold	Price-sensitive?	Behaviour under stress
Central banks (domestic)	Monetary policy	No	Buy more, usually
Foreign official reserves	Currency management, safety	No	May sell to defend a currency
Banks	Liquidity regulation; collateral	Partly	Constrained by leverage rules
Insurers	Match long liabilities	Partly	Forced sellers on downgrade
Pension funds	Match long liabilities	Partly	Can be forced sellers if levered (§13.4)
Mutual funds and ETFs	On behalf of end investors	Yes	Sell on redemptions
Hedge funds	Relative value, levered	Very	Forced to deleverage
Households	Income, safety	Yes	Usually stabilising

Two features of this table drive most of what happens in a crisis.

The largest holders are the least price-sensitive. Central banks and foreign official institutions have together held a very large share of Treasury debt. They do not buy because yields are attractive. So a change in their behaviour is a pure supply shock to everyone else.

The most price-sensitive holders are levered. Hedge funds running relative-value trades provide much of the market's day-to-day liquidity and are the first to withdraw when funding tightens. This is the asymmetry that turns a shock into a dislocation: the marginal buyer disappears precisely when the marginal seller appears (§3.5, §13.4).

12.3 Liability-driven demand

The most important non-economic source of bond demand is an accounting identity.

A defined-benefit pension fund owes a stream of payments decades into the future. Under modern accounting, those liabilities are discounted at a market interest rate, so when rates fall, the present value of the liabilities rises and the fund's deficit widens. The hedge is to own long-duration assets whose value rises at the same time — long government bonds and swaps.

This creates the defining feature of long-end demand: pension funds want long bonds most when long bonds are most expensive, because falling yields simultaneously widen their deficit and raise the price of the hedge. It is a structurally destabilising, momentum-amplifying demand function, and it explains persistent oddities such as the UK's inverted long end (§4.1).

Insurance regulation does something similar. Solvency-style frameworks require insurers to hold capital against duration mismatch, pushing them toward long bonds and toward selling anything downgraded — which is the forced-selling channel

of Ellul, Jotikasthira & Lundblad (2011). Bank liquidity rules similarly require holdings of high-quality liquid assets, of which government bonds are the primary example, creating a large regulatory bid that is insensitive to yield.

12.4 The shifting holder base

Two multi-decade shifts have changed the market's character.

The rise and partial retreat of official demand. Through the 2000s and 2010s, foreign official reserve accumulation and then central bank QE absorbed enormous quantities of duration. This is the leading candidate explanation for the negative term premium of §10.4: with price-insensitive buyers absorbing the supply, price-sensitive investors did not need to be paid to hold it. The subsequent reversal — QT, plus reserve managers diversifying — moves duration back to investors who must be compensated, and is a structural argument for a higher term premium. [Hypothesis] — the mechanism is well-supported; the magnitude of the reversal is not yet established.

The growth of funds and ETFs. A growing share of corporate credit is held through vehicles offering daily liquidity on an underlying asset that does not have it. This is a genuine structural fragility: Goldstein, Jiang & Ng (2017) document that corporate bond funds exhibit a concave flow-performance relationship, meaning bad performance triggers disproportionate outflows — the signature of a run incentive. [Fact] The mechanism is a first-mover advantage: redeeming early gets you out at today's stale price while the costs of liquidating fall on those who stay.

12.5 Index demand

A large share of bond money tracks an index, and bond indices have a design problem that equity indices do not.

An equity index weights companies by market capitalisation — by what the market thinks they are worth. A bond index weights issuers by amount of debt outstanding. So an indexed bond portfolio lends the most to whoever has borrowed the most. In the government sector this is arguably fine; in credit it selects toward the most levered issuers, and the maturity profile of the index is determined by issuers' funding choices rather than by any investor's needs.

The consequence is that the index has drifted with issuance. When corporate treasurers lock in low long-term rates, the index's duration extends, and every index-tracking fund's risk rises without any decision being made. Anyone benchmarked to a bond index should know that the benchmark's risk is set by borrowers. [Practice]

12.6 What to do with all this

Three usable conclusions.

Track the marginal buyer. The question “who has to buy this, and who has to sell it?” explains more short-horizon bond price action than any valuation model. Changes in central bank programmes, in regulatory treatment, and in index rules are genuine price events.

Expect dislocations where price-insensitive holders dominate. The TIPS–Treasury gap (§3.3), the CDS–bond basis (§8.5), and the on-the-run premium (§2.3) all persist because arbitraging them requires balance sheet that is scarce when the gap is widest. These are opportunities for patient unlevered capital and traps for levered capital.

Treat supply as a slow variable and positioning as a fast one. Debt issuance plans move yields over quarters; forced deleveraging moves them over days. Most large short-horizon bond moves are positioning, not fundamentals.

§12 Key takeaways

1. Supply and demand move bond prices because investors have maturity preferences and arbitrageurs have finite balance sheets. Preferred-habitat theory is the formal version, and it is well supported.
2. When the government tilts issuance long, long bonds subsequently earn higher excess returns — the market must be paid to absorb duration.
3. The largest holders of government bonds are the least price-sensitive, and the most price-sensitive are levered. That asymmetry is what turns a shock into a dislocation.
4. Pension liability hedging makes demand for long bonds strongest when long bonds are most expensive — a structurally momentum-amplifying demand function.
5. The decades-long absorption of duration by official buyers is the leading explanation for the negative term premium of the late 2010s. Its reversal is a structural argument for a higher one.
6. Corporate bond funds face a genuine run incentive: outflows respond disproportionately to bad performance, because redeeming early externalises the liquidation cost.

7. Bond indices weight by debt outstanding, so an indexed portfolio lends most to whoever borrowed most, and its duration is set by issuers rather than investors.
8. “Who must buy and who must sell?” explains more short-horizon price action than any valuation model.

Part V — How bonds behave

13. Sell-offs and rallies

13.1 Four kinds of sell-off

Identity 3 says a bond’s return is carry plus roll, minus duration times the yield change, plus convexity, minus credit losses. Since carry and roll are known in advance and convexity is small, essentially every large bond move is either a yield change or a credit loss — and yield changes come in three flavours, distinguished by which term of Identity 2 moved.

That gives four kinds of bond sell-off, and the whole of this section is the claim that they are genuinely different events requiring different responses.

Type	What moved	Typical trigger	Duration of the episode
1. Rate-path repricing	$r + \pi^e$	Inflation data, central bank shift	Months to years
2. Term-premium repricing	TP	Supply, positioning, policy uncertainty	Weeks to months
3. Credit repricing	EL + CRP	Recession fears, defaults, a sector shock	Months
4. Forced deleveraging	ℓ , and everything	Margin calls, redemptions, a funding freeze	Days

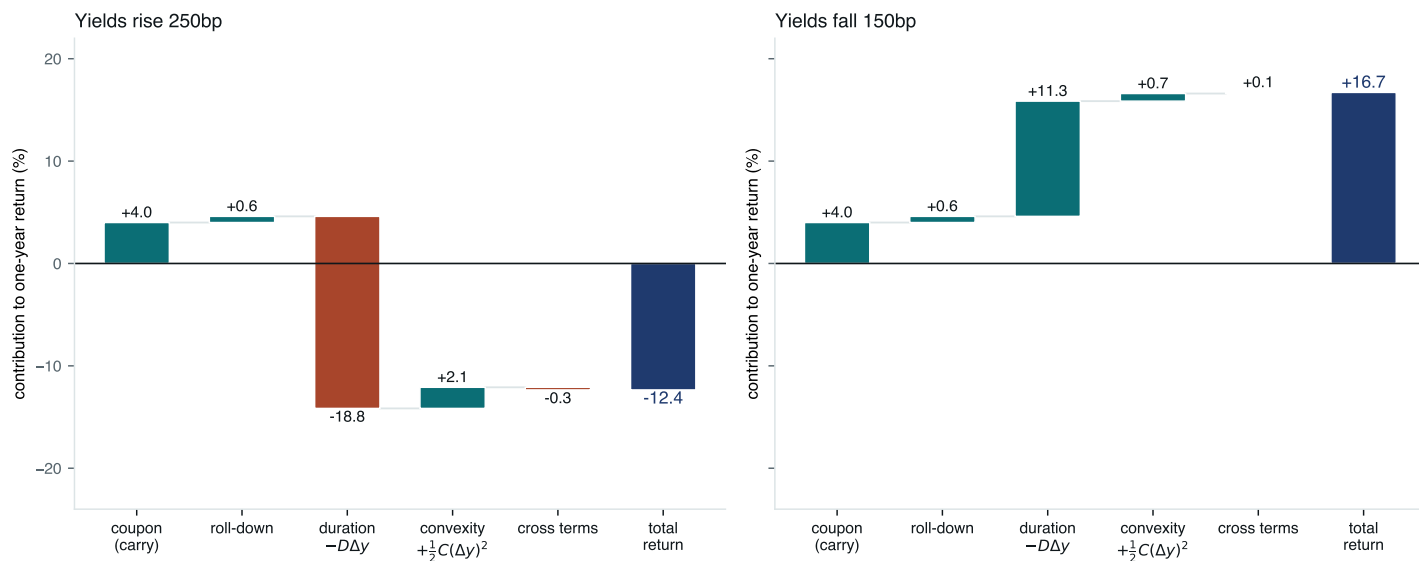
The diagnostic is what else moved. This is the most useful table in the section, because it lets you classify an episode while it is happening rather than afterwards:

Bonds	Equities	Currency	Credit spreads	Diagnosis
↓	↓	↑	modest ↑	Inflation / hawkish policy repricing. The classic 2022 pattern
↓	↑	↑	↓	Good-news growth repricing. Bonds fall because the economy is fine
↓	↓	↓	↑	Fiscal or credit repricing of the sovereign. The market has stopped treating the issuer as safe
↓	↓	mixed	↑ sharply	Liquidity event. Correlations break, everything is sold for cash
↑	↓	↑	↑	Flight to quality. Bonds doing the job they are held for

Row three is the one to internalise. In an advanced economy, government bonds falling while the currency falls is qualitatively different from bonds falling with a rising currency. The first is a solvency signal; the second is ordinary monetary tightening. The UK in September 2022 is the clean case (§13.4).

13.2 A sell-off, decomposed

Take the running 10-year bond and put it through a year in which yields rise 250 basis points — approximately the US experience in 2022 — and a year in which they fall 150.



A 10-year 4% bond bought at par, held one year, on a curve 8bp steeper between 9 and 10 years. The first two bars are known the day you buy; the third is the entire bet; the fourth is the consolation prize that makes the loss smaller than the gain.

The arithmetic: [Computed]

Component	Yields +250bp	Yields -150bp
Coupon (carry)	+4.00%	+4.00%
Roll-down	+0.60%	+0.60%
Duration, $-D \Delta y$	-18.76%	+11.26%
Convexity, $+\frac{1}{2}C(\Delta y)^2$	+2.06%	+0.74%
Cross terms	-0.26%	+0.11%
Total	-12.35%	+16.71%

(Cross terms is the small gap left when a second-order approximation — duration plus convexity — stands in for the bond's actual repricing over a full year rather than an instant; §7.4 shows the same kind of gap growing with the size of the yield move; A.3 has the expansion. It is never the story.)

Four readings that generalise.

The coupon is nearly irrelevant in a big year. Four points of income against nineteen points of price loss. The entire apparatus of “income investing” concerns the smallest term in the equation whenever anything interesting happens.

Convexity contributes two percentage points, unprompted and for free. It is the difference between a bad year and a slightly worse one, and it is why the duration-only approximation overstates losses.

The same bond makes 16.7% when yields fall 150. The asymmetry between the two columns — losing 12.35% on a 250bp rise, gaining 16.71% on a 150bp fall — is why “bonds are boring” is a claim about the recent past rather than about the instrument.

This is the whole explanation of 2022. Not a credit event, not a liquidity event. A rate-path repricing of unusual size, hitting portfolios whose duration had been extended by a decade of low coupons (§7.1: low coupons mean long duration). The US Aggregate index returned roughly -13% and long Treasuries far worse. [Fact] Nothing about it was mysterious; it was Identity 3 with a large Δy .

13.3 The episodes

Episode	Type	What happened	The lesson
1994	1	The Fed raised rates from 3% to 6% in a year; 10-year yields rose roughly 200bp	A well-telegraphed tightening can still be a shock if positioning is long
1998 LTCM	4	Convergence trades unwound; off-the-run spreads blew out; Treasuries rallied	Relative-value trades need capital exactly when capital vanishes
2008	3 then 4	Treasuries rallied hard; high-yield spreads reached extraordinary levels	Government bonds and credit are opposite trades in a crisis
2013 taper tantrum	2	A hint of reduced purchases moved 10-year yields well over 100bp higher within four months	Term premium can reprice violently on a change in expected supply
March 2020	4	Treasuries sold off during an equity crash; the Fed bought ~\$1tn	Even the safest market can lose its liquidity
2022	1	Inflation forced the fastest tightening in decades; bonds and equities fell together	The bond–equity hedge fails in inflation shocks (§14)
Sept 2022 UK	1, 3 then 4	A fiscal announcement triggered gilt losses, then LDI margin calls, then a doom loop	Leverage inside a “safe” strategy is the danger
March 2023	Rally	Bank failures drove one of the largest short-dated Treasury rallies on record	Flight to quality still works when the shock is not inflationary

The pattern across episodes: types 1 through 3 are repricings you can hold through; type 4 is a liquidity event that hurts leveraged holders and creates opportunity for unleveraged ones. Distinguishing them in real time is the practical skill, and the diagnostic table of §13.1 is the tool.

13.4 Three worth understanding in detail

1994: the well-telegraphed shock. The Federal Reserve began raising rates from 3% in February 1994 and reached 6% within a year. Nothing about the direction was secret. Bonds still had one of their worst years, because positioning had been built for the low-rate environment that preceded it, and because the pace was faster than the market had priced. The episode produced Orange County’s bankruptcy — a municipality that had levered a bond portfolio to enhance yield — and a series of derivative blow-ups at corporates who had sold rate volatility for income. The recurring lesson: the damage in a rate shock is concentrated in positions that were short volatility or levered, not in the bonds themselves.

March 2020: when the safe asset was sold. Covered in §3.5. The critical fact for an investor is that a Treasury position marked down during the sharpest equity decline in decades, for roughly two weeks, before resuming its normal behaviour. If your risk framework assumes government bonds rally in a crisis, it had a fortnight of being exactly wrong at the worst possible moment. The lesson: the flight-to-quality property of government bonds is reliable against economic shocks and unreliable against liquidity shocks, because in a liquidity shock the thing people want is not safety but cash — and the most saleable asset is the one that gets sold.

September 2022, UK gilts: leverage inside a hedge. This is the most instructive episode of the modern era because every component was individually reasonable.

UK pension funds had large, long-dated liabilities (§12.3). They hedged them by holding long gilts and, because they also wanted to own return-seeking assets, by using leveraged exposure — gilt repo and swaps — to get the duration with less capital. That is liability-driven investing, and as a hedging strategy it is correct.

On 23 September 2022 the government announced large unfunded tax cuts. Gilt yields rose sharply — an ordinary type-1 and type-3 repricing. But leveraged duration positions require collateral, and rising yields generated margin calls — on the order of 15% of notional for a 100 basis point rise on a 30-year position (A.16). Meeting them required selling assets; the most liquid asset was gilts; selling gilts pushed yields higher; which generated more margin calls. A hedge against falling

rates had become a forced seller into rising rates. The Bank of England intervened with emergency long-gilt purchases on 28 September, explicitly on financial-stability rather than monetary grounds, which is an unusual and revealing admission.

Three general lessons, all of which apply far beyond gilts:

- Leverage converts a price move into a solvency event. The underlying hedge was sound; the leverage on top of it was the mechanism of failure.
- A crowded hedge is not a hedge. When all holders of a position face the same trigger, their collective response moves the price against all of them.
- Liquidity is correlated with the thing you are hedging. The asset you must sell to meet a margin call is the asset whose price caused the call.

13.5 What makes bonds rally

The mirror image is shorter to state, because the causes are fewer.

- Expected policy easing. The dominant driver. Weak growth data, falling inflation, a dovish central bank shift.
- Flight to quality. Equity declines, geopolitical shocks, bank failures. This works when the shock is disinflationary — which most shocks are, but not all.
- Term premium compression. Central bank purchases, regulatory demand, a fall in rate volatility. Lower expected volatility directly reduces the compensation required for duration risk.
- Convexity hedging. When yields fall, mortgage portfolios shorten in duration and their hedgers must buy duration, pushing yields lower still (§5.6). A genuine amplification mechanism in both directions.

March 2023 is the clean modern example of the second. The failure of Silicon Valley Bank — itself caused by unhedged interest-rate risk in a bond portfolio, which is a tidy irony — triggered a flight to quality that moved two-year Treasury yields down by roughly 100 basis points in three days, among the largest such moves on record. [Fact] Bonds worked precisely as advertised, because the shock was a banking shock rather than an inflation shock.

13.6 What to watch

If you hold bonds, these are the variables that actually change your outcome, in descending order:

1. The inflation trajectory, because it determines whether the central bank is easing or tightening and whether bonds hedge equities or not.
2. The central bank's reaction function — not the next decision, but what the market believes about the path.
3. Real yields, which separate a repricing of growth from a repricing of inflation. Nominal yields rising with stable breakevens is a real-rate story and means something different from nominal yields rising with breakevens.
4. Positioning and leverage. Not observable directly, but visible in proxies: the CDS–bond basis, repo spreads, dealer inventories, futures positioning.
5. Credit spreads, as a cross-check. Government yields rising while credit spreads tighten is a growth story; both widening is a problem.

Notice that the issuer's fundamentals appear nowhere on this list for a developed-market government bond. That is correct, and it is the practical content of §3.1.

§13 Key takeaways

1. Every large bond move is a yield change or a credit loss. Yield changes come in three kinds — rate path, term premium, credit — plus a fourth category, forced deleveraging, that ignores fundamentals entirely.
2. Diagnose an episode by what else moved. Bonds and the currency falling together in an advanced economy is a solvency signal; bonds falling with a rising currency is ordinary tightening.
3. In a big year the coupon is almost irrelevant: four points of income against nineteen points of price. Income investing concerns the smallest term.
4. 2022 was not mysterious. It was Identity 3 with a large yield move, hitting portfolios whose duration had been extended by a decade of low coupons.
5. Repricings can be held through; liquidity events hurt the leveraged and create opportunities for everyone else.
6. Flight to quality works against economic shocks and fails against liquidity shocks, because what people want in a liquidity shock is cash, and the most saleable asset is the first sold.

7. The UK LDI episode is the definitive modern lesson: a sound hedge, made levered, became a forced seller into the move it was meant to protect against.
8. What matters for a government bond holder is inflation, the policy reaction function, real yields and positioning. The issuer's finances are nearly irrelevant until, suddenly, they are the whole story.

14. Bonds and equities

14.1 The two-shock framework

The relationship between bonds and stocks looks empirical and unstable. It is actually deducible from one observation, and the observation makes the instability predictable rather than mysterious.

A stock price is the present value of future profits. A bond price is the present value of future coupons. Both are present values, and they share a denominator. Write them schematically:

$$P^{\text{equity}} = \sum_t \frac{\mathbb{E}[\text{profit}_t]}{(1 + y_t + \text{ERP})^t}, \quad P^{\text{bond}} = \sum_t \frac{C_t}{(1 + y_t)^t}$$

ERP is the equity risk premium, the extra return equity investors demand over the bond yield for bearing the risk that profits are not fixed. The bond has a fixed numerator; the equity's numerator moves with the economy. So consider the two kinds of shock separately:

	DISCOUNT-RATE SHOCK rates rise, profits unchanged	CASH-FLOW (GROWTH) SHOCK growth falls, rates fall too
Equities	↓ (bigger denominator)	↓ (smaller numerator)
Bonds	↓ (bigger denominator)	↑ (smaller denominator)
Correlation	POSITIVE	NEGATIVE
Bonds as a hedge	fail	work
Term premium	positive (bonds are risky)	can be negative (bonds insure)

Whichever shock dominates determines the sign of the correlation. That is the whole framework, and it explains every regime in the historical record.

Notice the third row: the same logic determines the sign of the term premium (§10.4). Bonds that hedge your other risks are worth holding at a lower yield; bonds that amplify them require a premium. The bond–equity correlation and the term premium are the same question asked twice, which is a satisfying piece of internal consistency in this subject.

14.2 The historical record

The correlation is not a constant, and its regime shifts are large and long-lived. [Fact]

Era	Sign	Dominant shock	What it felt like
1960s–1990s	Positive	Inflation	Bonds and equities fell together; diversification came from elsewhere
~1998–2020	Negative	Growth and deflation fear	Bonds rallied in every equity sell-off; 60/40 looked like free lunch
2021–2023	Positive	Inflation	2022: both fell hard, simultaneously

The shift around 1998 is well documented, and Campbell, Sunderam & Viceira (2017) title it precisely: nominal bonds moved from being “inflation bets” to “deflation hedges.” [Fact] An entire generation of investment practice — the modern 60/40 portfolio, risk parity, the use of Treasuries as the standard equity hedge — was built during the negative-correlation regime and implicitly assumes it.

This matters more than almost anything else in this document for portfolio construction, because the diversification benefit of bonds is not a property of bonds. It is a property of the macroeconomic regime.

14.3 Why it flips

Campbell, Pflueger & Viceira (2020) give the mechanism, and it is the two-shock framework made rigorous. Two ingredients determine the sign:

Which shocks dominate. In an economy where the main disturbances are demand shocks — consumers spend less, investment falls — growth and inflation move together downward, so the central bank eases, bonds rally and equities fall. Negative correlation. In an economy where the main disturbances are supply shocks — energy prices, supply chains, wars — inflation rises while growth falls, so the central bank tightens, bonds fall and equities fall. Positive correlation.

How the central bank responds. A central bank that responds aggressively to inflation converts inflation shocks into real-rate shocks, which hurt both assets. One that accommodates lets inflation run, which hurts bonds more than equities, since equities have some claim on nominal revenues.

So the regime variable is inflation, and specifically whether inflation is a live concern. The empirical rule: when inflation is low and stable, bonds hedge equities; when inflation is the dominant macro risk, they do not. [Fact]

A complementary literature finds the correlation also has a flight-to-quality component that macro fundamentals do not capture — it moves with liquidity and risk appetite on horizons shorter than any macro variable changes (Baele, Bekaert & Inghelbrecht, 2010). [Fact] Both channels are real; the macro one sets the multi-year regime, the flight-to-quality one drives day-to-day comovement.

14.4 What 2022 actually demonstrated

In 2022 a US 60/40 portfolio had one of its worst years in modern history, with both legs down double digits. [Fact] The common reaction was that diversification had failed. A more precise statement is available.

Diversification did not fail; it was never designed for this shock. Bonds hedge growth risk. They have never hedged inflation risk, and the framework above says they cannot — an inflation shock raises the discount rate applied to both assets. In 1970s terms this was entirely familiar; it was novel only to people whose experience began in the 1990s.

The deeper point concerns equity duration. Equities are long-duration assets: a growth company's value is concentrated in distant cash flows, so its price is highly sensitive to the discount rate. In 2022, long-duration equities (technology, unprofitable growth) and long-duration bonds fell together for the same reason, in the same proportion to their durations. Investors who believed they were diversified across asset classes discovered they held one factor — duration — twice.

The right lesson is about the hedge for inflation, not about bonds. The assets that hedge inflation shocks are inflation-linked bonds, commodities, and to some degree short-duration equities with pricing power. A portfolio that holds nominal bonds as its only defensive asset is hedged against one of the two shocks that matter.

14.5 Credit is equity in disguise

The corporate bond's relationship with equities is different and simpler, and §5.5 already gave it: owning a corporate bond is owning a Treasury and having sold a put on the firm. So a corporate bond is long a rate instrument and short an equity-like exposure.

This has consequences that persistently surprise people:

- High-yield bonds behave like equities in a crisis. Correlations with equities rise sharply exactly when you need them not to, because the put you sold goes into the money precisely when equities fall.
- Investment-grade credit is mostly a rate exposure, with a modest equity component. High-yield is mostly an equity exposure with a modest rate component. These are different asset classes wearing the same label.
- A “balanced” portfolio of equities and high-yield bonds is not balanced. It is a concentrated bet on corporate health, expressed twice.
- Empirically, corporate bond returns load on factors closely related to equity risk, and a large part of the cross-section of credit returns is explained by downside risk and credit-quality factors (Bai, Bali & Wen, 2019). [Fact]

The practical implication for portfolio construction is to allocate to bonds by function, not by label. Government bonds are the growth hedge. Inflation-linked bonds are the inflation hedge. Credit is a return-seeking asset that belongs in the same mental bucket as equities, sized against the equity allocation rather than added to the “safe” side.

14.6 Practical guidance

- Ask which regime you are in before sizing bonds as a hedge. The operative variable is whether inflation is a live macro risk. When it is, expect bonds and equities to fall together and size accordingly.
- Do not estimate the bond–equity correlation from a long sample. A twenty-year average spans a regime shift and describes neither regime. A shorter window conditioned on the inflation environment is more honest, if noisier.
- Separate the growth hedge from the inflation hedge. Nominal government bonds do the first; inflation-linked bonds and real assets do the second. Holding only the first is a common and expensive gap.
- Count duration once, across the whole portfolio. Long-duration equities and long bonds are the same exposure.
- Put credit on the risky side of the ledger, and size it against equities. Its behaviour in the tail is equity-like by construction, not by accident.

§14 Key takeaways

1. Stocks and bonds are both present values sharing a discount rate. Discount-rate shocks move them together; cash-flow shocks move them oppositely. Whichever dominates sets the correlation's sign.
 2. The correlation was positive from the 1960s to the late 1990s, negative from around 1998 to 2020, and positive again in 2021–23. These are long regimes, not noise.
 3. The regime variable is inflation. When inflation is low and stable, bonds hedge equities; when it is the dominant risk, they cannot.
 4. The same logic sets the sign of the term premium. The bond–equity correlation and the term premium are one question asked twice.
 5. 2022 did not show that diversification failed. It showed that nominal bonds hedge growth shocks and have never hedged inflation shocks.
 6. Long-duration equities and long bonds are the same exposure. Investors who felt diversified in 2022 were holding duration twice.
 7. A corporate bond is a Treasury plus a short equity put. High-yield is an equity exposure with a rate component, not the reverse.
 8. Allocate bonds by function: government bonds as the growth hedge, inflation-linked bonds as the inflation hedge, credit on the risky side alongside equities.
-

15. Bonds and currencies

15.1 The identity that kills the yield pickup

Here is the fact that should be learned before any other in international fixed income, because it invalidates the most common reason people give for buying foreign bonds.

Suppose you are a dollar investor and Australian 10-year bonds yield 5% while US 10-year bonds yield 4%. A percentage point of extra yield, from an equally safe sovereign. You do not want the currency risk, so you hedge it by selling Australian dollars forward.

The hedge removes the yield pickup almost exactly. The forward exchange rate is not a forecast; it is set by arbitrage so that borrowing in one currency, converting, investing, and converting back gives the same return as investing at home. That relationship is covered interest parity:

$$\frac{\text{Fwd}}{S} = \frac{1 + i_{\text{dom}}}{1 + i_{\text{for}}}$$

where S and Fwd are the spot and forward exchange rates — not to be confused with the bond's face value F from the notation block — and i are short-term nominal money-market rates, not the real rate r used elsewhere in this document. If Australian short rates are above US short rates, the Australian dollar trades at a forward discount by exactly that difference, and selling it forward costs you exactly the rate differential.

Doing the algebra, the hedged return on a foreign bond is approximately

$$\underbrace{y^{\text{for}}}_{\text{foreign bond yield}} - \underbrace{(i^{\text{for}} - i^{\text{dom}})}_{\text{hedging cost}} = \underbrace{i^{\text{dom}}}_{\text{your own short rate}} + \underbrace{(y^{\text{for}} - i^{\text{for}})}_{\text{the foreign bond's premium over its own short rate}}$$

A currency-hedged foreign bond pays your domestic short rate plus the foreign bond's spread over its own short rate. The level of foreign yields is irrelevant; only the shape of the foreign curve relative to its own policy rate matters.

This is the single most useful piece of arithmetic in cross-border fixed income, and it reframes the question entirely. You do not buy foreign bonds for yield. You buy them because their curve is steeper than yours, or because their central bank is at a different point in its cycle and their bonds will therefore rally when yours do not. The diversification is across monetary cycles, not across yield levels. [Fact]

15.2 When covered interest parity breaks

CIP was treated as a near-identity before 2008. It is not one now.

Du, Tepper & Verdelhan (2018) document persistent, systematic deviations — the cross-currency basis — that are large enough to matter, widen at quarter-ends and year-ends, and correlate with bank regulatory constraints. [Fact] The mechanism is that the arbitrage requires a bank balance sheet, and post-crisis leverage rules make balance sheet costly. The regulation is the friction.

The consequence is directional and exploitable. For most major currencies the basis has run in a direction that penalises non-dollar investors hedging into dollars and rewards dollar investors hedging into those currencies. A dollar-based investor buying hedged Japanese or European bonds has often earned a few tenths of a percent above the CIP-implied return; a Japanese investor buying hedged Treasuries has paid it. [Fact] This is a real, persistent transfer, and it is the reason Japanese life insurers periodically retreat from hedged Treasury buying — an event that moves the US long end.

Note the pattern, now appearing for the third time: TIPS–Treasury (§3.3), CDS–bond basis (§8.5), cross-currency basis (here). Every persistent arbitrage-like gap in fixed income is the price of balance sheet. Once you see this, a whole class of apparent free lunches resolves into a single explanation.

15.3 Unhedged: the carry trade and its crash risk

If you do not hedge, you are making a bet that the exchange rate will not move as the forward implies. This is the uncovered interest parity question, and the empirical answer is one of the oldest anomalies in finance.

UIP says high-interest-rate currencies should depreciate by the interest differential, leaving expected returns equal. Fama (1984) showed the opposite: high-interest-rate currencies have historically depreciated less than the differential, and often appreciated. [Fact] Borrowing in low-rate currencies and lending in high-rate ones — the carry trade — has therefore earned a positive average return.

The catch is the shape of those returns. Carry trade returns are strongly negatively skewed: long stretches of steady gains punctuated by sharp losses, concentrated in periods of global risk aversion. Lustig, Roussanov & Verdelhan (2011) show the returns are compensation for exposure to a common risk factor rather than a free lunch. [Fact] The colloquialism — picking up nickels in front of a steamroller — is a fair description of the return distribution, and it is the same shape as credit (§5.5) and as bond carry trades (§11.5), which is not a coincidence: they are all short-volatility positions.

15.4 Hedged or unhedged: making the decision

The decision should be driven by how much risk the currency actually adds, and the answer depends sharply on maturity. Take a major exchange rate with 8.5% annualised volatility and bonds with typical yield volatility: [Computed]

Foreign bond	Bond return vol	Unhedged total vol	Share of variance from FX
2-year	1.5%	8.6%	97%
10-year	7.4%	11.2%	57%
30-year	15.6%	17.8%	23%

An unhedged foreign 2-year bond is, to a first approximation, not a bond position at all. It is a currency position with a small coupon attached, and it should be evaluated as one. At the long end the bond dominates, which is why unhedged long foreign bonds are at least arguably a bond investment.

The practical rules that follow:

- Hedge, by default. Currency risk is uncompensated for a passive holder over long horizons, and it multiplies the volatility of a low-volatility asset. [Practice]
- Hedging is nearly free in risk terms and roughly neutral in expected return, up to the cross-currency basis. The “cost of hedging” quoted by brokers is the interest differential, which you were never entitled to keep.
- If you want currency exposure, take it deliberately, sized as a currency allocation, not as a side-effect of a bond decision.
- The exception is emerging-market local debt, where the currency is the asset class and the yield partly compensates for it. Hedging EM local currency is expensive and partially self-defeating, since it strips out precisely the risk premium you came for. [Practice]

15.5 When a bond sell-off becomes a currency crisis

The most dangerous configuration in sovereign debt is when rates and the currency move together in the wrong direction, and the mechanism deserves stating because it is the practical content of §13.1’s diagnostic row three.

In an advanced economy, the normal response to rising yields is a stronger currency: higher rates attract capital. When yields rise and the currency falls at the same time, capital is leaving despite higher compensation, which means investors are worried about being repaid in real terms — through default, inflation, or depreciation. That is the signature of a sovereign risk repricing rather than a monetary one.

The escalation, in emerging markets with foreign-currency debt (§4.3), is a loop: depreciation raises the local-currency value of dollar debt, which worsens solvency, which drives more capital out, which deepens the depreciation. The self-reinforcing character is what makes these crises fast.

What breaks the loop is a credible policy response: large rate rises to defend the currency, IMF support, or fiscal correction. Each is costly, which is why prevention — borrowing in local currency, holding reserves, keeping the fiscal position credible — dominates cure. For an investor, the actionable version is: in a country with significant foreign-currency debt, the currency is the leading indicator of the bond, not the other way round. [Practice]

§15 Key takeaways

1. Hedging a foreign bond’s currency removes the yield pickup almost exactly, because the forward rate is set by the interest differential.
2. A hedged foreign bond pays your own short rate plus the foreign bond’s premium over its short rate. You are buying a curve shape and a monetary cycle, not a yield.
3. Covered interest parity has failed persistently since 2008. The cross-currency basis is the price of bank balance sheet, and it runs in a direction that rewards dollar investors hedging out and penalises the reverse.
4. TIPS–Treasury, CDS–bond basis, cross-currency basis: every persistent arbitrage-like gap in fixed income is the price of balance sheet.
5. Uncovered interest parity fails too, which is why the carry trade has positive average returns — with sharply negative skew, the same short-volatility shape as credit.
6. An unhedged foreign 2-year bond is 97% currency risk by variance. It is a currency position, not a bond position.
7. Hedge by default; take currency exposure deliberately if you want it. The exception is emerging-market local debt, where the currency is the asset class.
8. Yields rising while the currency falls is a sovereign risk signal. Where foreign-currency debt is significant, the currency leads the bond.

Part VI — When it goes wrong

16. Default, distress, and restructuring

16.1 What default actually means

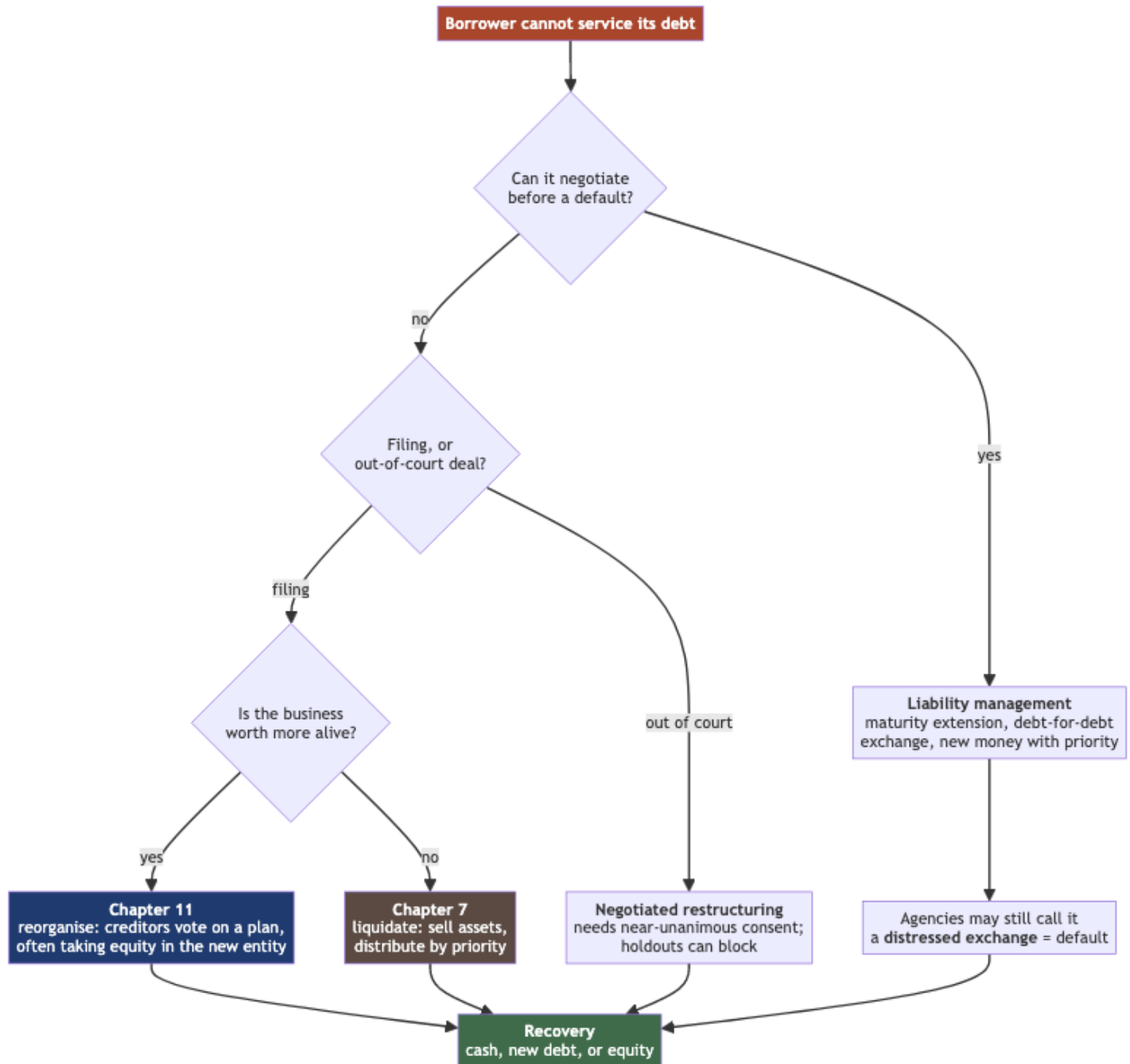
“Default” sounds binary and is not. Three different definitions are in active use and they disagree:

- Contractual default. The borrower breaches the indenture — a missed payment, a covenant violation, a bankruptcy filing. Missed payments usually have a grace period, typically 30 days.

- Rating agency default. The agencies additionally classify a distressed exchange — swapping old debt for new debt worth less — as a default, on the reasonable grounds that creditors took a loss under duress.
- CDS credit event. A narrower, contractually defined list: bankruptcy, failure to pay, and (in some contracts) restructuring, determined by an ISDA committee rather than a court.

The gaps between these matter. A company can restructure its debt at a substantial loss to creditors without triggering a CDS payout, if the restructuring is engineered to fall outside the contractual definition. Several high-profile disputes have turned on exactly this. If you are hedging credit risk with CDS, you are hedging the contractual definition, not the economic one. [Practice]

16.2 The resolution paths



Chapter 11 — the US reorganisation process — exists because a business is usually worth more running than dismantled. Management typically stays in control as “debtor in possession”, the company gets an automatic stay on collection, it can borrow new money that ranks ahead of everything existing (DIP financing), and creditors vote on a plan of reorganisation by class. Bondholders commonly emerge owning equity in the reorganised company — which is why distressed debt investing is in practice a form of private equity with a legal process attached.

Chapter 7 is liquidation, used when there is no viable business to preserve.

Out-of-court restructuring is faster and cheaper but requires near-unanimous consent for changes to payment terms under US law, so a small holdout can block a deal that a large majority wants. The workaround is the exchange offer with coercive features: offer new bonds, and simultaneously strip the covenants from the old ones so that not participating is worse than participating.

Liability management exercises are the modern dominant form, and they have changed the character of high-yield defaults. Rather than filing, a distressed borrower negotiates with a subset of creditors: extending maturities, moving valuable collateral to a new entity outside the reach of existing lenders, or issuing new debt that primes the existing stack. Creditors who are in the deal do well; those outside it do badly. The consequence is that the identity of your co-creditors and the precision of your documents now matter as much as the company's fundamentals. [Practice] This is a genuine deterioration in the position of a passive credit investor over the past decade.

16.3 Absolute priority, and how it bends

The absolute priority rule says junior claims receive nothing until senior claims are paid in full. It is the foundation of the capital stack in §5.2 and the reason seniority is priced.

It is routinely bent. Equity holders frequently receive something in a reorganisation even when creditors are impaired, for practical rather than legal reasons: management (aligned with equity) controls the process and information, the value of the firm is genuinely uncertain so there is room to argue, and creditors trade some value for a faster resolution. Deviations are well documented in the bankruptcy literature. [Fact]

The practical implications are unglamorous and important. Time is a cost: a Chapter 11 case runs from a few months if pre-negotiated to two years or more if contested, during which your claim earns nothing, so a 40% recovery two years out is worth substantially less than 40% today. Control is worth money: creditors who organise into a group with counsel systematically do better than those who do not. And valuation disputes are the substance of the process — the entire negotiation is about what the reorganised business is worth, since that determines where the value breaks.

16.4 What drives recovery

Recovery varies far more than average figures suggest, and the drivers are identifiable in advance:

Driver	Effect	Why
Debt ahead of you	Dominant	Value is consumed top-down; being behind a large secured loan can mean nothing
Asset tangibility	Large	Aircraft and real estate have buyers; brand and human capital do not
Industry distress	Large	Natural buyers are constrained exactly when assets come to market
Cycle timing	Large	Recoveries fall when defaults rise
Jurisdiction	Large	Creditor-friendly regimes recover more; process length varies widely
Covenant quality	Growing	Weak documents permit collateral to be moved before default

The second, third and fourth interact badly. Acharya, Bharath & Srinivasan (2007) show that industry distress drives recoveries down materially, and since industry distress is what causes defaults in the first place, the correlation is structural rather than incidental. [Fact] Any model using a constant recovery rate is not conservatively wrong; it is wrong in the direction of understating exactly the scenario you care about.

16.5 Sovereign restructuring

There is no bankruptcy court for countries, so the process is negotiation conducted in the shadow of litigation.

The mechanics. A government announces it cannot pay, proposes an exchange of old bonds for new ones with lower coupons and longer maturities, and bondholders decide whether to accept. The haircut is the loss in present value, which is usually much larger than the headline reduction in face value because maturity extension at below-market coupons destroys value even with full principal.

Collective action clauses are the institutional response to the holdout problem (§4.4). Modern sovereign bonds include clauses letting a supermajority bind all holders, and after Argentina's litigation saga these were strengthened to aggregate across bond series, preventing a holdout from acquiring a blocking position in a single small issue.

The official sector runs in parallel. The Paris Club coordinates bilateral government creditors, the IMF provides financing conditional on policy reform, and its lending rules effectively require a restructuring to restore sustainability. A complication of the past decade is that a large share of developing-country debt is now owed to creditors outside these frameworks, which has made recent restructurings markedly slower.

The costs are real and measurable. Cruces & Trebesch (2013) find that larger haircuts are followed by higher spreads and longer market exclusion, with effects persisting for years. [Fact] This is the empirical content of §4.4's claim that repayment is sustained by direct costs: countries face a genuine trade-off between relief today and access tomorrow, and they trade it off visibly.

16.6 Distressed investing

Defaulted and near-defaulted bonds trade, and a specialised industry buys them. The economics are worth understanding even if you never participate, because this industry is the marginal buyer that sets the price of your bond when it deteriorates.

The trade is a valuation claim plus a process claim: what is the enterprise worth, and where in the capital structure does that value run out — the fulcrum security, the most senior claim that does not get paid in full and therefore receives equity in the reorganised company. Buying the fulcrum is buying the post-reorganisation equity cheaply.

Three features shape the business. Returns depend heavily on legal and procedural skill rather than on credit analysis alone. Positions are illiquid and long-dated, which is why the capital is locked up. And the strategy is inherently capacity-constrained and contrarian: it works because forced sellers — index funds, insurers, ratings-constrained mandates — must sell what distressed buyers want to buy (§5.3). The returns are, in a real sense, payment for providing liquidity to the constrained.

16.7 If you hold a deteriorating bond

Practical guidance, in the order the decisions arise:

1. Decide early whether you are a holder or a seller. The worst outcome is selling after the price has fallen but before the process that would have recovered value. Forced sellers transfer value to distressed buyers; not being forced is an edge.
2. Read the documents before the price moves, not after. Seniority, collateral, guarantees and the covenant package determine your recovery, and they are knowable in advance.
3. Know your co-creditors. In a liability management exercise, being outside the deal is where the losses concentrate.
4. Value time. A recovery two years out discounted at a distressed rate is worth far less than its face figure.
5. Do not average down on a credit without a view on the fulcrum. Buying more of a bond that will be wiped out is the most common way to convert a manageable loss into a total one.

§16 Key takeaways

1. "Default" has three definitions — contractual, rating-agency, and CDS credit event — and they disagree. A CDS hedge covers the contractual definition, not the economic one.
2. Chapter 11 reorganises a business worth more alive than dead; creditors often emerge owning the equity, which makes distressed debt a form of private equity.
3. Liability management exercises now dominate high-yield defaults. The identity of your co-creditors and the precision of your documents matter as much as the company's fundamentals.
4. Absolute priority is routinely bent. Equity often receives something because management controls the process and valuation is genuinely disputable.
5. Recovery is driven above all by how much debt sits ahead of you, then by asset tangibility, industry distress and cycle timing — and the last two correlate with default rates against you.
6. Sovereign restructuring is negotiation in the shadow of litigation. Collective action clauses solve the holdout problem; larger haircuts buy relief today at the cost of access tomorrow.
7. Distressed investors are paid for supplying liquidity to constrained sellers. If you are the constrained seller, you are the source of their return.
8. Decide early whether you are a holder or a seller, and never average down on a credit without a view on where the value breaks.

Part VII — Investing

17. Building a bond portfolio

17.1 First, what job are the bonds doing?

Almost every bond allocation mistake comes from skipping this question. Bonds do at least four distinct jobs, and the right holding is different for each.

Job	What you need	What you do not need
Preserve capital / hold cash	Short maturity, high quality	Yield, credit, duration
Hedge equity risk	Long duration, government, nominal	Credit — it fails exactly when you need it
Match a known future liability	Duration matched to the liability; inflation-linked if the liability is real	Anything with optionality
Earn a return	Credit, spread, term premium	The pretence that this is the “safe” sleeve

The recurring error is holding one instrument and expecting it to do two jobs. A high-yield bond fund held as “the safe part of the portfolio” is a return-seeking asset with equity-like tail behaviour (§14.5). An intermediate corporate bond fund held as an equity hedge is half hedge and half the thing being hedged. Separating the jobs and holding a distinct instrument for each is the single highest-value structural decision in fixed income.

17.2 Individual bonds, funds, or ETFs

	Individual bonds	Mutual fund	ETF
Maturity date	Yes — pulls to par	No — constant duration	No, except target-maturity funds
Diversification	Poor unless you hold many	Good	Good
Transaction cost	High for retail (§2.3)	Embedded	Low on-screen; embedded in creation
Liquidity	Poor	Daily at NAV	Intraday on exchange
Tax control	Full	None — you inherit the fund’s gains	Better than a mutual fund
Best for	Liability matching; Treasuries	Credit, where diversification matters most	Trading, rebalancing, liquid exposure

For Treasuries, buying individual bonds is entirely reasonable — they are homogeneous, there is no credit analysis to do, diversification is irrelevant since every issue has the same credit, and retail platforms and direct auction purchase give decent execution.

For corporate credit, funds dominate for almost everyone. You need dozens of issuers to diversify idiosyncratic default risk, retail execution costs on individual corporate bonds are punitive (Edwards, Harris & Piwowar, 2007), and the analysis required per name is real work. The exception is a large enough portfolio to build genuine diversification directly.

17.3 The hold-to-maturity argument, and what is actually true

“I’ll just hold to maturity, so price moves don’t matter” is the most common defence of individual bonds over funds. It is half right, and the correct half is more interesting than the argument usually made.

What is wrong with it. Holding to maturity does not make you whole. If you own a 4% bond and rates go to 7%, you will receive your 100 back — but you spent ten years earning 4% in a 7% world. The opportunity loss is real; you have simply chosen not to mark it. The accounting differs; the economics do not.

What is right, and better. A bond fund recovers from a rate shock too, and there is a precise result about how long it takes. Leibowitz, Bova & Kogelman (2014) show that the annualised return of a portfolio held at a constant duration D converges,

in both mean and dispersion, around its starting yield over a horizon of about $2D - 1$ years, across a wide range of rate paths. [Fact]

The mechanism is the immunisation logic of §7.3 applied year after year: a rate rise costs you price now and pays you a higher reinvestment rate afterwards, and over the right horizon the two cancel. Which horizon is “right” depends on the shape of the rate path, and seeing that is more useful than the headline. Simulating a portfolio that rolls a 7-year par bond every year — Macaulay duration 6.17, so $2D - 1 \approx 11$ — from a starting yield of 4.04% (effective annual): [Computed]

Rate path	3 years	6 years ($\approx D$)	11 years ($\approx 2D-1$)	15 years
+300bp at once, then flat	0.92%	3.98%	5.39%	5.85%
-200bp at once, then flat	6.25%	4.11%	3.15%	2.84%
Rising 25bp every year	3.00%	3.39%	4.06%	4.59%
Falling 25bp every year	5.13%	4.76%	4.15%	3.67%
Random walk, 80bp a year: mean	4.07%	4.05%	4.03%	4.04%
Random walk: dispersion (sd)	1.99%	1.03%	0.77%	0.98%

Three patterns, each worth carrying away.

A one-off shock is neutralised at the duration. Both one-time moves bring the annualised return back to within about a tenth of a percentage point of the starting yield at six years. That is the immunisation horizon of §7.3 appearing again, now for a portfolio rather than a single bond.

A steady trend takes about $2D - 1$ years. When yields keep moving, every year delivers a fresh price change, so the reinvestment effect needs longer to catch up; both trend rows are back within about a tenth of a percentage point of the starting yield by eleven years, which is where the published result lives.

Across random paths the starting yield is the forecast, and it is sharpest near $2D - 1$. The mean annualised return equals the starting yield at every horizon. The dispersion around it falls from 2.0% at three years to its minimum of about 0.75% at ten to eleven years, and then widens again, because over very long horizons the yield you reinvest at has itself drifted far from where you started.

The consequences are the practical payoff of this whole document’s machinery:

- A bond fund’s starting yield is the best simple forecast of its return over one to two durations. For a typical aggregate bond fund with a duration near six, that is roughly six to eleven years. This is the closest thing fixed income has to the equity market’s “starting valuation predicts long-run return” regularity, and it is considerably more reliable.
- A rate rise is good news for a long-horizon bond investor. In the first row the +300bp shock produces the highest 11- and 15-year returns in the table, because every subsequent coupon is reinvested at the higher rate. A sell-off is a mark-to-market loss for a short-horizon investor and a gift to a long-horizon one.
- The individual-bond-versus-fund distinction is mostly psychological. Both recover; the individual bond does so via a maturity date, the fund via reinvestment. If the psychological difference stops you from selling at the bottom, it has real value — but it is a behavioural benefit, not a financial one.

17.4 Bond ETFs

Bond ETFs hold illiquid assets and offer intraday liquidity, which sounds like an obvious contradiction and mostly is not one.

The mechanism: authorised participants can create or redeem ETF shares in kind, exchanging a basket of bonds for shares. If the ETF trades above the value of its holdings, they create shares and sell them; if below, they redeem. This arbitrage keeps the ETF near fair value — when the underlying can be traded.

In March 2020, several corporate bond ETFs traded at discounts of several percent to their reported net asset value, which was widely reported as a failure. [Fact] The better interpretation is the opposite. Corporate bond NAVs are computed from matrix-pricing models and stale quotes on bonds that had not traded; the ETF price was a live, executable market. The ETF was not wrong about the bonds; the NAV was wrong about the bonds. [Contested] — this reading is now widely held but not universal, and the discounts were larger than pure staleness explains, which points to genuine liquidity premia as well.

The durable lessons: an ETF's premium or discount is information about the underlying market's liquidity, not a mispricing to arbitrage; ETFs have become a price discovery mechanism for the underlying market; and daily-dealing funds holding illiquid assets carry a genuine first-mover advantage on redemption (Goldstein, Jiang & Ng, 2017), which is a reason to prefer ETFs over mutual funds for illiquid credit — you exit at a market-clearing price rather than externalising your liquidation cost onto the people who stayed.

17.5 Ladders, barbells and bullets

Three ways to distribute a portfolio across maturities:

Structure	What it is	Duration	Convexity	Suits
Bullet	Everything at one maturity	Concentrated	Lowest	Matching a single liability
Ladder	Equal amounts at 1, 2, 3, ... years	Middling, self-maintaining	Middling	Ongoing cash needs; simplicity
Barbell	Very short plus very long	Same as a bullet, if matched	Highest	A view on curve shape; wanting convexity

The non-obvious point: a barbell and a bullet with the same duration do not have the same risk. The barbell has materially more convexity, because convexity grows much faster with maturity than duration does — from the 2-year to the 30-year, duration rises about ninefold and convexity about ninetyfold (§7.2) — so the long leg contributes disproportionately. A duration-matched barbell therefore outperforms a bullet on any large parallel move in either direction — and underperforms if the curve flattens or steepens against it, and yields slightly less, since convexity is priced (§7.4).

A ladder is the retail workhorse and deserves its reputation. It produces a predictable schedule of maturing principal, it automatically reinvests a slice each year at prevailing rates so it adapts to the rate environment, and it requires no view. Its duration is roughly half the longest rung, and it maintains that duration automatically. A ten-year ladder and a constant-duration-5 bond fund are, over a full cycle, close economic substitutes — with the ladder costing more to build and offering better behavioural properties.

17.6 Costs and taxes

Costs. For government bonds, execution is cheap and fees should be near zero. For credit, the dominant costs are the bid-ask spread and, for retail, dealer markup — and these can exceed a year of the spread you were trying to capture. The practical rule: in fixed income, the cost of trading is a larger share of the expected return than in equities, so turnover is more expensive and buy-and-hold is more attractive than intuition suggests.

Taxes. Bonds are tax-inefficient in a way equities are not, because coupon income is taxed annually at income rates regardless of whether you spend it, while equity returns are largely deferred capital gains. Five specifics:

- Discount and premium bonds are not tax-equivalent, even at the same yield. How the pull to par (§6.5) is taxed depends on the jurisdiction. Where it counts as a capital gain taxed below coupon income — UK gilts held by individuals are the clean case — a low-coupon bond bought below par beats a high-coupon one after tax. In the US, market discount is generally taxed as ordinary income unless it is very small, and that threshold alone moves municipal bond prices (Ang, Bhansali & Xing, 2010). Either way, bonds of the same issuer and maturity can trade at slightly different yields because of it.
- Bonds belong in tax-sheltered accounts where the choice exists. This is one of the few genuinely free improvements available in portfolio construction.
- Municipal bonds are exempt from US federal income tax, which is why they yield less; the comparison to taxable bonds must be on a tax-equivalent basis, and the right adjustment is more subtle than dividing by one minus your rate (Ang, Bhansali & Xing, 2010).
- TIPS generate phantom income: the inflation adjustment to principal is taxed in the year it accrues, though you receive it only at maturity. In a taxable account a TIPS can generate a tax bill exceeding its cash coupon.
- Treasury interest is exempt from US state and local tax, which is worth real money in high-tax states and is part of why corporate spreads look wider than pure credit risk justifies (§9.5).

17.7 Active or passive

The case for indexing is weaker in bonds than in equities, for a specific structural reason rather than because bond managers are more skilled: the index itself is poorly designed (§12.5). Weighting by debt outstanding lends most to the most indebted, and lets issuers set your duration.

But the usual evidence for active management in bonds is contaminated. Active bond funds have on average beaten their benchmarks more often than active equity funds — and the returns are substantially explained by systematically holding more credit risk and more duration than the benchmark, which is a risk premium harvest rather than skill. [Contested] Once returns are adjusted for credit and duration exposure, the outperformance shrinks considerably.

The pragmatic position: use cheap passive exposure for government bonds, where the index problem is mild and the securities are homogeneous. Consider active or alternative weighting for credit, where the index's flaws are real and where avoiding deteriorating issuers has value — but pay for it at a price that reflects what it actually is, and check whether the manager's returns come from skill or from running a permanently higher risk exposure. [Practice]

17.8 Putting it together

A workable default framework, to be adapted rather than copied:

1. Decide the job (§17.1). Write down which of the four you want, in what proportion.
2. Set duration from the horizon, not from a rate view. If you have a genuine liability, match it. If you are hedging equities, duration is the hedge and longer is more efficient per dollar. If you do not know, a duration near half your investment horizon is a defensible starting point — it puts your horizon at about $2D - 1$, where the starting yield is the most reliable forecast (§17.3) — and the intermediate part of the curve is where carry per unit of duration is best (§7.5).
3. Choose nominal or real by the nature of your liability. If your spending is real, inflation-linked bonds are your risk-free asset and nominal bonds are the speculation (§3.3).
4. Size credit against equities, not against government bonds. It belongs in the risk budget with the risk assets (§14.5).
5. Hedge foreign currency unless you specifically want the exposure (§15.4).
6. Put it in the right account and keep costs near zero for the government portion.
7. Rebalance mechanically. The rebalancing itself buys after sell-offs, which is the behaviour §17.3 says is rewarded.

§17 Key takeaways

1. Ask which of four jobs the bonds are doing — cash, equity hedge, liability match, or return-seeking — and hold a distinct instrument for each. One instrument doing two jobs does both badly.
2. Individual Treasuries are fine; individual corporate bonds are usually not, because retail execution costs are punitive and diversification needs dozens of names.
3. “Hold to maturity” does not make you whole — you still spent years earning a below-market rate. The accounting differs; the economics do not.
4. A constant-duration bond portfolio's return is anchored to its starting yield. A one-off shock is neutralised in about D years, a steady trend in about $2D - 1$, and dispersion across random paths is smallest near $2D - 1$ — for a typical aggregate fund, six to eleven years. It is fixed income's version of “valuation predicts long-run return”, and more reliable than the equity version.
5. A rate rise is therefore good for a long-horizon bond investor. The 2022 sell-off raised expected returns for anyone whose horizon exceeded their portfolio's duration.
6. A bond ETF's discount to NAV in a crisis is usually information about stale NAVs, not a mispricing — and ETFs avoid the first-mover problem that daily-dealing mutual funds have.
7. A duration-matched barbell has more convexity than a bullet, so it wins on large parallel moves either way and loses on curve reshaping.
8. Bonds are tax-inefficient: put them in sheltered accounts, and know that TIPS generate taxable income you have not received.
9. Index passively in governments; scrutinise credit indices, and check whether an active manager's outperformance is skill or a permanently higher risk exposure.

18. Failure modes

The document's warnings, collected in one place. They are grouped by the kind of mistake, because the fixes cluster the same way: misreading a number, mismeasuring a risk, building the wrong structure, or being forced to act at the wrong time.

18.1 Misreading a number

Mistake	Mechanism	Fix	§
Treating yield to maturity as expected return	YTM assumes coupons reinvest at YTM; a 30-year 4% bond realises 2.7%–6.0% depending on the path	Model reinvestment, or use a zero-coupon bond	§6.4
Reading breakeven inflation as an inflation forecast	It is expectation plus an inflation risk premium minus a liquidity discount; in 2008 it implied a decade of deflation	Use it as market-implied compensation; compare with surveys	§3.3
Treating a credit spread as extra return	Roughly half a high-yield spread is expected loss	Size off spread minus expected loss	§9.5
Treating ratings as probabilities	Through-the-cycle ordinal rankings, with an issuer-pays conflict	Use for eligibility and coarse sorting only	§2.4
Comparing a TIPS yield to a nominal yield	Different units — one is real, one nominal	Compare real with real, or add breakevens	§3.3
Quoting an OAS without its assumptions	OAS depends on the volatility and prepayment model	Always ask what model produced it	§8.4
Reading a nominal spread off a steep curve	Coupon differences manufacture tens of basis points	Use Z-spread	§8.2
"The market prices an X% default chance"	Only $\lambda(1 - R)$ is observable, and the implied rate is risk-neutral	State the recovery assumption; do not confuse risk-neutral with real-world	§9.2, §9.4

18.2 Mismeasuring a risk

Mistake	Mechanism	Fix	§
Analytic duration on a callable bond or MBS	The formula assumes cash flows independent of yield; there they are not	Effective duration by repricing under shifted curves	§7.6
Duration without convexity on long bonds	Duration alone predicts –52% on a 300bp sell-off; the truth is –37%	Include the convexity term for long bonds and large moves	§7.4
Spread duration without spread level	Spreads move proportionally, so a 500bp bond carries ~5× the risk of a 100bp bond at equal duration	Risk-budget on duration times spread	§7.7
Constant recovery assumptions	Recoveries fall when defaults rise, so tails are worse than modelled	Make recovery cycle-dependent, or stress it explicitly	§9.3
Estimating the bond–equity correlation on a long sample	The sample spans a regime change and describes neither regime	Condition on the inflation environment	§14.2
Counting duration once per asset class	Long-duration equities and long bonds are the same exposure	Aggregate duration across the whole portfolio	§14.4

Mistake	Mechanism	Fix	\$
Trusting empirical duration on high yield	It reverses in inflation-driven sell-offs, which is when it matters	Do not rely on spread/rate offsets for hedging	\$7.7
Treating liquidity as a property of a security	It is a state of the intermediation system; Treasuries were illiquid in March 2020	Stress liquidity by scenario, not by instrument	\$3.5

18.3 Building the wrong structure

Mistake	Mechanism	Fix	\$
Reaching for yield	Yield is compensation for risk; the instinct is systematically punished, and is measurable in fund behaviour	Decompose any yield advantage into which of Identity 2's terms it comes from	\$1.2, \$1.4
Holding credit as "the safe sleeve"	Credit is a short put on the firm; its tail behaviour is equity-like	Size credit against equities in the risk budget	\$14.5
One instrument doing two jobs	An intermediate corporate fund is half hedge, half the thing being hedged	One instrument per job	\$17.1
Unhedged foreign bonds	An unhedged foreign 2-year is 97% currency risk by variance	Hedge by default; take currency risk deliberately if wanted	\$15.4
Owning only nominal bonds as the defensive asset	Nominal bonds hedge growth shocks, never inflation shocks	Hold inflation-linked bonds for the other shock	\$14.4
Ignoring index duration drift	Bond indices weight by debt outstanding, so issuers set your duration	Monitor benchmark duration as a decision, not a given	\$12.5
Buying agency MBS "for the spread"	The spread is option premium for options written to homeowners	Judge it on OAS, and know you are short convexity	\$5.6, \$7.6
Assuming a AAA tranche is AAA-safe	Tranching concentrates correlation risk in the senior tranche	Ask what correlation assumption supports the rating	\$5.6

18.4 Being forced to act

This group is the most expensive, because these failures convert a temporary price move into a permanent loss.

Mistake	Mechanism	Fix	\$
Levering a hedge	Margin calls force selling into the move you were hedging; UK LDI, September 2022	Stress the collateral path, not just the hedge ratio	\$13.4
Running convergence trades on borrowed money	Gaps widen when capital is scarce; LTCM 1998, March 2020	Assume the gap doubles before it closes	\$2.3, \$8.5
Selling at the bottom of a rate shock	A sell-off raises long-horizon returns; selling converts it to a loss	Set the horizon at purchase; rebalance mechanically	\$17.3

Mistake	Mechanism	Fix	\$
Averaging down on a deteriorating credit	Without a view on the fulcrum security you may be buying what gets wiped out	Form a view on where value breaks, or do not add	\$16.7
Being the constrained seller	Forced sellers transfer value to distressed and opportunistic buyers	Avoid mandates that force sales; know your triggers	\$5.3, \$16.6
Trading credit frequently	Transaction costs are a larger share of expected return than in equities	Low turnover; use ETFs for tactical exposure	\$17.6

18.5 The three that cost the most

If the list is too long to remember, these are the ones that have done the most damage, and each is an instance of the same underlying error — treating a conditional property as an unconditional one:

1. Believing bonds are safe without asking “safe against what”. Safe against default is not safe against rates, and not safe against inflation. 2022 cost holders of the safest bonds in the world roughly an eighth of their money (§13.2).
2. Believing bonds hedge equities unconditionally. They hedge growth shocks. The hedge fails in inflation shocks, and it fails by construction, not by accident.
3. Adding leverage to something safe. Every major bond market accident of the past three decades — 1994, LTCM, March 2020, UK LDI 2022 — was leverage applied to an instrument or strategy whose unlevered risk was genuinely small. Small risk levered is not small risk.

19. Synthesis

19.1 The framework, restated

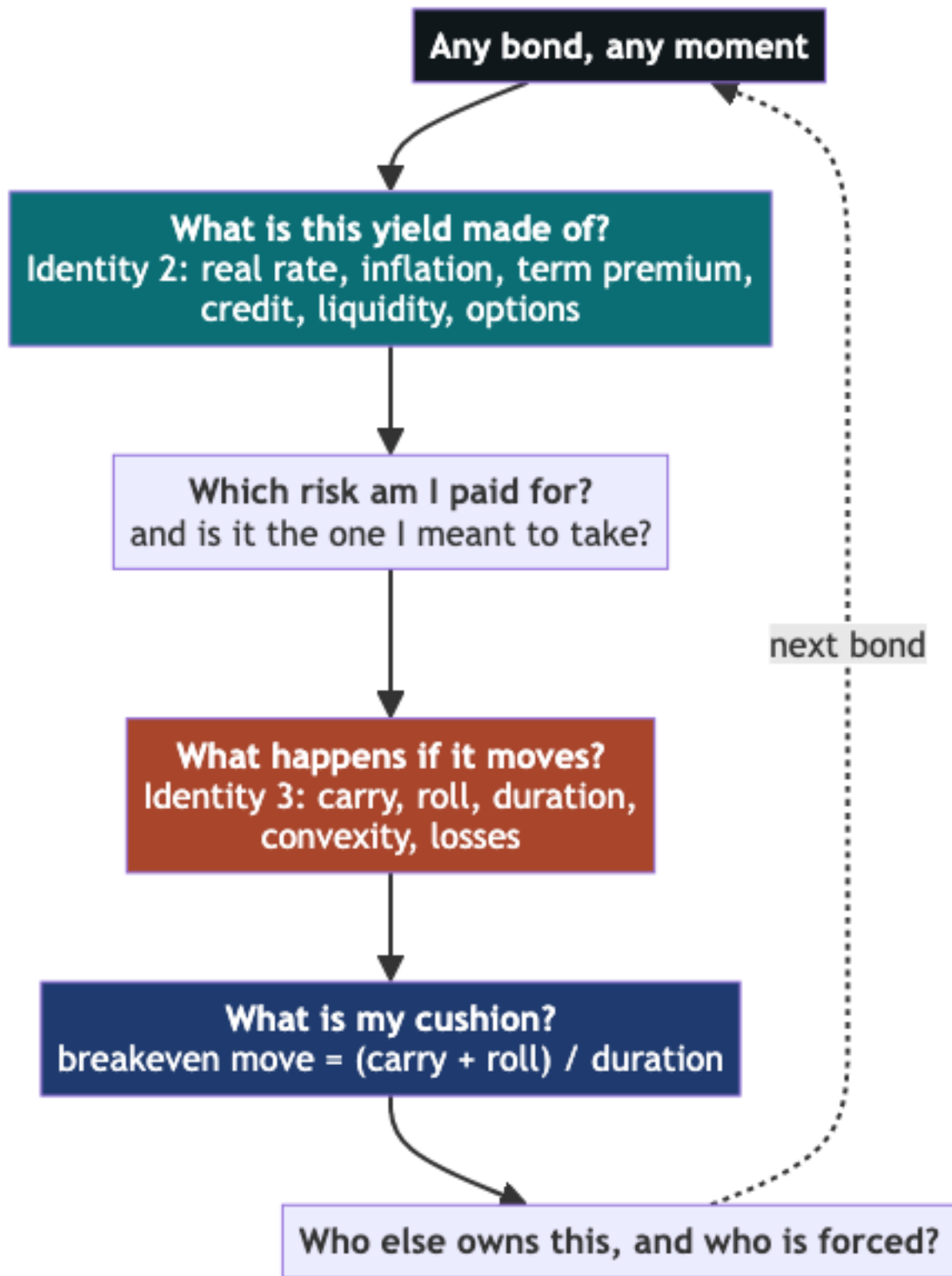
Everything in this document is three identities and a habit.

Identity 1 — price. A bond is a schedule of promised payments; its price is that schedule discounted, weighted by the chance of being paid. Only two things move it: the discount rate, or the belief that the payments arrive.

Identity 2 — yield. The quoted yield decomposes into a real rate, expected inflation, a term premium, a credit spread, a liquidity premium and an option cost. Every bond market on earth is that sum with a different subset of terms switched on. Which terms are present tells you what market you are in; how large they are tells you what you are being paid; which one is moving tells you what is happening.

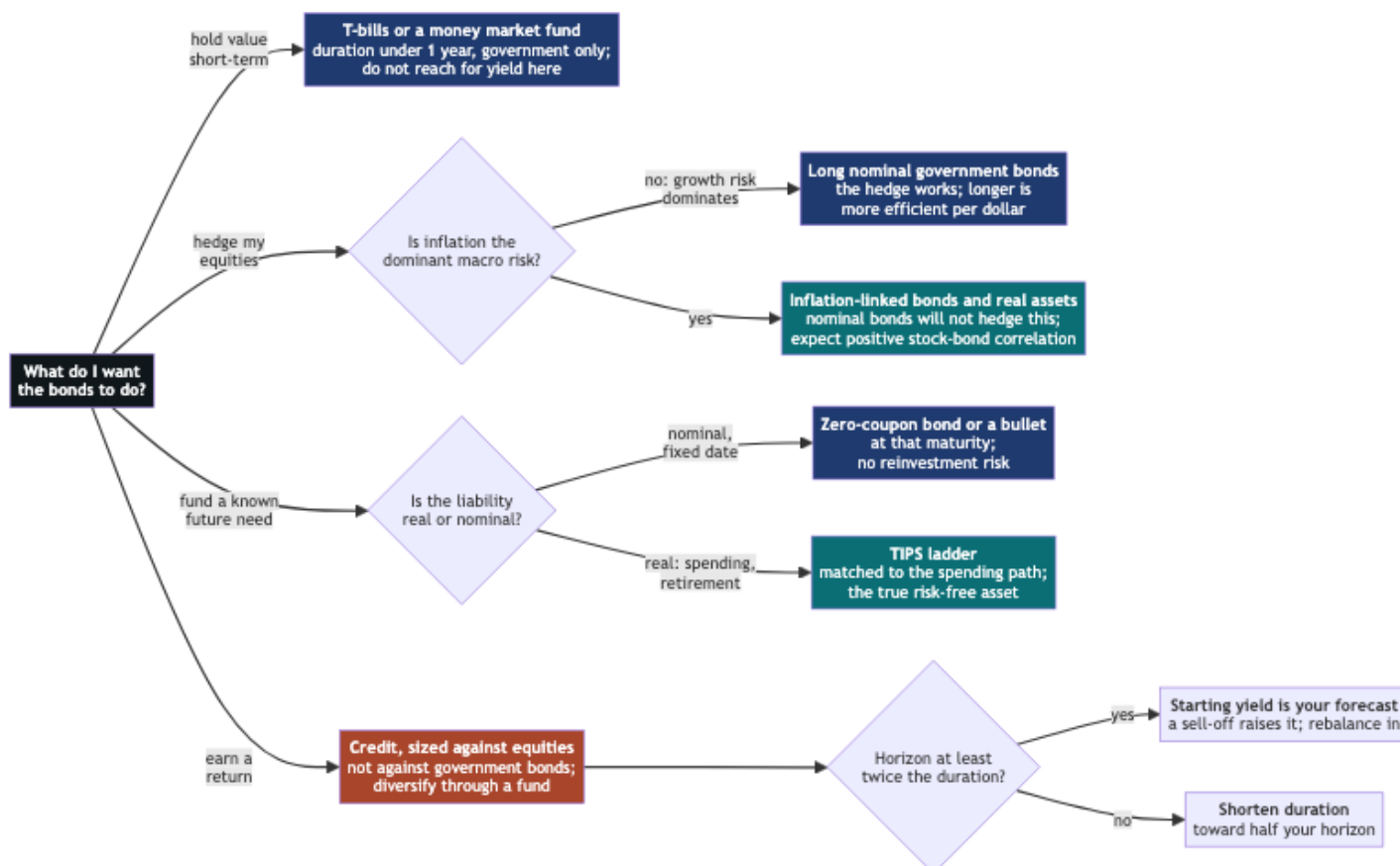
Identity 3 — return. Your return is carry plus roll-down, minus duration times the yield change, plus convexity, minus credit losses. The first two are known the day you buy; the third is the bet; the fourth is a small gift unless you have sold options, in which case it is a small tax; the fifth is what the borrower failed to pay.

The habit is to decompose before judging. Confronted with any yield, ask which of Identity 2’s terms it is made of. Confronted with any move, ask which term moved. Confronted with any position, ask which term you are being paid to bear and whether that is the risk you meant to take.



The loop is the point. It is a small number of questions that apply unchanged to a Treasury bill, a Brazilian local-currency bond, and a distressed second-lien loan.

19.2 A decision tree



Whatever the branch: hedge foreign currency, hold bonds in tax-sheltered accounts, count duration once across the whole portfolio, and never lever the safe sleeve.

19.3 A roadmap, if you are starting from scratch

Staged, with a gate at each stage. The early stages are unglamorous, and skipping them is the usual cause of expensive surprises.

Stage 1 — get the arithmetic right. Build a bond pricer: cash flow schedule, discount factors, yield solver, accrued interest, duration and convexity by both analytic formula and finite difference. Gate: your analytic and numerical durations agree to four decimal places for a bullet, and disagree for a callable — and you can explain why.

Stage 2 — build a curve. Bootstrap zero rates from bills and coupon bonds, fit a Nelson–Siegel or spline curve, and compute forwards and par yields from it. Gate: your curve reprices the bonds you built it from to within a basis point, and your forward rates are smooth rather than saw-toothed.

Stage 3 — compute spreads properly. Z-spread against your curve, then the comparison against nominal spread. Gate: you can reproduce the §8.2 result — two bonds with identical Z-spreads showing very different nominal spreads on a steep curve.

Stage 4 — decompose returns. Take a real bond over a real period and attribute its return into carry, roll, duration, convexity and residual. Gate: the residual is small, and you can explain what drives it when it is not.

Stage 5 — add credit. The credit triangle, survival curves from CDS or bond spreads, and the risk-neutral/real-world distinction. Gate: you can state what recovery assumption your implied default probabilities rest on, and how the answer changes if it is wrong.

Stage 6 — only now, form views. Term premium estimates, curve trades, credit selection, tactical duration. Gate: you can state which term of Identity 2 each view is a view about.

19.4 Ten things I would tell someone starting today

1. Ask “safe against what?” every time someone says a bond is safe. Safe from default, safe from rates and safe from inflation are three different properties, and no single instrument has all three.

2. Learn Identity 2 and use it constantly. Every yield you see is a sum. Decompose it before forming any opinion about whether it is attractive.
3. Yield is not return. It is a price quoted in a convenient unit, and it silently assumes a reinvestment path you will not get.
4. Duration is the only number that matters most of the time, and it has three meanings that are the same number. Learn all three.
5. Compute the cushion. Carry plus roll, divided by duration, tells you how wrong you can be before it costs you. It is the fastest way to size a position honestly.
6. Bonds hedge growth, not inflation. If you hold nominal bonds as your only defensive asset, you are hedged against one of the two shocks that matter.
7. Credit is a short put. Put it on the risky side of your portfolio, size it against equities, and expect its correlations to rise when you need them not to.
8. Never lever the safe part. Every large bond accident in thirty years was safe things, levered.
9. A rate rise is good news if your horizon is longer than your portfolio's duration — comfortably so beyond twice the duration. This inverts the instinct that makes people sell at the bottom, and it is the most valuable single fact in this document for a long-horizon investor.
10. Ask who is forced. Forced buyers and forced sellers explain more short-horizon bond price action than any model, and being un-forced is the most durable edge available to a private investor.

19.5 What we do not know

Honesty about the boundaries, because several of this document's most useful claims sit on softer ground than their prominence suggests.

The term premium is not measured, it is modelled. Every estimate depends on an assumed model of expectations, and reasonable models disagree by enough to change the sign in some periods. Statements about whether the long end is "cheap" rest on this foundation and should be held loosely.

We do not know why credit spreads are as wide as they are. The credit spread puzzle (§9.5) has candidate resolutions but no consensus decomposition. Since that decomposition determines how much of a spread is return rather than expected loss, this is not an academic gap — it is the central uncertainty in sizing a credit allocation.

Bond return predictability is real and weaker than published. The Fama–Bliss/Campbell–Shiller/Cochrane–Piazzesi results survive, but the statistical inference in this literature has known problems, and out-of-sample performance of these predictors has been notably worse than in-sample.

The correlation regime is not forecastable, only classifiable. We can say with confidence that the bond–equity correlation depends on whether inflation is the dominant shock. We cannot say when the regime will change, and the mechanism gives no timing.

Structural change may be underway and we will only know afterwards. The holder base of government debt has shifted from price-insensitive to price-sensitive (§12.4), debt stocks are larger, and intermediation capacity has not grown in proportion. These point toward higher term premia and more frequent liquidity events. Whether that is a regime change or a phase is genuinely unknown. [Hypothesis]

What is solid is the machinery. Identity 1 is arithmetic. Identity 3 is a Taylor expansion. The credit triangle is a no-arbitrage condition. Duration's three meanings are a theorem. The decomposition in Identity 2 is a definition, and definitions cannot be wrong — only the estimates of its components can be, and knowing which component you are uncertain about is most of what it means to understand this asset class.

20. References

Grouped by kind, because the kinds are read differently. Every entry carries a link where a findable one exists; freely readable copies are preferred over paywalled publisher pages, and working-paper versions are linked where the published article is behind a paywall.

20.1 If you only read five things

1. Tuckman & Serrat, *Fixed Income Securities* — the best single technical reference, and the one to own.
2. Merton (1974) — twelve pages that make credit intelligible forever.
3. Campbell, Pflueger & Viceira (2020) — why bonds sometimes hedge equities and sometimes do not.
4. Ilmanen, *Expected Returns*, chapters on bonds — the best available synthesis of what each part of a yield has historically paid.
5. Leibowitz, Bova & Kogelman (2014) — the duration-targeting result, which answers the question every bond investor actually has.

20.2 Books

Modern technical references

- Tuckman, B. & Serrat, A. (2022). *Fixed Income Securities: Tools for Today's Markets*, 4th ed. Wiley. — The standard practitioner-academic bridge. Careful on conventions, curve construction and relative value.
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- Fabozzi, F. J. (ed.) (2021). *The Handbook of Fixed Income Securities*, 9th ed. McGraw-Hill. — Encyclopaedic reference rather than a book to read through; the place to look up an instrument you have not met.
- Duffie, D. & Singleton, K. (2003). *Credit Risk: Pricing, Measurement, and Management*. Princeton University Press. — The standard treatment of reduced-form credit modelling.
- Ilmanen, A. (2011). *Expected Returns: An Investor's Guide to Harvesting Market Rewards*. Wiley. — The most useful single book on what each risk premium has actually paid. The bond chapters are excellent.

Historical and institutional

- Homer, S. & Sylla, R. (2005). *A History of Interest Rates*, 4th ed. Wiley. — Four thousand years of interest rates. Read it to stop believing any particular rate level is normal.
- Garbade, K. (2012). *Birth of a Market: The U.S. Treasury Securities Market from the Great War to the Great Depression*. MIT Press. — How the world's most important market was actually constructed.
- Stigum, M. & Crescenzi, A. (2007). *Stigum's Money Market*, 4th ed. McGraw-Hill. — The reference on repo, bills and short-term funding plumbing.
- Reinhart, C. & Rogoff, K. (2009). *This Time Is Different: Eight Centuries of Financial Folly*. Princeton University Press. — Sovereign default across centuries. [Contested] — the associated debt-threshold empirical work was subject to a well-known replication dispute; the historical narrative is the durable contribution.
- Sturzenegger, F. & Zettelmeyer, J. (2006). *Debt Defaults and Lessons from a Decade of Crises*. MIT Press. — The best account of how sovereign restructurings actually work.
- Cochrane, J. (2023). *The Fiscal Theory of the Price Level*. Princeton University Press. — A complete alternative account of what determines inflation, and therefore nominal yields. [Contested]

Foundational classics

- Macaulay, F. R. (1938). *Some Theoretical Problems Suggested by the Movements of Interest Rates, Bond Yields and Stock Prices in the United States since 1856*. NBER. — Where duration comes from.
- Fisher, I. (1930). *The Theory of Interest*. Macmillan. — The real/nominal decomposition, and still the clearest statement of it.
- Hicks, J. R. (1939). *Value and Capital*. Oxford University Press. — The original liquidity-preference argument for a positive term premium.

20.3 Yields, curves and term premia

- Redington, F. M. (1952). "Review of the Principles of Life-Office Valuations." *Journal of the Institute of Actuaries* 78(3), 286–340. [DOI] — Immunisation: the result that duration is the horizon where price and reinvestment risk cancel (§7.3).
- Fama, E. & Bliss, R. (1987). "The Information in Long-Maturity Forward Rates." *American Economic Review* 77(4). — The first clean evidence against the expectations hypothesis.

- Campbell, J. & Shiller, R. (1991). "Yield Spreads and Interest Rate Movements: A Bird's Eye View." *Review of Economic Studies* 58(3), 495–514. — The regression whose coefficient has the wrong sign.
- Cochrane, J. & Piazzesi, M. (2005). "Bond Risk Premia." *American Economic Review* 95(1), 138–160. [DOI] — One tent-shaped factor prices the whole curve's risk premium.
- Adrian, T., Crump, R. & Moench, E. (2013). "Pricing the Term Structure with Linear Regressions." *Journal of Financial Economics* 110(1), 110–138. — The term premium estimates the New York Fed publishes; see the data page.
- Kim, D. & Wright, J. (2005). "An Arbitrage-Free Three-Factor Term Structure Model and the Recent Behavior of Long-Term Yields." *FEDS* 2005-33. — The other standard term premium series.
- Litterman, R. & Scheinkman, J. (1991). "Common Factors Affecting Bond Returns." *Journal of Fixed Income* 1(1), 54–61. — Level, slope, curvature. [paywalled]
- Nelson, C. & Siegel, A. (1987). "Parsimonious Modeling of Yield Curves." *Journal of Business* 60(4), 473–489. — The functional form everyone still uses.
- Svensson, L. (1994). "Estimating and Interpreting Forward Interest Rates: Sweden 1992–1994." NBER Working Paper 4871. — The standard extension of Nelson–Siegel.
- Gürkaynak, R., Sack, B. & Wright, J. (2007). "The U.S. Treasury Yield Curve: 1961 to the Present." *Journal of Monetary Economics* 54(8), 2291–2304. — The Fed's daily fitted curve dataset, and the standard research input.
- Gürkaynak, R., Sack, B. & Wright, J. (2010). "The TIPS Yield Curve and Inflation Compensation." *American Economic Journal: Macroeconomics* 2(1), 70–92. — Why breakevens are not forecasts (§3.3).
- Estrella, A. & Hardouvelis, G. (1991). "The Term Structure as a Predictor of Real Economic Activity." *Journal of Finance* 46(2), 555–576. — Curve inversion and recessions.
- Estrella, A. & Mishkin, F. (1998). "Predicting U.S. Recessions: Financial Variables as Leading Indicators." *Review of Economics and Statistics* 80(1), 45–61. — The follow-up that established which spread works best.
- Ilmanen, A. (1995). "Time-Varying Expected Returns in International Bond Markets." *Journal of Finance* 50(2), 481–506. — Bond risk premia are predictable outside the US too.

Term-structure models

- Vasicek, O. (1977). "An Equilibrium Characterization of the Term Structure." *Journal of Financial Economics* 5(2), 177–188. — The first tractable one-factor model.
- Cox, J., Ingersoll, J. & Ross, S. (1985). "A Theory of the Term Structure of Interest Rates." *Econometrica* 53(2), 385–407. — The general-equilibrium version, with rates that cannot go negative.
- Ho, T. & Lee, S. (1986). "Term Structure Movements and Pricing Interest Rate Contingent Claims." *Journal of Finance* 41(5), 1011–1029. — The first model calibrated to fit today's curve exactly.
- Hull, J. & White, A. (1990). "Pricing Interest-Rate-Derivative Securities." *Review of Financial Studies* 3(4), 573–592. — The workhorse for pricing rate options.
- Heath, D., Jarrow, R. & Morton, A. (1992). "Bond Pricing and the Term Structure of Interest Rates: A New Methodology." *Econometrica* 60(1), 77–105. — Model the whole forward curve's evolution; the general framework.
- Black, F. (1976). "The Pricing of Commodity Contracts." *Journal of Financial Economics* 3(1–2), 167–179. — The formula for options on forwards, and through it the market's convention for quoting interest rate options. [paywalled]
- Duffie, D. & Kan, R. (1996). "A Yield-Factor Model of Interest Rates." *Mathematical Finance* 6(4), 379–406. — The affine class, which contains almost everything used in practice.
- Piazzesi, M. (2010). "Affine Term Structure Models." In *Handbook of Financial Econometrics*, 691–766. — The standard survey. [paywalled]

20.4 Credit

- Merton, R. (1974). "On the Pricing of Corporate Debt: The Risk Structure of Interest Rates." *Journal of Finance* 29(2), 449–470. — Equity is a call on the firm; risky debt is a risk-free bond minus a put (§5.5). The foundation of everything in credit.
- Black, F. & Cox, J. (1976). "Valuing Corporate Securities: Some Effects of Bond Indenture Provisions." *Journal of Finance* 31(2), 351–367. — Merton extended to default before maturity and to covenants.
- Leland, H. (1994). "Corporate Debt Value, Bond Covenants, and Optimal Capital Structure." *Journal of Finance* 49(4), 1213–1252. — Endogenous default and optimal leverage in one framework.
- Jarrow, R. & Turnbull, S. (1995). "Pricing Derivatives on Financial Securities Subject to Credit Risk." *Journal of Finance* 50(1), 53–85. — The reduced-form alternative: default as a jump with an intensity.
- Duffie, D. & Singleton, K. (1999). "Modeling Term Structures of Defaultable Bonds." *Review of Financial Studies* 12(4), 687–720. — The reduced-form framework in its standard form.
- Duffee, G. (1999). "Estimating the Price of Default Risk." *Review of Financial Studies* 12(1), 197–226. — Early evidence that spreads contain a large risk premium.
- Elton, E., Gruber, M., Agrawal, D. & Mann, C. (2001). "Explaining the Rate Spread on Corporate Bonds." *Journal of Finance* 56(1), 247–277. — The decomposition: expected loss is a small part; taxes and a systematic premium are

large parts.

- Collin-Dufresne, P., Goldstein, R. & Martin, J. S. (2001). "The Determinants of Credit Spread Changes." *Journal of Finance* 56(6), 2177–2207. — Structural variables explain surprisingly little of spread changes; a common factor explains most.
- Huang, J.-Z. & Huang, M. (2012). "How Much of the Corporate-Treasury Yield Spread Is Due to Credit Risk?" *Review of Asset Pricing Studies* 2(2), 153–202. — The canonical statement of the credit spread puzzle.
- Amato, J. & Remolona, E. (2003). "The Credit Spread Puzzle." *BIS Quarterly Review*, December 2003. — The skewness-and-undiversifiability explanation, clearly argued.
- Chen, L., Collin-Dufresne, P. & Goldstein, R. (2009). "On the Relation Between the Credit Spread Puzzle and the Equity Premium Puzzle." *Review of Financial Studies* 22(9), 3367–3409. — The most persuasive resolution: one mechanism explains both.
- Giesecke, K., Longstaff, F., Schaefer, S. & Strebulaev, I. (2011). "Corporate Bond Default Risk: A 150-Year Perspective." *Journal of Financial Economics* 102(2), 233–250. — The long-run record of defaults against spreads. Essential calibration.
- Altman, E. (1968). "Financial Ratios, Discriminant Analysis and the Prediction of Corporate Bankruptcy." *Journal of Finance* 23(4), 589–609. — The Z-score; the origin of quantitative default prediction.
- Altman, E. & Kishore, V. (1996). "Almost Everything You Wanted to Know about Recoveries on Defaulted Bonds." *Financial Analysts Journal* 52(6), 57–64. — The reference recovery statistics.
- Altman, E., Brady, B., Resti, A. & Sironi, A. (2005). "The Link between Default and Recovery Rates." *Journal of Business* 78(6), 2203–2228. — Recoveries fall when defaults rise, quantified.
- Acharya, V., Bharath, S. & Srinivasan, A. (2007). "Does Industry-wide Distress Affect Defaulted Firms? Evidence from Creditor Recoveries." *Journal of Financial Economics* 85(3), 787–821. — Why the correlation between default rates and recoveries is structural.
- Asquith, P., Mullins, D. & Wolff, E. (1989). "Original Issue High Yield Bonds: Aging Analyses of Defaults, Exchanges, and Calls." *Journal of Finance* 44(4), 923–952. — The paper that established that high-yield default rates cumulate far higher than annual rates suggest.
- Becker, B. & Milbourn, T. (2011). "How Did Increased Competition Affect Credit Ratings?" *Journal of Financial Economics* 101(3), 493–514. — Competition among agencies made ratings more favourable, not more accurate.
- Ellul, A., Jotikasthira, C. & Lundblad, C. (2011). "Regulatory Pressure and Fire Sales in the Corporate Bond Market." *Journal of Financial Economics* 101(3), 596–620. — The forced-selling effect at the investment-grade boundary.
- Greenwood, R. & Hanson, S. (2013). "Issuer Quality and Corporate Bond Returns." *Review of Financial Studies* 26(6), 1483–1525. [DOI] — The composition of issuance predicts credit returns.
- Bai, J., Bali, T. & Wen, Q. (2019). "Common Risk Factors in the Cross-Section of Corporate Bond Returns." *Journal of Financial Economics* 131(3), 619–642. — A factor model for credit, with downside risk central.
- Kelly, B., Palhares, D. & Pruitt, S. (2023). "Modeling Corporate Bond Returns." *Journal of Finance* 78(4), 1967–2008. — A modern, high-dimensional treatment of the credit cross-section.

20.5 Market structure, liquidity and funding

- Duffie, D., Gârleanu, N. & Pedersen, L. H. (2005). "Over-the-Counter Markets." *Econometrica* 73(6), 1815–1847. [DOI] — Search and bargaining: why your outside option determines your price.
- Edwards, A., Harris, L. & Piwowar, M. (2007). "Corporate Bond Market Transaction Costs and Transparency." *Journal of Finance* 62(3), 1421–1451. — Retail investors pay far more, and more on small trades.
- Bessembinder, H., Maxwell, W. & Venkataraman, K. (2006). "Market Transparency, Liquidity Externalities, and Institutional Trading Costs in Corporate Bonds." *Journal of Financial Economics* 82(2), 251–288. — The TRACE natural experiment.
- Goldstein, M., Hotchkiss, E. & Sirri, E. (2007). "Transparency and Liquidity: A Controlled Experiment on Corporate Bonds." *Review of Financial Studies* 20(2), 235–273. — The controlled version of the same experiment.
- Hendershott, T. & Madhavan, A. (2015). "Click or Call? Auction versus Search in the Over-the-Counter Market." *Journal of Finance* 70(1), 419–447. — Electronic request-for-quote against telephone search.
- Bao, J., Pan, J. & Wang, J. (2011). "The Illiquidity of Corporate Bonds." *Journal of Finance* 66(3), 911–946. — Measuring the illiquidity component of the spread.
- Dick-Nielsen, J., Feldhütter, P. & Lando, D. (2012). "Corporate Bond Liquidity Before and After the Onset of the Subprime Crisis." *Journal of Financial Economics* 103(3), 471–492. — How much of the crisis spread widening was liquidity.
- Feldhütter, P. (2012). "The Same Bond at Different Prices." *Review of Financial Studies* 25(4), 1155–1206. — Identical claims trading at different prices, and why.
- Amihud, Y. & Mendelson, H. (1991). "Liquidity, Maturity, and the Yields on U.S. Treasury Securities." *Journal of Finance* 46(4), 1411–1425. — The on-the-run premium, measured.
- Longstaff, F. (2004). "The Flight-to-Liquidity Premium in U.S. Treasury Bond Prices." *Journal of Business* 77(3), 511–

526. — Treasuries against agency bonds: the pure liquidity premium.
- Ang, A., Bhansali, V. & Xing, Y. (2010). “Taxes on Tax-Exempt Bonds.” *Journal of Finance* 65(2), 565–601. — How the tax treatment of market discount moves municipal bond prices.
 - Duffie, D. (1996). “Special Repo Rates.” *Journal of Finance* 51(2), 493–526. — Why a bond in demand to short trades rich.
 - Hu, G. X., Pan, J. & Wang, J. (2013). “Noise as Information for Illiquidity.” *Journal of Finance* 68(6), 2341–2382. [DOI] — Deviations from a smooth curve as a measure of arbitrage capital.
 - Gorton, G. & Metrick, A. (2012). “Securitized Banking and the Run on Repo.” *Journal of Financial Economics* 104(3), 425–451. — How a funding market freezes.
 - Malvey, P. & Archibald, C. (1998). “Uniform-Price Auctions: Update of the Treasury Experience.” US Treasury, Office of Market Finance. — The Treasury’s own evaluation of the switch to uniform-price auctions (§2.2).
 - Lou, D., Yan, H. & Zhang, J. (2013). “Anticipated and Repeated Shocks in Liquid Markets.” *Review of Financial Studies* 26(8), 1891–1912. — The Treasury auction cycle’s price footprint.
 - He, Z., Nagel, S. & Song, Z. (2022). “Treasury Inconvenience Yields during the COVID-19 Crisis.” *Journal of Financial Economics* 143(1), 57–79. [DOI] — When the convenience yield inverted.
 - Vissing-Jorgensen, A. (2021). “The Treasury Market in Spring 2020 and the Response of the Federal Reserve.” *Journal of Monetary Economics* 124, 19–47. — What broke, and what fixed it.
 - Duffie, D. (2020). “Still the World’s Safe Haven? Redesigning the U.S. Treasury Market After the COVID-19 Crisis.” Brookings. — The structural diagnosis and the proposed fixes.
 - Barth, D. & Kahn, R. J. (2025). “Hedge Funds and the Treasury Cash-Futures Basis Trade.” *Journal of Monetary Economics* 155. — Sizing and mechanics of the levered basis trade.
 - van Binsbergen, J., Diamond, W. & Grotteria, M. (2022). “Risk-Free Interest Rates.” *Journal of Financial Economics* 143(1), 1–29. — A risk-free rate measured from options, above the Treasury yield.
 - Fleckenstein, M., Longstaff, F. & Lustig, H. (2014). “The TIPS–Treasury Bond Puzzle.” *Journal of Finance* 69(5), 2151–2197. [DOI] — A very large, very persistent violation of the law of one price.
 - Shleifer, A. & Vishny, R. (1997). “The Limits of Arbitrage.” *Journal of Finance* 52(1), 35–55. — Why arbitrageurs who invest other people’s money cut positions exactly when mispricing is widest.
 - Schrimpf, A. & Sushko, V. (2019). “Beyond LIBOR: A Primer on the New Benchmark Rates.” *BIS Quarterly Review*, March 2019. — What replaced LIBOR and why it matters for discounting.

20.6 Macro, policy, supply and demand

- Laubach, T. & Williams, J. (2003). “Measuring the Natural Rate of Interest.” *Review of Economics and Statistics* 85(4), 1063–1070. — The standard r^* estimator.
- Holston, K., Laubach, T. & Williams, J. (2017). “Measuring the Natural Rate of Interest: International Trends and Determinants.” *Journal of International Economics* 108, S59–S75. — The decline is global and synchronised.
- Rachel, Ł. & Summers, L. (2019). “On Secular Stagnation in the Industrialized World.” *Brookings Papers on Economic Activity* 2019(1), 1–76. — Why private-sector r^* fell further than the observed rate.
- Modigliani, F. & Sutch, R. (1966). “Innovations in Interest Rate Policy.” *American Economic Review, Papers and Proceedings*. — The original preferred-habitat argument.
- Vayanos, D. & Vila, J.-L. (2021). “A Preferred-Habitat Model of the Term Structure of Interest Rates.” *Econometrica* 89(1), 77–112. [DOI] — Preferred habitat made rigorous; the theoretical basis for QE.
- Greenwood, R. & Vayanos, D. (2014). “Bond Supply and Excess Bond Returns.” *Review of Financial Studies* 27(3), 663–713. [DOI] — More long-maturity supply, higher subsequent long-bond returns.
- Greenwood, R., Hanson, S. & Stein, J. (2015). “A Comparative-Advantage Approach to Government Debt Maturity.” *Journal of Finance* 70(4), 1683–1722. — How a government should choose its maturity profile, and what that does to the market.
- Krishnamurthy, A. & Vissing-Jorgensen, A. (2012). “The Aggregate Demand for Treasury Debt.” *Journal of Political Economy* 120(2), 233–267. — The convenience yield, measured.
- Krishnamurthy, A. & Vissing-Jorgensen, A. (2011). “The Effects of Quantitative Easing on Interest Rates.” *Brookings Papers on Economic Activity* 2011(2), 215–287. — Which channels QE actually works through.
- Gagnon, J., Raskin, M., Remache, J. & Sack, B. (2011). “The Financial Market Effects of the Federal Reserve’s Large-Scale Asset Purchases.” *International Journal of Central Banking* 7(1), 3–43. — The first systematic event study of QE.
- D’Amico, S. & King, T. (2013). “Flow and Stock Effects of Large-Scale Treasury Purchases.” *Journal of Financial Economics* 108(2), 425–448. — Separating the temporary flow effect from the permanent stock effect.
- Blanchard, O. (2019). “Public Debt and Low Interest Rates.” *American Economic Review* 109(4), 1197–1229. [DOI] — What $r < g$ means for the fiscal cost of debt. [Contested]
- Sargent, T. & Wallace, N. (1981). “Some Unpleasant Monetarist Arithmetic.” *Federal Reserve Bank of Minneapolis*

- Quarterly Review 5(3). — The original fiscal dominance argument.
- Jiang, Z., Lustig, H., Van Nieuwerburgh, S. & Xiaolan, M. (2024). “The U.S. Public Debt Valuation Puzzle.” *Econometrica* 92(4), 1309–1347. [DOI] — The market value of US debt against the present value of surpluses. Uncomfortable arithmetic.
 - Goldstein, I., Jiang, H. & Ng, D. (2017). “Investor Flows and Fragility in Corporate Bond Funds.” *Journal of Financial Economics* 126(3), 592–613. — The concave flow-performance relationship and the run incentive.
 - Choi, J. & Kronlund, M. (2018). “Reaching for Yield in Corporate Bond Mutual Funds.” *Review of Financial Studies* 31(5), 1930–1965. — Funds reach for yield to attract flows, especially when rates are low; the tilt earns higher raw returns and inflows but negative risk-adjusted returns.
 - Koont, N., Ma, Y., Pástor, L. & Zeng, Y. (2025). “Steering a Ship in Illiquid Waters: Active Management of Passive Funds.” *Review of Financial Studies* 38(10), 2887–2935. — How bond ETFs actually manage liquidity mismatch.

20.7 Cross-asset, currency and sovereign

- Campbell, J., Sunderam, A. & Viceira, L. (2017). “Inflation Bets or Deflation Hedges? The Changing Risks of Nominal Bonds.” *Critical Finance Review* 6(2), 263–301. [DOI] — Documenting the sign change in bonds’ equity beta.
- Campbell, J., Pflueger, C. & Viceira, L. (2020). “Macroeconomic Drivers of Bond and Equity Risks.” *Journal of Political Economy* 128(8), 3148–3185. [DOI] — Why the correlation flips, and what the monetary policy rule has to do with it. The single most useful paper in §14.
- Baele, L., Bekaert, G. & Inghelbrecht, K. (2010). “The Determinants of Stock and Bond Return Comovements.” *Review of Financial Studies* 23(6), 2374–2428. — Macro fundamentals explain less of the comovement than liquidity and risk appetite.
- Fama, E. (1984). “Forward and Spot Exchange Rates.” *Journal of Monetary Economics* 14(3), 319–338. — The forward premium puzzle; the failure of uncovered interest parity.
- Lustig, H., Roussanov, N. & Verdelhan, A. (2011). “Common Risk Factors in Currency Markets.” *Review of Financial Studies* 24(11), 3731–3777. [DOI] — Carry returns as compensation for a common risk factor.
- Du, W., Tepper, A. & Verdelhan, A. (2018). “Deviations from Covered Interest Rate Parity.” *Journal of Finance* 73(3), 915–957. [DOI] — The cross-currency basis, and why regulation is the friction.
- Eaton, J. & Gersovitz, M. (1981). “Debt with Potential Repudiation: Theoretical and Empirical Analysis.” *Review of Economic Studies* 48(2), 289–309. — The reputational theory of sovereign repayment.
- Bulow, J. & Rogoff, K. (1989). “Sovereign Debt: Is to Forgive to Forget?” *American Economic Review* 79(1). — Reputation alone is not enough; direct costs must sustain repayment.
- Arellano, C. (2008). “Default Risk and Income Fluctuations in Emerging Economies.” *American Economic Review* 98(3), 690–712. — The modern quantitative sovereign default model.
- Aguiar, M. & Gopinath, G. (2006). “Defaultable Debt, Interest Rates and the Current Account.” *Journal of International Economics* 69(1), 64–83. — Default driven by trend growth shocks.
- Cole, H. & Kehoe, T. (2000). “Self-Fulfilling Debt Crises.” *Review of Economic Studies* 67(1), 91–116. — The standard model of a run on a sovereign.
- Cruces, J. & Trebesch, C. (2013). “Sovereign Defaults: The Price of Haircuts.” *American Economic Journal: Macroeconomics* 5(3), 85–117. — Bigger haircuts, higher subsequent spreads, longer exclusion.
- Longstaff, F., Pan, J., Pedersen, L. H. & Singleton, K. (2011). “How Sovereign Is Sovereign Credit Risk?” *American Economic Journal: Macroeconomics* 3(2), 75–103. [DOI] — Global risk appetite dominates country fundamentals.
- Eichengreen, B., Hausmann, R. & Panizza, U. (2003). “Currency Mismatches, Debt Intolerance and Original Sin.” NBER Working Paper 10036. — Why borrowing in someone else’s currency is so dangerous.
- Jordà, Ò., Knoll, K., Kuvshinov, D., Schularick, M. & Taylor, A. (2019). “The Rate of Return on Everything, 1870–2015.” *Quarterly Journal of Economics* 134(3), 1225–1298. — The long-run return record for bonds alongside every other asset. The best available calibration of long-horizon expectations.

20.8 Corporate finance and capital structure

- Modigliani, F. & Miller, M. (1958). “The Cost of Capital, Corporation Finance and the Theory of Investment.” *American Economic Review* 48(3), 261–297. — The irrelevance benchmark that every later theory departs from.
- Jensen, M. & Meckling, W. (1976). “Theory of the Firm: Managerial Behavior, Agency Costs and Ownership Structure.” *Journal of Financial Economics* 3(4), 305–360. — Why the conflict between shareholders and creditors shapes debt contracts.
- Jensen, M. (1986). “Agency Costs of Free Cash Flow, Corporate Finance, and Takeovers.” *American Economic Review* 76(2), 323–329. — The disciplining role of debt.
- Myers, S. (1977). “Determinants of Corporate Borrowing.” *Journal of Financial Economics* 5(2), 147–175. — Debt overhang: why too much debt kills good investment.

- Chava, S. & Roberts, M. (2008). “How Does Financing Impact Investment? The Role of Debt Covenants.” *Journal of Finance* 63(5), 2085–2121. — Covenant violations transfer control and change behaviour, measurably.
- Bradley, M. & Roberts, M. (2015). “The Structure and Pricing of Corporate Debt Covenants.” *Quarterly Journal of Finance* 5(2). — What covenants are worth in yield terms.

20.9 Mortgages and securitised credit

- Gabaix, X., Krishnamurthy, A. & Vigneron, O. (2007). “Limits of Arbitrage: Theory and Evidence from the Mortgage-Backed Securities Market.” *Journal of Finance* 62(2), 557–595. [DOI] — Prepayment risk is priced because the marginal investor is specialised and constrained.
- Hanson, S. (2014). “Mortgage Convexity.” *Journal of Financial Economics* 113(2), 270–299. — MBS hedging as a driver of Treasury yield dynamics.
- Malkhozov, A., Mueller, P., Vedolin, A. & Venter, G. (2016). “Mortgage Risk and the Yield Curve.” *Review of Financial Studies* 29(5), 1220–1253. — The same mechanism, in the term structure.

20.10 Practitioner research

Directly usable, and produced by people who sell the thing. Discount accordingly, but do not ignore — much of the implementation evidence exists nowhere else.

- Leibowitz, M., Bova, A. & Kogelman, S. (2014). “Long-Term Bond Returns under Duration Targeting.” *Financial Analysts Journal* 70(1), 31–51. — The $2D - 1$ convergence result of §17.3. The most practically useful bond paper on this list.
- Ben Dor, A., Dynkin, L., Hyman, J., Houweling, P., van Leeuwen, E. & Penninga, O. (2007). “DTS (Duration Times Spread).” *Journal of Portfolio Management* 33(2), 77–100. — Spreads move proportionally, so risk scales with duration times spread. [paywalled] [Contested] — authors were at a bank selling credit analytics; the result has nonetheless replicated widely.
- Asvanunt, A. & Richardson, S. (2017). “The Credit Risk Premium.” *Journal of Fixed Income* 26(3), 6–24. — A long-history estimate of what credit has actually paid over duration-matched Treasuries. [Contested] — AQR authors; the firm runs credit strategies.
- Israel, R., Palhares, D. & Richardson, S. (2018). “Common Factors in Corporate Bond Returns.” *Journal of Investment Management* 16(2). — Carry, defensive, momentum and value in credit. [Contested] — same interest disclosure.
- Houweling, P. & van Zundert, J. (2017). “Factor Investing in the Corporate Bond Market.” *Financial Analysts Journal* 73(2), 100–115. — An independent replication of credit factors. [Contested] — Robeco authors.
- Koijen, R., Moskowitz, T., Pedersen, L. H. & Vrugt, E. (2018). “Carry.” *Journal of Financial Economics* 127(2), 197–225. — Carry as a unified concept across asset classes, including bonds. Academic, with practitioner co-authors.
- Frazzini, A. & Pedersen, L. H. (2014). “Betting Against Beta.” *Journal of Financial Economics* 111(1), 1–25. — Includes the fixed-income evidence that low-duration bonds have had better risk-adjusted returns, attributed to leverage aversion.
- Fama, E. & French, K. (1993). “Common Risk Factors in the Returns on Stocks and Bonds.” *Journal of Financial Economics* 33(1), 3–56. — The TERM and DEF factors: the original bond risk factors.

20.11 Critiques and cautions

A bibliography listing only a field’s successes is propaganda. These are the papers that constrain how confidently the rest should be read.

- Bauer, M. & Hamilton, J. (2018). “Robust Bond Risk Premia.” *Review of Financial Studies* 31(2). — Much of the bond-return predictability literature does not survive proper small-sample inference. Read alongside §11.2.
- Stambaugh, R. (1999). “Predictive Regressions.” *Journal of Financial Economics* 54(3), 375–421. — The finite-sample bias that inflates a predictive slope when a persistent predictor’s shocks are correlated with returns.
- Huang, J.-Z. & Huang, M. (2012). (above) — Structural credit models do not explain observed spreads. The foundational critique of §5.5’s framework.
- Collin-Dufresne, Goldstein & Martin (2001). (above) — Structural variables explain little of spread changes, which is a harder problem than explaining levels.
- Becker & Milbourn (2011). (above) — The ratings on which a large part of the institutional bond market’s plumbing depends respond to competitive pressure.
- Amato & Remolona (2003). (above) — An argument that the puzzle is really about the difficulty of diversifying a negatively-skewed portfolio, not about mispricing.

- He, Nagel & Song (2022), Vissing-Jorgensen (2021), Duffie (2020) (above) — Three independent demonstrations that the safe-asset property of Treasuries is conditional on intermediation capacity.

20.12 Data and official sources

- FRED — Federal Reserve Bank of St. Louis. Yields, spreads, breakevens, and index option-adjusted spreads. The default starting point.
 - Federal Reserve fitted yield curves — the Gürkaynak–Sack–Wright daily zero curve, updated.
 - NY Fed term premium estimates — the ACM series of §10.4.
 - US Treasury auction results and debt data — issuance calendar, auction statistics, and the outstanding debt profile.
 - BIS debt securities statistics — the standard source for international bond market size and composition.
 - FINRA TRACE — US corporate bond transaction reporting; the underlying data for most microstructure research.
 - Rating agency default studies — Moody's and S&P publish annual default and recovery studies. These are the reference series for every calibration in §9, and they are free with registration.
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Appendix A — Concepts and prerequisites

This appendix collects the concepts the main text uses without fully developing. Each entry has the same four parts — what the concept is, its formal definition, why it appears in this document, and where to go deeper — and the entries are ordered so that each leans only on earlier ones. A reader meeting an unfamiliar term in the main text can look it up in the index below; a reader without a finance background can read the appendix straight through as a build-up.

Symbols defined inside an entry are local to it unless they appear in the notation block at the front of the document.

Concept	Used in	Concept	Used in
A.1 Compounding conventions	§6.6, §9.2	A.18 Interest rate swaps and OIS	§3.1, §8.2
A.2 Internal rate of return	§1.2, §6.3	A.19 Covered and uncovered interest parity	§15.1–§15.3
A.3 Taylor expansion	§1.4, §7.4, §13.2	A.20 Credit default swaps	§8.5, §16.1
A.4 Law of one price and replication	§3.3, §6.2, §15.1	A.21 Uniform-price and pay-as-bid auctions	§2.2
A.5 Convenience yield	§1.5, §3.1	A.22 Over-the-counter markets	§2.3, §17.2
A.6 Call and put options	§1.3, §5.5, §7.6	A.23 ETF creation and redemption	§17.4
A.7 Volatility and option value	§5.5, §8.4	A.24 Reserves and the policy rate	§10.5
A.8 Long and short optionality	§5.5, §7.4, §15.3	A.25 State-space models and the Kalman filter	§10.2
A.9 Hazard rates and survival	§1.4, §9.2	A.26 Debt dynamics: $r - g$	§10.6
A.10 Systematic risk and factor models	§9.5, §14.5, §15.3	A.27 Multiple equilibria	§4.2, §13.4
A.11 Risk-neutral probabilities	§9.4, §9.5	A.28 Principal components of the curve	§7.7, §11.4
A.12 Default correlation	§5.6, §9.1	A.29 Affine term-structure models	§10.4, §11.4
A.13 Distance to default	§5.5, §9.7	A.30 Modigliani–Miller and the trade-off theory	§1.5, §5.1
A.14 Discriminant analysis and the Z-score	§9.7	A.31 Agency costs and debt overhang	§1.5, §5.1
A.15 Repurchase agreements	§3.4	A.32 Absolute priority and Chapter 11	§16.2–§16.3
A.16 Margin and mark-to-market	§3.4, §13.4	A.33 Predictive regressions and their traps	§11.2
A.17 Bond futures and the basis	§3.4, §8.5	A.34 Risk contributions and risk parity	§14.2, §14.4

Part 1 — Time, prices and arbitrage

A.1 Compounding conventions

What it is. An interest rate is not a number until you know how often it compounds. “4% a year” paid as 2% every six months turns 100 into 104.04 after a year, because the second half-year earns interest on the first half-year’s interest. Markets quote rates under different conventions — annual for most European government bonds, semi-annual for US Treasuries and corporates, simple interest for money-market instruments — and comparing two rates quoted under different conventions is comparing numbers in different units. Continuous compounding is the limit of compounding ever more often. It is rarely quoted, but it is the natural convention for mathematics, because growth factors become exponentials and exponentials multiply by adding their exponents.

Formally. A rate y compounded m times a year grows 1 unit to $(1 + y/m)^{mt}$ over t years. Its effective annual rate is $(1 + y/m)^m - 1$, and the continuously compounded rate y_c with $e^{y_c t} = (1 + y/m)^{mt}$ is $y_c = m \ln(1 + y/m)$. So

4% semi-annual is 4.04% effective annual and 3.96% continuous. Under continuous compounding the discount factor is $Z(t) = e^{-z_c(t)t}$, and rates over consecutive periods simply add in the exponent.

Why it appears here. The notation block writes $Z(t)$ with annual compounding; §6.3 and §6.6 work with semi-annual yields and warn about converting between conventions; the survival probability $e^{-\lambda t}$ behind the credit triangle (§9.2) is a continuous-time object.

Going deeper. Sections 1 and 3 of Simple and Log Returns — a continuously compounded rate is a log return. Tuckman & Serrat (2022) on quoting conventions.

A.2 Internal rate of return

What it is. The single constant interest rate that, used to discount every cash flow of an investment, makes their present value equal to its price. It compresses a whole schedule of payments into one number, so that investments with different schedules can be compared. The cost of the compression is an assumption hidden inside it: that one rate applies to every period, and in particular that every interim cash flow can be reinvested at that same rate until the end.

Formally. For a price P paid for cash flows C_t , the IRR is the root k of $\sum_t C_t(1+k)^{-t} - P = 0$. When every cash flow is positive, as for a bond, the left-hand side falls steadily as k rises, from $\sum_t C_t - P$ at $k = 0$ towards $-P$, so there is exactly one root (negative only if the price exceeds the undiscounted sum of the payments). Investments whose cash flows change sign more than once can have several IRRs, and then the number is ambiguous. Terminal wealth equals $P(1+k)^T$ exactly when every interim cash flow earns k until T .

Why it appears here. Yield to maturity is a bond's IRR (§6.3), and the reinvestment assumption is the whole reason a yield is not an expected return (§1.2, §6.4).

Going deeper. Tuckman & Serrat (2022) on yield measures.

A.3 Taylor expansion, and why the return identity is one

What it is. Any smooth function can be approximated near a point by its value there, its slope and its curvature — a straight line corrected by a parabola. Duration is the slope of price against yield, convexity is the curvature, and carry is the slope against time. Identity 3 is nothing more than this approximation applied to a bond's price, which is why it is accurate for small moves and increasingly wrong for large ones.

Formally. For a price that depends on yield and time,

$$P(y + \Delta y, t + \Delta t) \approx P + \frac{\partial P}{\partial t} \Delta t + \frac{\partial P}{\partial y} \Delta y + \frac{1}{2} \frac{\partial^2 P}{\partial y^2} (\Delta y)^2.$$

Dividing by P and using $D = -\frac{1}{P} \frac{\partial P}{\partial y}$ and $\mathcal{C} = \frac{1}{P} \frac{\partial^2 P}{\partial y^2}$ gives $\frac{\Delta P}{P} \approx \frac{1}{P} \frac{\partial P}{\partial t} \Delta t - D \Delta y + \frac{1}{2} \mathcal{C} (\Delta y)^2$. Held at an unchanged yield, a bond's price drift plus its coupon income earns exactly the yield, which supplies the carry term $y \Delta t$. Roll-down enters through Δy : on a sloped curve, the yield at which the bond is valued changes as its maturity shortens even if the curve stays put. The terms the expansion drops — third derivatives, and the interaction between time and yield — are the “cross terms” of §13.2.

Why it appears here. Identity 3 (§1.4); duration and convexity (§7.2, §7.4); the return attribution of §13.2; the remark in §19.5 that Identity 3 is a Taylor expansion.

Going deeper. Tuckman & Serrat (2022) on DV01, duration and convexity.

A.4 The law of one price, replication and the limits of arbitrage

What it is. Two assets that deliver identical cash flows in every possible future must cost the same today. If they did not, you could buy the cheap one, sell the dear one, and keep the difference with no risk — an arbitrage — and the presumption that such opportunities are competed away is the backbone of fixed-income and derivatives pricing. Replication is the constructive version: build a portfolio of instruments you can already price that reproduces a security's payoffs, and the security must be worth what the portfolio costs. The limits of arbitrage are the catch. Arbitrage needs capital, financing and patience, and a trade that is riskless at maturity can lose heavily along the way — so gaps can open wide, and stay open, when capital is scarce.

Formally. If a payoff X equals $\sum_i w_i X_i$ in every state, the law of one price requires $\text{price}(X) = \sum_i w_i \text{price}(X_i)$. A coupon bond is the portfolio $\sum_t C_t \times$ (zero-coupon bond maturing at t), so $P = \sum_t C_t Z(t)$ — Identity 1 without default. The limits-of-arbitrage argument of Shleifer & Vishny (1997) is that arbitrageurs invest other people's money and face withdrawals after losses, so they are forced to cut positions exactly when the mispricing is widest.

Why it appears here. Coupon bonds as portfolios of zeros, and STRIPS (§3.2, §6.2); the TIPS–Treasury gap (§3.3); the CDS–bond basis (§8.5); covered interest parity and its failure (§15.1–§15.2); the recurring conclusion that persistent gaps are the price of balance sheet (§15.2).

Going deeper. Shleifer & Vishny (1997); Gabaix, Krishnamurthy & Vigneron (2007) for the mortgage-market evidence; section 3.2 of Dealer Hedging and Gamma Exposure for replication in options.

A.5 Convenience yield

What it is. The part of an asset's return that comes not from its cash flows but from the services it provides by being held. For a Treasury those services are the ability to post it as collateral, to satisfy a liquidity regulation, to park cash safely, and to sell it in size in almost any market. Holders receive these services in kind, so they accept a lower cash yield than the cash flows alone would justify. The term comes from commodity markets, where holding physical inventory carries a similar non-cash benefit.

Formally. If y^{obs} is the observed yield and y^{cf} is the yield the same cash flows would command without the services — for example on an equally safe but less money-like claim — the convenience yield is $y^{\text{cf}} - y^{\text{obs}}$. Krishnamurthy & Vissing-Jorgensen (2012) measure it from the spread of the highest-quality corporate bonds over Treasuries and find that it shrinks as the supply of Treasuries grows: a downward-sloping demand curve for safety and liquidity.

Why it appears here. It explains why the Treasury curve is not a clean risk-free curve (§1.5, §3.1) and why practitioners discount with OIS instead (§8.2); in March 2020 it briefly turned negative (§3.5).

Going deeper. Krishnamurthy & Vissing-Jorgensen (2012); He, Nagel & Song (2022); van Binsbergen, Diamond & Grotteria (2022).

Part 2 — Options

A.6 Call and put options

What it is. An option is a right without an obligation. A call is the right to buy an asset at a fixed price, the strike; a put is the right to sell at the strike. The buyer pays a premium up front; the seller keeps it and carries the obligation. Because the holder exercises only when it pays, the payoff is one-sided — and that asymmetry is the source of both an option's value and its characteristic risk.

Formally. At expiry, with the underlying worth S_T and strike K , a call pays $\max(S_T - K, 0)$ and a put pays $\max(K - S_T, 0)$. An option is in the money when immediate exercise would pay and out of the money when it would not; the part of its value above what immediate exercise would yield is its time value. For European options on an asset that pays nothing before expiry, put–call parity holds: $\text{Call} - \text{Put} = S - K Z(T)$. Merton's model (§5.5) is parity with the firm as the underlying: equity is a call on the assets V with strike F , debt is $F Z(T)$ minus a put, and since equity plus debt equals V , the identity $\text{Call} - \text{Put} = V - F Z(T)$ follows.

Why it appears here. Convertible bonds (§1.3, §5.4); credit as a short put on the firm (§5.5); callable bonds, where the issuer holds a call on its own debt (§2.5, §7.6); mortgages, where each homeowner holds a prepayment option (§5.6).

Going deeper. Sections 2.1–2.6 of Dealer Hedging and Gamma Exposure, which build option vocabulary from zero.

A.7 Volatility and option value

What it is. Because an option's payoff is floored on one side, more uncertainty about the underlying helps the holder: bigger moves in the favourable direction raise the payoff, and bigger moves in the other direction cannot push it below zero. So an option's value rises with the volatility of what it is written on. Volatility cannot be observed directly — only estimated from history or implied from option prices — which makes it the central, and most argued-over, pricing input.

Formally. For an option on a forward price F with strike K , expiry T and lognormal volatility σ , the Black (1976) formula values the call at $Z(T) [F N(d_1) - K N(d_2)]$, with $d_{1,2} = [\ln(F/K) \pm \frac{1}{2}\sigma^2 T]/(\sigma\sqrt{T})$ and N the standard normal distribution function. Its sensitivity to σ is positive for calls and puts alike. For an option on a bond, the bond's price volatility is approximately its duration times the volatility of its yield — the conversion used to price the callable bond in the §7.6 figure.

Why it appears here. Credit spreads widen with the firm's asset volatility (§5.5); option-adjusted spreads depend on the assumed volatility of rates (§8.4); the callable bond of §7.6.

Going deeper. Sections 3.1–3.3 and 3.7 of Dealer Hedging and Gamma Exposure; Hull & White (1990) for options on interest rates.

A.8 Long and short optionality: convexity, gamma and skew

What it is. Owning an option makes your gains accelerate when the market moves your way and your losses slow down when it does not: you are long convexity. Selling one does the reverse — you collect a small premium in quiet times and absorb large losses in violent ones. Bond convexity and option gamma are the same object, the curvature of value against the thing that moves it. A short-convexity position has a return distribution with a long left tail — negative skewness — which is why carry trades, credit and mortgage-backed securities share a family resemblance.

Formally. For a value $V(x)$, $\Delta V \approx V' \Delta x + \frac{1}{2} V'' (\Delta x)^2$, and $V'' > 0$ is long convexity. If x moves with zero mean and variance σ^2 a year, the curvature term contributes about $\frac{1}{2} V'' \sigma^2$ a year on average — for a bond, $\frac{1}{2} \mathcal{C} \sigma_y^2$ of return with σ_y the yield's volatility. That is why convexity is priced: a more convex bond yields less, and a short-convexity instrument must yield more (§7.4, §7.6). Skewness is $\mathbb{E}[(X - \mu)^3] / \sigma^3$; selling insurance-like payoffs makes it negative. Negative skew also diversifies slowly: averaging many independent positions shrinks variance like one over their number, but losses from a shared shock do not average away (A.12).

Why it appears here. Convexity helps and is priced (§7.4); negative convexity in callables and mortgages (§7.6); credit as a short put (§5.5); carry trades in bonds and currencies (§11.5, §15.3); the diversification argument in the credit spread puzzle (§9.5).

Going deeper. Sections 3.5 and 4.4–4.5 of Dealer Hedging and Gamma Exposure: the identity that a hedged option earns its gamma times the variance surprise is the general form of convexity's payoff.

Part 3 — Probability and credit

A.9 Hazard rates, Poisson arrivals and survival

What it is. A way of modelling an event that can strike at any moment without warning — a default, a machine failure. The hazard rate is the probability per unit of time that the event happens now, given that it has not happened yet. A constant hazard makes the event memoryless: a borrower that has survived five years is exactly as likely to default in the coming year as a new one. In reduced-form credit models, default is the first arrival of a Poisson process with this intensity.

Formally. With hazard $\lambda(t)$ and default time T_D , $\Pr(T_D \in (t, t + dt] \mid T_D > t) = \lambda(t) dt$, and the survival probability is $S(t) = \Pr(T_D > t) = \exp(-\int_0^t \lambda(u) du)$, which is $e^{-\lambda t}$ for a constant hazard. The exponential comes from chaining: surviving to t means surviving each of many short intervals in turn, and $\prod (1 - \lambda \Delta t) \rightarrow e^{-\lambda t}$ as the intervals shrink. The probability of default within one year is $1 - e^{-\lambda}$, close to λ when λ is small — 1.66% for a hazard of 1.67% a year.

Why it appears here. The survival term $Q(t)$ of Identity 1 (§1.4); the credit triangle (§9.2); reduced-form models (§9.7).

Going deeper. Duffie & Singleton (2003); Jarrow & Turnbull (1995) and Duffie & Singleton (1999) for the pricing framework; the Poisson process appears in section 9 of Stochastic Processes.

A.10 Systematic risk and factor models

What it is. Not every risk is paid. A risk that can be diversified away — one company's bad luck, which washes out across a large portfolio — earns no premium in equilibrium, because anyone who bears it can shed it for free. A risk that cannot be diversified away — a recession that hurts almost everything at once — must be paid, because someone has to hold it. Factor models make this operational: they write each asset's return as exposures to a few common sources of risk plus an idiosyncratic remainder, and say that expected excess returns compensate only the common exposures.

Formally. A linear factor model writes an asset's excess return as $R_i = \alpha_i + \sum_k \beta_{ik} f_k + e_i$, where the f_k are common factors, the β_{ik} are the asset's exposures to them, and e_i is a residual uncorrelated with the factors and across assets. If the factors capture all priced risk, then $\mathbb{E}[R_i] = \sum_k \beta_{ik} \gamma_k$, with γ_k the premium per unit of exposure to factor k , and $\alpha_i = 0$: the residual earns nothing, however volatile it is. For bonds, Fama & French (1993) proposed two such factors: a term factor, long government bonds minus bills, and a default factor, long corporate bonds minus long government bonds.

Why it appears here. Why a credit spread contains a premium beyond expected loss (§9.4–§9.5); why sovereign spreads move with global risk appetite (§4.3); the claims that credit, carry trades and currency carry are compensation for exposure to common risk rather than free money (§14.5, §15.3).

Going deeper. Fama & French (1993); Bai, Bali & Wen (2019) for credit; Lustig, Roussanov & Verdelhan (2011) for currencies.

A.11 Risk-neutral and real-world probabilities

What it is. Two different sets of probabilities describe the same future. Real-world (physical) probabilities describe how often things actually happen. Risk-neutral probabilities are the ones under which every price equals its expected payoff discounted at the risk-free rate. They differ because investors are not indifferent to risk: a payoff that arrives in a recession, when money is scarce, is worth more than the same payoff in a boom. Risk-neutral probabilities fold that valuation into the probabilities themselves, overweighting bad states. They are the right tool for pricing and the wrong tool for forecasting.

Formally. Absent arbitrage there is a positive stochastic discount factor M with $\text{price}(X) = \mathbb{E}^{\mathbb{P}}[MX]$ for every payoff X . Defining $\mathbb{Q}$ by $d\mathbb{Q}/d\mathbb{P} = M/\mathbb{E}^{\mathbb{P}}[M]$ gives $\text{price}(X) = Z \mathbb{E}^{\mathbb{Q}}[X]$, where $Z = \mathbb{E}^{\mathbb{P}}[M]$ is the risk-free discount factor. Where M is high — bad times — $\mathbb{Q}$ puts more weight than $\mathbb{P}$. Defaults cluster in bad times, so the risk-neutral default probability exceeds the real-world one, and the gap is the credit risk premium expressed as a probability.

Why it appears here. Default probabilities implied from spreads are risk-neutral, which is why they exceed historical default rates (§9.2, §9.4); the survival term $Q(t)$ in Identity 1 is of this kind (§1.4); the credit spread puzzle asks how large the gap should be (§9.5).

Going deeper. Duffie & Singleton (2003); Chen, Collin-Dufresne & Goldstein (2009) on why the gap is large; section 3.2 of Dealer Hedging and Gamma Exposure on how replication leads to risk-neutral pricing.

A.12 Default correlation and the one-factor model

What it is. Borrowers fail together because they share an economy. Default correlation does not change a portfolio's expected loss — that is just the sum of the individual expected losses — but it transforms the distribution around it. With independent defaults, a large portfolio loses almost exactly its expected loss every year. With correlated defaults it loses little in most years and a great deal in a few. Tranching (§5.6) slices exactly this distribution, which is why a senior tranche's safety is a bet on correlation.

Formally. In the one-factor Gaussian model, borrower i defaults if $\sqrt{\rho} M + \sqrt{1-\rho} \varepsilon_i < N^{-1}(\text{PD})$, where M is a common factor, the ε_i are independent, both are standard normal, and ρ is the asset correlation. Given M , defaults are independent with probability $\text{PD}(M) = N([N^{-1}(\text{PD}) - \sqrt{\rho} M]/\sqrt{1-\rho})$, so in a large portfolio the default rate is $\text{PD}(M)$ — random, and driven entirely by the common factor. With a 2% default probability: [Computed]

Asset correlation	Median year	1-in-100 year	1-in-1,000 year
0.0	2.0%	2.0%	2.0%
0.1	1.5%	8.2%	12.8%
0.2	1.1%	12.9%	22.6%
0.3	0.7%	17.6%	33.3%

Raising the correlation makes the typical year better and the bad year far worse — the signature of a risk that averaging cannot remove. The same model underlies the bank capital rules for credit risk.

Why it appears here. The fourth number of credit risk (§9.1); tranching and the 2007–08 failure (§5.6); the co-movement of default and recovery rates (§9.3).

Going deeper. Duffie & Singleton (2003).

A.13 Distance to default

What it is. The Merton model of §5.5 turned into a single comparable number: how many standard deviations the value of a firm's assets sits above the level at which it could no longer cover its debt. A firm with assets far above its debt and stable asset values is many standard deviations from default; a highly levered firm in a volatile business is only a few.

Formally. With asset value V , debt F due at horizon T , expected asset growth μ and asset volatility σ_V , $DD = [\ln(V/F) + (\mu - \frac{1}{2}\sigma_V^2)T]/(\sigma_V\sqrt{T})$, and under the model's lognormal assumption the default probability is $N(-DD)$. Neither V nor σ_V is observed; both are backed out of the equity price and equity volatility, using the fact that equity is a call on V . Commercial implementations map DD to default frequencies empirically rather than trusting the normal tail.

Why it appears here. §5.5 names it as a direct consequence of the Merton view, and §9.7 lists the structural approach among the ways default risk is estimated.

Going deeper. Merton (1974); Duffie & Singleton (2003).

A.14 Discriminant analysis and the Z-score

What it is. A classic statistical recipe for sorting cases into two groups: find the weighted combination of measurements that best separates, say, firms that went bankrupt from firms that did not, and score new firms by that combination. Altman's Z-score applies it to five accounting ratios and is still a standard first screen for distress.

Formally. Linear discriminant analysis chooses weights a to maximise the distance between the two groups' average scores relative to the spread within groups: $a \propto \Sigma_W^{-1}(\bar{x}_1 - \bar{x}_0)$, where $\bar{x}_1, \bar{x}_0$ are the groups' mean ratio vectors and Σ_W their pooled within-group covariance matrix. A firm's score is $a^\top x$, and a threshold classifies it. The weights are proportional to the coefficients of a linear regression of the group label on the ratios. Altman (1968) combines working capital, retained earnings, operating earnings and sales, each scaled by total assets, with the market value of equity relative to total liabilities.

Why it appears here. The accounting-score row of the §9.7 table.

Going deeper. Altman (1968); section 3.2 of Foundations of Econometrics on linear projection, of which discriminant analysis is a special case.

Part 4 — Instruments and plumbing

A.15 Repurchase agreements

What it is. A repo is a short-term loan secured by a bond, written as a sale plus an agreed repurchase. The cash lender holds the bond as collateral and lends somewhat less than its value; the gap, the haircut, protects the lender if the bond falls in price. Most repo is general collateral: any bond of a class will do, and the rate sits close to the overnight policy rate. When a particular bond is in heavy demand, typically to cover short sales, it trades special: cash lenders accept a lower rate to get that bond specifically.

Formally. Against a bond worth P with haircut h , the cash lent is $P(1 - h)$ and the repurchase price is $P(1 - h)(1 + r_{\text{repo}}d/360)$ for a d -day term at the money-market convention. The maximum leverage is roughly $1/h$: a 2% haircut supports a position about fifty times the equity behind it. Specialness is the general collateral rate minus the special rate; a bond expected to stay special earns its owner that saving, and is priced higher for it (Duffie, 1996).

Why it appears here. Financing and leverage in the Treasury market (§3.4); the basis trade (§3.4); LTCM and March 2020 (§2.3, §3.5); funding costs in the CDS–bond basis (§8.5).

Going deeper. Stigum & Crescenzi (2007); Duffie (1996); Gorton & Metrick (2012) on how repo funding can run.

A.16 Margin, collateral and mark-to-market

What it is. Derivative and repo positions are revalued every day, and the side that has lost must post cash or bonds to the side that has gained — variation margin. This keeps the credit exposure between counterparties small, but it converts price moves into immediate demands for cash. A position can be economically sound, even perfectly hedged against the liabilities it exists to match, and still fail because it cannot find the cash for today's call.

Formally. Variation margin paid on a day equals the fall in the position's value, $-\Delta V$; for a rate position that is about its DV01 times the yield move in basis points. Initial margin is an extra buffer sized to cover a severe move. For scale: a long position of 1 billion in 30-year bonds or receive-fixed swaps at a 4% yield loses about 155 million if yields rise 100 basis points (174 million by duration alone), and the cash must be found at once, whatever has happened to the liabilities being hedged. [Computed]

Why it appears here. The UK LDI episode, where rising yields generated collateral calls that forced gilt sales (§13.4); leverage in repo and futures (§3.4); LTCM (§2.3).

Going deeper. Duffie (2020) on dealer balance sheets and market functioning.

A.17 Bond futures, cheapest-to-deliver and the basis

What it is. A Treasury futures contract obliges the seller to deliver a government bond on a future date at a price fixed today. The seller may choose which bond to deliver from a basket of eligible issues, each scaled by a conversion factor; in practice one issue is cheapest to deliver, and the futures price tracks it. The basis is the gap between a bond's cash price and its futures-implied price. The basis trade buys the cash bond, finances it in repo and sells the future, earning the difference between the financing rate implied by the prices and the actual repo rate.

Formally. For a deliverable bond i with conversion factor CF_i and futures price F , $\text{basis}_i = P_i^{\text{clean}} - F \cdot CF_i$. The cheapest-to-deliver issue is the one with the highest implied repo rate — the return from buying it now and delivering it into the future. The basis trade earns roughly the implied repo rate minus the actual repo rate over the holding period, which is small, so it is run at high leverage.

Why it appears here. The cash–futures basis trade and its role in March 2020 (§3.4); the delivery option as a template for the cheapest-to-deliver option in CDS (§8.5).

Going deeper. Tuckman & Serrat (2022) on note and bond futures; Barth & Kahn (2025).

A.18 Interest rate swaps and OIS

What it is. An agreement to exchange a fixed interest rate for a floating one on a notional amount that never changes hands. One side pays, say, 4% a year fixed; the other pays whatever the floating rate turns out to be. The fixed rate that makes the swap worth nothing at inception is the swap rate, and swap rates by maturity form the market's other main yield curve. In an overnight indexed swap the floating leg compounds an overnight rate — SOFR, €STR, SONIA — so the curve carries almost no bank credit risk.

Formally. Paying a fixed rate s on accrual fractions α_i against a floating leg worth par at inception, the swap is fair when $s \sum_i \alpha_i Z(t_i) = 1 - Z(T)$, so

$$s_T = \frac{1 - Z(T)}{\sum_i \alpha_i Z(t_i)},$$

the par-yield formula of §6.2. A receive-fixed swap is economically a fixed-rate bond financed by a floating-rate note, so its DV01 is close to that of a par bond of the same maturity.

Why it appears here. OIS as a cleaner discount curve (§3.1, §8.2); the I-spread and the asset swap spread (§8.2); inflation swaps in the TIPS–Treasury comparison (§3.3); pension liability hedging (§12.3, §13.4).

Going deeper. Schrimpf & Sushko (2019); Tuckman & Serrat (2022) on swaps.

A.19 Covered and uncovered interest parity

What it is. Two statements about how interest rates and exchange rates connect. Covered parity is an arbitrage relation: investing abroad and locking in the exchange rate home with a forward contract must earn the same as investing at home. Uncovered parity is a hypothesis: investing abroad without the hedge should earn the same as at home on average, because high-rate currencies should depreciate by the rate difference. The first held almost exactly until 2008 and has deviated persistently since; the second has failed for as long as it has been tested.

Formally. With S and Fwd the spot and forward prices of one unit of foreign currency in domestic units, and $i_{\text{dom}}, i_{\text{for}}$ the nominal interest rates for the forward's term, covered parity is $\text{Fwd}/S = (1 + i_{\text{dom}})/(1 + i_{\text{for}})$. A persistent deviation is measured by the cross-currency basis, the adjustment to one currency's rate needed to restore equality. Uncovered parity is $\mathbb{E}[S_T]/S = (1 + i_{\text{dom}})/(1 + i_{\text{for}})$. The standard test regresses the realised change in the exchange rate on the forward premium; parity implies a slope of one, and estimates have typically been below zero (Fama, 1984).

Why it appears here. The identity that hedging removes a foreign bond's yield pickup (§15.1); the cross-currency basis (§15.2); the carry trade (§15.3).

Going deeper. Du, Tepper & Verdelhan (2018); Fama (1984); Lustig, Roussanov & Verdelhan (2011).

A.20 Credit default swaps

What it is. Insurance against a borrower's default, traded as a derivative. The protection buyer pays a regular premium; if a defined credit event occurs, the seller pays the buyer the loss — face value minus the recovery value of the defaulted debt,

fixed by an industry auction. Whether a credit event has happened is decided not by a court but by a committee of dealers and investors convened under the industry's standard contract terms, the ISDA Determinations Committee — which is why the CDS definition of default is contractual, and narrower than the economic one. Anyone can buy protection, whether or not they own the bonds, so CDS are used both to hedge and to take views.

Formally. With survival probabilities $S(t)$ (A.9) and recovery R , the premium leg is worth $s \sum_i \alpha_i Z(t_i) S(t_i)$ and the protection leg $(1 - R) \sum_i Z(t_i) [S(t_{i-1}) - S(t_i)]$. Setting them equal with a flat hazard over short periods gives $s \approx \lambda(1 - R)$ — the credit triangle of §9.2, derived from the contract itself. Contracts now trade with standardised coupons, such as 100 or 500 basis points, plus an upfront payment that makes up the difference from the fair spread.

Why it appears here. CDS as the cleanest credit measure, and the CDS–bond basis (§8.5); the three definitions of default and why a CDS hedges only the contractual one (§16.1).

Going deeper. Duffie & Singleton (2003).

A.21 Uniform-price and pay-as-bid auctions

What it is. Two ways to sell a fixed quantity to many bidders. In a pay-as-bid (discriminatory) auction each winner pays its own bid, so bidders shade their bids below their true value to avoid overpaying — and worry about the winner's curse, winning precisely because they valued the bond more than everyone else. In a uniform-price auction every winner pays the same market-clearing price, which weakens the incentive to shade. The US Treasury has used the uniform format for all its auctions since 1998.

Formally. Bids — a yield and a quantity — are sorted from the lowest yield upward, and the clearing (stop-out) yield y^* is the highest yield needed to fill the offering. In a uniform-price auction every accepted bid is filled at y^* , with bids exactly at y^* pro-rated; in pay-as-bid, each is filled at its own yield. Theory does not rank the two formats' revenue in general.

Why it appears here. The primary market for government bonds (§2.2).

Going deeper. Malvey & Archibald (1998) on the Treasury's own experiment.

A.22 Over-the-counter markets and search frictions

What it is. On an exchange, orders meet in a central book. In an over-the-counter market you trade bilaterally: you ask dealers for prices, one at a time or through an electronic request, and bargain. Prices then depend on who you are and what alternatives you have — an investor who can easily call ten dealers gets a better price than one who can call one — and a dealer holding inventory charges for the risk and the balance sheet it uses. Search frictions are the costs of finding and negotiating with a counterparty.

Formally. In the model of Duffie, Gârleanu & Pedersen (2005), investors meet dealers at a rate that reflects their search technology and split the gains from trade by bargaining. The bid–ask spread an investor pays rises with the dealer's bargaining power and falls with how quickly the investor could find another dealer, so better-connected investors face tighter spreads for the same bond.

Why it appears here. Why bonds trade through dealers rather than on exchanges, and why retail investors pay more (§2.3, §17.2); the value of dealer balance sheet in a crisis (§3.5).

Going deeper. Duffie, Gârleanu & Pedersen (2005); Hendershott & Madhavan (2015); section 1.2 of Dealer Hedging and Gamma Exposure on what a market maker sells.

A.23 ETF creation, redemption and NAV

What it is. A bond ETF's shares trade on an exchange all day, while the bonds it owns trade over the counter, if at all. The link between the two is a group of large dealers, authorised participants, who can deliver a basket of bonds to the fund in exchange for new shares (creation) or hand shares back in exchange for bonds (redemption). When the ETF's price rises above the value of its holdings they create and sell shares; when it falls below, they buy shares and redeem. The net asset value (NAV) is the fund's estimate of its holdings' value — for bonds, largely matrix prices: estimates for bonds that did not trade, interpolated from the prices of comparable bonds that did.

Formally. The premium is $(\text{ETF price} - \text{NAV})/\text{NAV}$. Arbitrage keeps it within the cost of creating or redeeming, which includes the bid–ask spread on the underlying bonds; when those costs spike, the band widens. If the NAV is stale, a large “discount” can simply mean the ETF price has moved and the NAV has not.

Why it appears here. ETF discounts in March 2020 (§17.4); the run incentive in daily-dealing mutual funds (§12.4).

Going deeper. Koont, Ma, Pástor & Zeng (2025); Goldstein, Jiang & Ng (2017) for the mutual fund comparison.

Part 5 — Macro, policy and the curve

A.24 Central bank reserves and the policy rate

What it is. Reserves are the deposits commercial banks hold at the central bank — the money banks use to pay one another. Modern central banks set overnight interest rates largely by choosing what they pay on those reserves: no bank will lend overnight for less than it can earn risk-free at the central bank, so the rate on reserves acts as a floor under market rates. Quantitative easing creates reserves, because the central bank pays for the bonds it buys by crediting the reserve accounts of the sellers’ banks.

Formally. In the US “ample reserves” framework, the Federal Reserve announces a target range for the federal funds rate and steers market rates into it with two administered rates: the interest it pays on banks’ reserve balances, and the rate on an overnight reverse repo facility open to a wider set of institutions such as money market funds. Short-term market rates trade close to these administered rates for as long as reserves remain plentiful.

Why it appears here. How the policy rate anchors the front of the curve, and how QE and QT work (§10.5).

Going deeper. Stigum & Crescenzi (2007) on money-market plumbing.

A.25 State-space models and the Kalman filter

What it is. A framework for estimating something you cannot see from things you can. The unobserved quantity — the natural real rate, say — evolves over time by a simple rule; the observed data — output and inflation — depend on it plus noise. The Kalman filter updates the best estimate of the hidden quantity each period, weighting the new data by how informative they are relative to the uncertainty already carried forward.

Formally. A state equation $\xi_{t+1} = \Phi \xi_t + w_t$ and an observation equation $\zeta_t = \Lambda \xi_t + v_t$, with independent Gaussian noises w_t and v_t . The filter alternates a prediction, $\hat{\xi}_{t|t-1} = \Phi \hat{\xi}_{t-1|t-1}$, with a correction, $\hat{\xi}_{t|t} = \hat{\xi}_{t|t-1} + G_t (\zeta_t - \Lambda \hat{\xi}_{t|t-1})$, where the gain G_t is large when the data are precise relative to the prior. When the observations carry little information about the state — as output and inflation do about the natural rate — the gain is small, the estimate moves slowly, and its confidence band is wide.

Why it appears here. The Laubach–Williams and Holston–Laubach–Williams estimates of r^* (§10.2); many term-premium models are estimated the same way (§10.4).

Going deeper. Laubach & Williams (2003); Appendix A.25 of Foundations of Econometrics.

A.26 Government debt dynamics: $r - g$

What it is. A government’s debt, measured against the size of the economy, grows with the interest it pays and shrinks as the economy grows. If the interest rate on the debt exceeds the growth rate, debt compounds faster than the economy and must eventually be stabilised with budget surpluses. If growth exceeds the rate, the debt ratio shrinks by itself, and the government can run modest deficits indefinitely without the ratio rising.

Formally. With debt-to-GDP b , real interest rate r on the debt, real growth rate g and primary surplus s (the budget balance excluding interest) as a share of GDP, $b_{t+1} = b_t (1 + r)/(1 + g) - s_{t+1}$. The same identity holds with nominal rates on both sides, since inflation cancels. The ratio is stable when $s = b (r - g)/(1 + g)$. At a debt ratio of 100% and 2% real growth: [Computed]

$r - g$	Primary balance that holds debt/GDP steady
-1 point	deficit of 0.98% of GDP
0	balanced budget excluding interest
+2 points	surplus of 1.96% of GDP

Why it appears here. Blanchard’s argument about low rates and fiscal dominance (§10.6), and why the euro-area sovereigns of §4.2 were so sensitive to their own borrowing rate.

Going deeper. Blanchard (2019); Jiang, Lustig, Van Nieuwerburgh & Xiaolan (2024); Sargent & Wallace (1981).

A.27 Multiple equilibria and self-fulfilling crises

What it is. Sometimes an outcome is set by what people expect rather than by fundamentals alone, because the expectations change the fundamentals. If investors fear a government will default, they demand a high yield; the high yield raises its interest bill; the bill makes default likelier — so the fear is justified. If they are calm, the yield stays low and the calm is justified too. Both outcomes are self-consistent, and which one prevails is a coordination problem. A credible backstop can select the good outcome without ever being used, because once investors believe in it the bad outcome stops being self-consistent.

Formally. The sovereign's yield must solve a fixed point, $y = r_f + \lambda(y)(1 - R)$ — the credit triangle of §9.2 with r_f the rate on safe debt — where the default intensity λ rises with the interest burden and so with y itself. If $\lambda(y)$ is steep enough over some range, the equation has two stable solutions, one with a low yield and little default risk and one with a high yield and a lot, separated by an unstable one. Bank runs have the same structure, as does the first-mover advantage in daily-dealing bond funds.

Why it appears here. The euro-area crisis and the ECB's "whatever it takes" (§4.2); the doom loop in the UK LDI episode (§13.4); the run incentive in bond funds (§12.4).

Going deeper. Cole & Kehoe (2000), the standard model of self-fulfilling debt crises.

A.28 Principal components of the yield curve

What it is. Yields at different maturities move together, but not identically. Principal component analysis finds the few independent patterns of movement that account for most of the variation — here a level shift in which everything moves together, a change in slope in which short and long rates move oppositely, and a change in curvature in which the middle moves against the ends. Because these few patterns explain nearly everything, a whole curve's risk can be summarised by three numbers.

Formally. Collect changes in yields at n maturities and compute their covariance matrix Σ . Its eigendecomposition $\Sigma = V \Omega V^\top$ gives orthogonal directions — the columns of V , the components' loadings — and their variances, the diagonal entries ω_k of Ω . Component k explains the share $\omega_k / \sum_j \omega_j$ of total variance. For yield curves the first loading vector is nearly flat, the second changes monotonically across maturities, and the third is hump-shaped.

Why it appears here. Level, slope and curvature (§11.4); why key-rate durations beyond three points add little (§7.7).

Going deeper. Litterman & Scheinkman (1991); Appendix A.28 of Portfolio Construction and the Covariance Matrix on principal component analysis in general.

A.29 Affine term-structure models

What it is. A family of models in which a handful of unobserved factors drive the short-term interest rate, and every bond yield is a linear ("affine") function of those factors, with coefficients pinned down by the requirement that there be no arbitrage across maturities. The linearity makes pricing tractable. Because the model specifies both how the factors actually evolve and how they are priced, it can split each yield into an expected-rate part and a term premium.

Formally. With a state vector X_t following a Gaussian autoregression and a short rate $r_t = \delta_0 + \delta_1^\top X_t$, the price of a zero-coupon bond with n periods to maturity takes the form $\exp(A_n + B_n^\top X_t)$, so its yield, $-(A_n + B_n^\top X_t)/n$, is linear in the state. The coefficients A_n and B_n follow recursions set by the factor dynamics under the pricing (risk-neutral) measure. The term premium is the model yield minus the average of expected future short rates computed under the real-world dynamics. The one-factor model of Vasicek (1977), in which the short rate mean-reverts, $dr_t = \kappa(\theta - r_t) dt + \sigma dW_t$, is the simplest member.

Why it appears here. The Kim–Wright and Adrian–Crump–Moench term premium estimates (§10.4), and the modelling lineage of §11.4.

Going deeper. Piazzesi (2010); Duffie & Kan (1996); Stochastic Processes for the Brownian motion W_t .

Part 6 — Corporate finance and distress

A.30 Modigliani–Miller and the trade-off theory

What it is. The baseline result of corporate finance: in a world without taxes, bankruptcy costs or information problems, how a firm splits its financing between debt and equity cannot change its total value — slicing a pie differently does not make it bigger. Everything interesting about corporate debt comes from the ways the real world departs from that baseline. The

main departures give the trade-off theory: debt adds value through the tax deductibility of interest and destroys it through the expected costs of financial distress, and a firm settles where the two balance at the margin.

Formally. Without frictions, $V_{\text{levered}} = V_{\text{unlevered}}$. With a corporate tax rate ϕ and permanent debt D_{debt} , the tax shield adds ϕD_{debt} . The trade-off theory writes $V_{\text{levered}} = V_{\text{unlevered}} + \text{PV}(\text{tax shields}) - \text{PV}(\text{distress costs})$ and puts optimal leverage where the marginal tax benefit equals the marginal expected distress cost; Leland (1994) derives this inside a structural credit model.

Why it appears here. Why companies issue debt (§1.5) and how much (§5.1).

Going deeper. Modigliani & Miller (1958); Leland (1994).

A.31 Agency costs and debt overhang

What it is. Managers, shareholders and creditors want different things, and the conflicts cost money. Two matter most to bondholders. Asset substitution: shareholders of a heavily indebted firm gain from gambling, because they keep the upside while creditors absorb the downside. Debt overhang: they may turn down a profitable investment because most of its benefit would go to making the existing debt safer. Covenants exist largely to contain these incentives.

Formally. Both follow from the Merton view (§5.5, A.6). Equity is a call on the firm's assets, so its value rises with asset volatility — shifting into riskier projects pays shareholders at creditors' expense. For overhang: a project costing I , funded by the shareholders, raises firm value by $\Delta V > I$, but part of that gain, ΔD_{debt} , accrues to creditors as their claim becomes safer. Shareholders go ahead only if $\Delta V - \Delta D_{\text{debt}} > I$, so projects with $I < \Delta V < I + \Delta D_{\text{debt}}$ are rejected even though they create value.

Why it appears here. The costs of debt (§1.5, §5.1); the value of covenants (§2.1, §5.2); event risk from debt-funded buybacks (§5.1).

Going deeper. Jensen & Meckling (1976); Myers (1977); Chava & Roberts (2008) on covenants in action.

A.32 The absolute priority rule and Chapter 11

What it is. The rules that decide who gets what when a US company reorganises. Creditors are grouped into classes by the nature of their claims; the company proposes a plan stating what each class receives; the classes vote. The absolute priority rule says a dissenting senior class must be paid in full before any junior class receives anything. The class whose claim is only partly covered by the reorganised firm's value — the fulcrum — typically receives most of the new equity.

Formally. A class accepts a plan if creditors holding at least two-thirds of the amount and more than half of the number of claims voting in that class approve. A plan can be confirmed over a dissenting class — a cramdown — if at least one impaired class accepts and the plan is “fair and equitable” to the dissenter, which for unsecured creditors means absolute priority. New debtor-in-possession financing can be granted priority over existing claims to keep the business running.

Why it appears here. The resolution paths, and why absolute priority bends in practice (§16.2–§16.3); the fulcrum security in distressed investing (§16.6).

Going deeper. Acharya, Bharath & Srinivasan (2007) on how the process shows up in recoveries; Sturzenegger & Zettelmeyer (2006) for the sovereign contrast, where none of this machinery exists.

Part 7 — Statistics and portfolios

A.33 Predictive regressions and their traps

What it is. Regressing future returns on something known today — the slope of the curve, a forward spread — to see whether it forecasts them. The method is simple and the inference is treacherous in exactly the setting of §11.2: the predictors move slowly, the samples are short, and multi-year returns sampled every month overlap, so consecutive observations share most of their information. Each of these makes the evidence look stronger than it is.

Formally. The regression is $r_{t \rightarrow t+h} = a + b x_t + \varepsilon_{t+h}$. With overlapping h -period returns the errors are serially correlated and naive standard errors understate the uncertainty; autocorrelation-robust corrections help but are unreliable in short samples. When x_t is persistent and its innovations are correlated with returns, $\hat{b}$ is biased in finite samples — the bias of Stambaugh (1999), approximately $-(\sigma_{uv}/\sigma_v^2)(1 + 3\phi)/T$ for an autoregressive predictor with persistence ϕ over T observations, where σ_{uv} is the covariance between the return and predictor innovations and σ_v^2 the predictor innovation's variance. In-sample R^2 also rises mechanically with the horizon even when predictability is modest.

Why it appears here. The expectations-hypothesis tests, and the critique of bond-return predictability (§11.2).

Going deeper. Section 6.6 and Appendix A.22 of Simple and Log Returns; section 4.5 of Foundations of Econometrics on autocorrelation-robust standard errors; Bauer & Hamilton (2018).

A.34 Risk contributions, risk parity and the 60/40 portfolio

What it is. A portfolio’s capital weights say where the money is; its risk contributions say where the volatility comes from, and the two can be very different. The classic 60/40 portfolio — 60% equities, 40% bonds — is balanced by capital but carries almost all of its risk in equities, because equities are far more volatile. Risk parity sizes assets so that each contributes equally to risk, which for a stock–bond portfolio means holding far more bonds, usually with leverage to reach a useful return. Both designs depend on the stock–bond correlation, and risk parity depends on it heavily.

Formally. With weights w and covariance matrix Σ , asset i contributes $w_i(\Sigma w)_i$ to the portfolio variance $w^\top \Sigma w$, and the contributions sum to the total. With 16% equity volatility and 6% bond volatility: [Computed]

Stock–bond correlation	60/40 volatility	Equity share of risk
−0.3	9.17%	101%
0.0	9.90%	94%
+0.3	10.57%	89%

At a negative correlation the bond sleeve’s contribution is itself negative — it subtracts risk — which is the regime §14.2 describes. Equal risk contributions from two assets require weights inversely proportional to their volatilities: 27% equities and 73% bonds here, before leverage.

Why it appears here. The 60/40 portfolio and risk parity as products of the negative-correlation regime (§14.2), and what 2022 did to them (§14.4).

Going deeper. Section 4.5 of Portfolio Construction and the Covariance Matrix on risk-based portfolios.